Model-Directed Attention and the Persistence of Wrong Mental Models: Evidence from Firm Pricing*

Yi Han David Huffman Yiming Liu

November 13, 2026

Click to see the latest version

Abstract

We document a mechanism through which wrong mental models and beliefs survive in data-rich environments: by remaining silent about the patterns they deem irrelevant, wrong models direct attention away from the patterns that refute them. We examine this in firm pricing under consumer intertemporal substitution (IS), where shifts in purchase timing add to the sales spike on the discount day and produce dips before and after. A simple model in which lower prices lead to higher sales is silent about the dips and attributes the spike entirely to new demand. In survey and administrative data from 13,000 gas station managers, more than 20 percent completely neglect IS, and those holding the simple model perceive demand as more elastic and set lower prices, which leads to lower profits. Online experiments provide causal evidence. Holding the data fixed, subjects assigned the simple model are less likely to recall the dips and more likely to neglect IS. Two interventions further support the mechanism. Prompting attention to the dips lowers neglect of IS, and encountering a second model dislodges the incumbent, but only if it draws attention to data features neglected by the incumbent.

*Han: Renmin University of China; yihanecon@ruc.edu.cn. Huffman: Cornell University; dbh234@Cornell.edu. Liu: Humboldt University of Berlin and WZB; yiming.liu@hu-berlin.de.

1 Introduction

Traditional economic theory assumes decision makers hold a correct model of the economic environment, and rationally update beliefs about model parameters to the true values, given sufficient data. An important advance has been theories that weaken the first half of this assumption, and show how beliefs need not converge to the true values, no matter how much data arrive, if the model itself is wrong and held fixed (Berk, 1966; Esponda and Pouzo, 2016; Spiegler, 2016; Heidhues et al., 2018, 2021; Fudenberg et al., 2021); see Bohren and Hauser (2025) for a review. This has led in turn to a growing literature that allows for sophistication in terms of decision makers potentially revising a wrong "mental model," but also proposes various mechanisms for why revision might not occur (e.g., Ortoleva, 2012; Cho and Kasa, 2015; Schwartzstein and Sunderam, 2021; Fudenberg and Lanzani, 2023; Ba, 2026).

This paper provides the first direct test of one such mechanism, model-directed attention, which can lead a decision maker to keep a wrong mental model even when the disconfirming data are right in front of them. Formalized by Gagnon-Bartsch et al. (2026), the idea is that a mental model directs attention towards the patterns it deems relevant, while the decision maker chooses models based on how well they fit the attended data. One prediction is that a specific, and potentially broad, class of wrong models can persist in the face of data: simple wrong models that explain the patterns they deem relevant well, but are silent about the patterns they cannot explain. This has important implications, as such "blind spot" models are likely pervasive, and data availability is not sufficient to correct wrong beliefs or prevent mistakes. A second prediction concerns encountering another model: even an incomplete or wrong second model can dislodge the initial one, and lead to the correct model, if it broadens attention to data features the initial model cannot explain and thereby lowers its fit on the attended data. This implies that communicating models may correct beliefs more effectively than data, and helps even when it comes from others with wrong models, as long as models are diverse.

We test these two sets of predictions in the field and in the lab. In the field we measure the mental models and economic decisions of roughly 13,000 managers in an organization. Online, we run experiments with roughly 1,400 Prolific subjects. These data sources allow us to do what is otherwise difficult: hold the data fixed in an environment in which we know the correct model, test consequences of wrong mental models for beliefs and high-stakes decisions, assign the initial mental model, attention, and subsequent model exogenously, and measure causal effects on attention to specific data features, model revision, and beliefs.

Our investigation necessarily focuses on a particular application of model-directed attention, but one that is of central importance in economics: mental models about how consumer behavior responds to price. This concerns one of the most fundamental causal relationships in economics, which is a key input into pricing decisions. A key feature of how consumers

respond to temporary price cuts is intertemporal substitution (IS): regular customers shift the timing of purchases they would have made anyway, and this shift contributes to the sales spike during a pre-announced discount and creates dips in sales before and after it. We hypothesize that some decision makers instead have an overly simple model in which low prices lead to high sales and consumers react only to the contemporaneous price, so that the entire sales spike during a discount is attributed to the low price while the model is silent about the dips. These dips in sales are data features that can disconfirm the simple wrong model, if they are noticed.

Section 2 introduces our main measure of whether a decision maker's mental model neglects consumer intertemporal substitution, the *detergent task*. Decision makers see hypothetical sales data for an online detergent seller, a product for which consumers are known to substitute intertemporally (Hendel and Nevo, 2006b). They know the data were generated from underlying rules but are not told the rules. The table repeats the same noise-free pattern twice: sales are 100 on regular days, 202 on discount days, and below 100 on the days just before and after each discount, while the competitor's price stays constant. Every decision maker thus sees the same data, which have an objectively correct interpretation. Decision makers then predict daily sales if the discount price were charged every day. The simple model attributes the discount-day spike entirely to new demand and predicts 202; the correct model recognizes that part of the spike consists of purchases shifted from surrounding days and predicts 165. Predicting 202 is our main indicator of neglecting intertemporal substitution.

We then provide a conceptual framework, building on Gagnon-Bartsch et al. (2026), in which the simple model persists through model-directed attention. Under the simple model every regular-price day is the same, so there is no reason to attend to more than one, and the dips go unnoticed. On the data the decision maker attends to, the simple model fits as well as the correct one and can persist in the long run. The framework yields three further predictions. First, directing attention to the dips dislodges the simple model. Second, holding the simple model and thus neglecting intertemporal substitution leads a decision maker to overestimate the elasticity of demand and to set lower prices. Third, encountering a second model has asymmetric effects: the correct model survives encountering the simple one, but the simple model can be dislodged by encountering the correct model, or even another wrong model, provided it draws attention to the dips.

Section 3 provides evidence from the field. We collaborate with a firm that operates more than 20,000 gas stations, each run by a manager with substantial influence over the station's fuel prices. Managers are incentivized based on station profits, so they care about setting price optimally. Intertemporal substitution is plausible in fuel consumption: drivers can delay refueling, for example by choosing another way to commute, and can stockpile when prices are low because they usually have empty space in the tank. We go beyond plausibility by testing for intertemporal substitution directly in a region where we have

transaction-level data. Specifically, we employ an event study focusing on stations before and after they introduce a weekly discount day. Because the discount falls on the same day each week, customers can anticipate it, and we find that loyal customers of a station concentrate their purchases on the discount day while the share of their purchases on the surrounding days falls. In a survey sent to all station managers, we presented the detergent task, and elicited managers' beliefs about the price elasticity of fuel demand and about the profitability of fuel price cuts. We then match these measures to panel data on the prices the managers set at their stations.

We find that a non-trivial number of managers completely neglect intertemporal substitution, in a way that is consistent with model-directed attention. Specifically, roughly 22 percent of managers predict 202 in the detergent task, that is, they predict that sales would be the same when the price is low every day as when it is low for only one day. This wrong model is persistent in the sense that it survives being shown two cycles of noise-free data that refute it. Also, it is only weakly negatively correlated with years of experience as a station manager, another proxy for exposure to data. We also find evidence consistent with model-directed attention. Managers who predict 202 are less likely than other managers to give an explanation of the data that reveals they noticed the pre-discount dip or the post-discount dip, while their explanations mention the discount-day spike as often as those of managers who predict 165, the correct answer.

We next show that the mental model elicited with the detergent task is empirically relevant for how managers understand and price fuel. First, it generalizes to how managers interpret lasting price cuts at gas stations. When shown station data around a lasting fuel discount and asked to predict daily sales had the cut started earlier, managers who neglect or underestimate intertemporal substitution are significantly less likely to shift the initial sales spike to the earlier start date. Second, the detergent measure predicts managers' perceived price elasticity of fuel demand, both when all managers evaluate the same typical station and when they evaluate their own station. Across both scenarios, greater neglect of intertemporal substitution is associated with perceiving fuel demand as more elastic. We then examine the actual fuel prices managers set at their stations, controlling for district-by-month fixed effects, station and market conditions (e.g. location and number of local competitors), and manager characteristics (cognitive skills, strategic sophistication, and demographics). Managers who predict 202 charge significantly lower fuel prices than managers who predict 165. Evaluating this price gap with a station-level demand model estimated from exogenous cost shifters, we find that the lower prices of managers who predict 202 imply 3.3 percent lower daily fuel profit at the average station relative to managers who predict 165.

Section 4 tests the predictions of the conceptual framework in Section 2 and provides causal evidence on model-directed attention. We run two online experiments with an independent sample from Prolific, Study 1 and Study 2, and again use the detergent task.

Because the framework predicts that a mental model directs attention, we hold the data fixed while exogenously assigning subjects different peer models, motivated by the idea that decision makers such as managers can learn their models from peers. To obtain these peer models, we ran an initial pilot in which subjects saw the same kind of detergent data as the station managers and explained how price relates to sales in their own words. We selected representative descriptions written by subjects who predicted 202 and 165 as the wrong model and the correct model, respectively.

Study 1 ($N = 549$) tests model-directed attention by randomly assigning subjects a mental model before they see the detergent data. In two treatments, subjects receive either the correct model, which explains the dips by customers shifting the timing of their purchases, or the simple (wrong) model, in which a low price leads to high sales. As in the manager survey, subjects make a prediction of daily sales under a permanent low price, which identifies whether they adopt the assigned model. In addition, they state their confidence in their prediction, and explain the data in their own words. To measure attention, we then ask subjects to recall the sales data they have just seen, given the date and the corresponding price. The random assignment works: being assigned the wrong model leads 74 percent of subjects to predict 202, compared to 32 percent among subjects assigned the correct model. For recall, our theory predicts that while holders of the correct model recall both dips, holders of the wrong model ignore both dips and recall that sales were the same when the price was high. This is what we find, both for median recall and for the share recalling each pattern. Among subjects assigned the wrong model, median recalled sales are 202 on the discount day and 100 on every regular-price day, whereas among subjects assigned the correct model they fall below 100 on the dip days. Almost everyone recalls the discount-day spike, at 94 percent when assigned the wrong model compared to 88 percent when assigned the correct model. Being assigned the wrong model, however, significantly lowers the share of subjects that recall the pre-discount dip, from 41 to 27 percent, and the post-discount dip, from 55 to 27 percent.

In our framework, the wrong model is not refuted by the data despite its poor fit because it directs attention away from the dips. One way to reduce its fit and thus potentially dislodge it is to prompt attention to the dips. In a third treatment, we add attention prompts to the wrong model. Specifically, we ask subjects to explain the sales patterns around each discount, such as two days and one day before the discount. This makes subjects realize that sales are different on these two days even though they have the same price, something the simple wrong model cannot explain. By just these prompts, without providing new information or offering an explanation for the dips, we restore recall of the pre-discount and post-discount dips from 27 and 27 percent to 49 and 59 percent, and, more importantly, reduce the share predicting 202 relative to the wrong-model treatment from 74 to 63 percent, a difference significant at the 5 percent level.

Study 2 ($N = 896$) tests whether encountering a second peer model can dislodge the

model a subject has already adopted. In stage 1, as in Study 1, subjects are randomly assigned either the simple (wrong) model or the correct model, see the detergent data, and predict daily sales under a permanent low price. In stage 2, subjects encounter the other model and make a second prediction, with those assigned the wrong model first now reading the correct model and vice versa. Our theory predicts an asymmetric effect of encountering the second model. For subjects who adopt the wrong model in stage 1, encountering the correct model directs attention to the previously ignored dips, which reduces the fit of the wrong model and can dislodge it. By contrast, subjects who adopt the correct model in stage 1 have already noticed the dips, and encountering the wrong model should not move them toward the wrong model. This is what we find when subjects make their second prediction. Among subjects assigned the wrong model first, encountering the correct model reduces the share predicting 202 from 72 to 37 percent, compared to a change from 28 to 24 percent among those assigned the correct model first. Even after both groups have seen both models, however, the correction is incomplete. In stage 2, 37 percent of subjects assigned the wrong model first still predict 202, compared to 24 percent of those assigned the correct model first. This order effect is consistent with some subjects not entertaining the second model, and remaining at 202 in stage 2 is indeed more common among subjects who were more confident in their initial prediction.

Study 2 also tests the prediction that even an incomplete second model can dislodge the simple wrong model and lead to the correct model, provided it directs attention to the dips. The peer-written description used in the correct-model treatment tests this prediction directly because it is incomplete. Specifically, it explains the spike and the dips by customers postponing purchases and stocking up, without mentioning that the low price also brings new demand. Taken literally, this description corresponds to a model with intertemporal substitution only and implies daily sales of 100 under a permanent low price. In our framework, however, subjects who adopted the wrong model in stage 1 have already noticed the discount-day spike of 202. Once the second model directs their attention to the dips, neither the simple wrong model nor the intertemporal-substitution-only model can explain both the spike and the dips, whereas the correct model can. Consistent with this prediction, among subjects assigned the wrong model first, encountering this incomplete model raises the share predicting 165 by 21 percentage points (from 14 to 35 percent), compared to only 6 percentage points (from 8 to 13 percent) for predicting 100.

Section 5 concludes with a discussion of implications and directions for future research. The mechanism we demonstrate allows wrong models to persist even when data that refute them arrive exogenously. This mechanism also makes it possible to identify persistent wrong models ex ante: simple wrong models that fit well the patterns they explain, but are silent about the rest. Such simple wrong models are likely to be pervasive and to lead to large mistakes, and we discuss examples from a range of settings.

We inform a theoretical literature in economics on why wrong models are not revised

(see Appendix A for a survey of related literature in psychology). One explanation is that endogenous actions generate data that fail to refute the wrong model (Fudenberg and Levine, 1993; Cho and Kasa, 2015; Lanzani, 2025; Ba, 2026), or that agents abandon an initial model only after sufficiently unlikely anomalies (Ortoleva, 2012). Others are that alternative models fail to yield higher payoffs than the wrong model (Fudenberg and Lanzani, 2023; He and Libgober, 2025), that discovering or estimating more complex models is costly (Aragones et al., 2005; Hong et al., 2007; Montiel Olea et al., 2022), and that the correct model lies outside the consideration set (Karni and Vierø, 2013; Galperti, 2019; Schwartzstein and Sunderam, 2021). Our design rules out all but the last. Disconfirming data arrive exogenously and contradict the deterministic wrong model with certainty, while the wrong model yields lower payoffs for both managers and subjects. Persistence also depends on model order when subjects see both models, holding model complexity fixed, and attention prompts dislodge the wrong model without simplifying the correct model. Lack of consideration of the correct model is not ruled out by design and likely plays some role in our data, but it cannot explain three findings: the assigned model shapes attention, attention prompts that supply no alternative model nonetheless dislodge the wrong model, and placing the correct model in the consideration set does not fully dislodge a wrong initial model.

Our results complement a theoretical literature on how decision makers who know the correct model can still hold biased beliefs. Under rational inattention, a decision maker trades the precision of beliefs against the cost of attention (Sims, 2003; Maćkowiak et al., 2023); with costly cognition, one may neglect variables whose payoff consequences are small relative to the cost of tracking them (Gabaix, 2014, 2019).[1] In our field setting, persistent wrong beliefs do not appear to reflect small benefits of paying attention, or a decision that the correct model is too complicated to apply, and our online setting uses treatments that are orthogonal to these mechanisms. For managers, neglecting intertemporal substitution is costly in terms of foregone profits, and managers who predict 165 appear to take IS into account by revising estimates of fuel elasticity downwards, and charging higher prices. In the online experiments, the data, task and incentives are fixed across treatments, holding constant the calculus and complexity, with only the assigned model varying. The assigned model affects attention, and beliefs about consumer response to price. Rational inattention can match this pattern only if the assigned model shifts a prior over which features are informative governs attention, which is essentially the mechanism we test.[2]

We also contribute to the experimental literature on misspecified and competing mental

[1] Decision makers may also shade answers toward priors when uncertain about their own reasoning (Enke and Graeber, 2023).

[2] Motivated reasoning (Bénabou and Tirole, 2016; Golman et al., 2017; Zimmermann, 2020) could also produce the bias we document, if for some reason the DM prefers to believe that consumers do not engage in IS, but it is unclear why managers would prefer the simple model's belief to the profits that accurate beliefs bring, and Prolific subjects are rewarded for accuracy and have no reason to change their motives because of a randomly assigned peer's explanation.

models (for a review, see Ambuehl et al., 2026). Fit appears to govern adoption: narratives that explain the data better persuade more (Barron and Fries, 2024), and many subjects reject a model contradicted by the data when a rival is shown alongside it (Ambuehl and Thysen, 2025). Yet wrong models can persist: randomly assigned false narratives distort choices even though subjects see everything (Charles and Kendall, 2025), and subjects given the parameters of an updating problem end up further from the Bayesian benchmark after 200 rounds of feedback than subjects not given them (Esponda et al., 2024), consistent with reduced uncertainty lowering the perceived need for attention (Gagnon-Bartsch et al., 2026).[3] We test model-directed attention directly, in three ways. First, we randomly assign the model with the data held fixed, as Charles and Kendall (2025) do for narratives, but we differ in measuring attention pattern by pattern, abd testing whether the wrong model lowers recall of the patterns it deems irrelevant but not of those it explains.[4] Second, we manipulate attention while holding the model fixed, using prompts that point to the patterns the wrong model deems irrelevant. Third, we test the predicted asymmetry in the effect of encountering a second model.

We also complement field evidence on misspecified mental models. Previous work shows that households hold heterogeneous mental models of the macroeconomy and of inflation (Andre et al., 2022, 2026a). Most households and some financial professionals also expect stale news about a company's earnings to predict its future stock returns (Andre et al., 2026b). In the same setting we study, Han et al. (2025a) show that gas station managers with lower cognitive skills under-appreciate how competitors respond to price cuts. Hanna et al. (2014) show that seaweed farmers fail to optimize seaweed pod size even though their own plots generate data on its value, and change their choices only after receiving a summary that highlights the relationship and recommends a size. These studies elicit the models people hold through hypothetical scenarios, open-ended explanations, and direct belief questions, and trace where they come from, for example showing the media to be an important source of macroeconomic narratives (Andre et al., 2026a). Our study is different in examining the mechanism of model-directed attention: Directly measuring how a model affects attention to specific data features, showing the ability of a pure attention intervention to trigger model revision, and how the impact of encountering a new model depends on whether it widens attentional focus.

Our paper has implications for a literature on behavioral IO. Many studies examine

[3]Models formed from data are also imperfect and sticky (Fréchette et al., 2025; Kendall and Oprea, 2024; Grass et al., 2026; Esponda et al., 2023), and subjects neglect displayed information on sample selection (Esponda and Vespa, 2018; Enke, 2020), alternative causes (Graeber, 2023) and correlation (Enke and Zimmermann, 2019). In these designs the experimenter chooses which feature goes unnoticed; in ours, the model is randomized and what goes unnoticed follows from it.

[4]Prior work measures attention more coarsely, through feedback viewing (Esponda et al., 2024), chart opening (Ambuehl and Thysen, 2025) or the weight placed on data (Barron and Fries, 2024). Closest to ours, Esponda et al. (2024) find in an online appendix that subjects with base-rate neglect recall signal-outcome frequencies with larger errors, in the direction of their beliefs.

behavioral biases of consumers and how firms rationally exploit them (for a survey, see Heidhues and Kőszegi 2018), but there has been less work on biases of managers about consumers. A notable exception is Strulov-Shlain (2023), who finds that firms' pricing decisions are consistent with partial unawareness of consumers' left-digit bias (see also List et al. 2023).[5] We add evidence, from directly eliciting managers' mental models, that some firm managers neglect a basic feature of consumer behavior, intertemporal substitution. Neglecting it is associated with believing that demand is more elastic and with a greater tendency to view price cuts as good for profits. We also provide evidence on a psychological mechanism that can explain why this mistake persists.

Our findings also bear on a literature that asks why firms offer periodic, temporary price discounts. Studies of many product markets find that some consumers shift purchases to discount periods, presumably in part because they are more price sensitive (e.g., Boizot et al. 2001; Pesendorfer 2002; Hendel and Nevo 2006a, 2006b). These findings have led to an explanation of discounts as dynamic price discrimination. Our evidence on how managers think about intertemporal substitution shows that some managers do not underestimate it and may use discounts to price discriminate. Other managers underestimate it and more often view price cuts as good for profits, consistent with overestimating how effective discounts are at attracting new customers. At least some discounts may therefore reflect managers' systematic misperception of consumer behavior rather than dynamic price discrimination.

Finally, our results connect to a literature on how communication of models shapes learning. In theories of model persuasion and competition, a decision maker evaluate competing models by how well they fit the data and adopts the best-fitting one (Schwartzstein and Sunderam, 2021; Montiel Olea et al., 2022). Thus a second model can dislodge an incumbent model only if it fits the data better. Under model-directed attention, however, encountering a second model also changes which data features are noticed. A second model can help even when it is incomplete or fits no better than the incumbent, by directing attention to patterns that then point to the correct model. Consistent with this mechanism, encountering an incomplete description of the correct model in Study 2 moves many subjects from the wrong model to the correct prediction rather than to the prediction the description literally implies. This helps explain why groups of diverse problem solvers can outperform groups of the individually best ones as their different perspectives explore different parts of a problem (Hong and Page, 2004), and why randomly assigned meetings among managers of young Chinese firms raise firm performance, partly through learning from peers (Cai and Szeidl, 2018). Exchanging models can correct beliefs that data alone do not, even when the model communicated is itself incomplete or wrong.

[5] A nascent literature on "behavioral firms" documents mistakes in firm behavior and manager beliefs (e.g., Hortaçsu and Puller 2008; List and Mason 2011; Goldfarb and Xiao 2011, 2024; Hanna et al. 2014; DellaVigna and Gentzkow 2019; Hortaçsu et al. 2019; Tadelis et al. 2023). For a survey of an older tradition of bounded rationality in IO, see Ellison (2006).

2 Conceptual Framework

We adapt the theory of channeled attention of Gagnon-Bartsch et al. (2026) to a pricing environment. The framework explains how a decision maker's mental model directs attention, why a wrong model can persist despite data that refute it, and what dislodges it. Proofs are in Appendix B.

2.1 The data the decision maker sees.

For concreteness, we build the framework around the hypothetical detergent-retailing data shown to managers and Prolific subjects in our study. Table 1 presents the data as seen by the respondents. The data table shows that Retailer A charges a regular price of 5.9 per kg in local currency and offers an announced discount to 5.2 on one day each week. Competitor B keeps its price at 5.9 and does not respond. The table covers two weeks. Respondents were told that the survey/experiment designers created the data according to a set of rules, but were not told what the rules were.

We used the following data-generating process (DGP) for Retailer A's daily sales q_t at price p_t:

$$q_t = D^*(p_t) + \delta^*(d_t), \tag{1}$$

where $D^*(p)$ is demand at price p and d_t is the signed distance to the nearest discount day. The term $\delta^*(d_t)$ represents intertemporal substitution (IS): customers shift purchases toward the announced discount from surrounding days. We set δ^* to zero outside $d \in \{-1, 0, 1, 2\}$ and require it to sum to zero over each discount cycle, so purchases are shifted from the day before and the two days after a discount to the discount day. Note that if the price were fixed over time, there would be nothing to wait for or stock up on, so there would be no substitution and sales would not vary across days.

On baseline days, when Retailer A charges the regular price and there is no intertemporal substitution, sales in Table 1 are 100. Around each discount, sales fall to 90 on the day before ($d_t = -1$), rise to 202 on the discount day ($d_t = 0$), and fall to 82 and 91 on the two days after ($d_t = 1, 2$). In Equation 1, demand is $D^*(5.9) = 100$ at the regular price and $D^*(5.2) = 165$ at the discount price. Intertemporal substitution is $\delta^*(-1) = -10$, $\delta^*(0) = 37$, $\delta^*(1) = -18$, and $\delta^*(2) = -9$.

2.2 Mental models of consumer behavior and impacts on attention.

In our environment, a mental model specifies which factors matter for sales and how they interact to produce the level of sales. In the theory of model-directed attention, the key assumption is that the mental model a decision maker holds determines which features of the data they pay attention to: they attend only to the features that the model deems

Table 1: Laundry detergent sales data presented to managers and subjects

Day	Retailer A's price	Competitor B's price	Retailer A's sales volume	Retailer A's profit
Day 1	5.9	5.9	100	160
Day 2	5.9	5.9	100	160
Day 3	5.9	5.9	100	160
Day 4	5.9	5.9	90	144
Day 5	5.2	5.9	202	182
Day 6	5.9	5.9	82	131
Day 7	5.9	5.9	91	146
Day 8	5.9	5.9	100	160
Day 9	5.9	5.9	100	160
Day 10	5.9	5.9	90	144
Day 11	5.2	5.9	202	182
Day 12	5.9	5.9	82	131
Day 13	5.9	5.9	91	146
Day 14	5.9	5.9	100	160

Notes: This table shows the data managers see in the survey and Prolific subjects see in online experiments. The data-generating process (DGP) involves both genuine demand response to prices and intertemporal substitution as specified in Equation 1. Both groups are told that the data are representative and cover all possible situations. They are also told that there is no noise, and the table shows all the variables can affect sales.

relevant.

The *correct model* specifies that sales depend on both the current price and intertemporal substitution around discounts: $q_t = D(p_t) + \delta(d_t)$, where demand and substitution are learned from the data. Because both price and distance to a discount matter under this model, a holder of the correct model attends to all five distinct sales values in Table 1: 100, 90, 202, 82, and 91. Baseline sales give $D(5.9) = 100$. The three dips measure purchases shifted toward the discount: $(100-90)+(100-82)+(100-91) = 37$. Subtracting these shifted purchases from discount-day sales gives $D(5.2) = 165$. The attended data therefore identify demand at both prices and all substitution parameters.

The *simple wrong model* omits intertemporal substitution and specifies that sales depend only on the current price: $q_t = D(p_t)$. Because price alone determines sales under this model, every regular-price day is the same and one sales value at each price pins down demand. A holder of the simple wrong model notices baseline sales of 100 (as on Day 1) and discount-day sales of 202, and interprets them as a two-entry price–sales schedule with $D(5.9) = 100$ and $D(5.2) = 202$. The dip days are in full view, but the decision maker does not notice that their sales (90, 82, and 91) differ from 100, because there are no need to pay attention to two days with the same price according to the simple wrong model. As a result, the decision maker misses the data features that would refute the model.

Assumption 1 (Model-directed attention)**.** *The decision maker's model directs attention towards the patterns it explains and away from distinctions it rules out. Sales values that provide no additional information under the model are not separately noticed.*

Assumption 2 (Model fit)**.** *The decision maker gives up the model if and only if the data* as noticed *are impossible under it, or become infinitely less likely under it than under the correct model.*

Under these assumptions, only the incumbent model directs attention, and the decision maker judges model fit using the noticed data. Following Gagnon-Bartsch et al. (2026), Assumption 2 takes the correct model as a natural benchmark for comparison.[6] Because the correct model fully explains the data in this deterministic environment, a wrong model can persist only if it also fully explains the data it directs attention to. When it does, it fits the noticed data as well as any alternative, including the correct model.

2.3 Predictions

Here we derive the predictions of the framework. All proofs are in Appendix B.

Prediction 1 (Attentional stability of the simple wrong model)**.** *Under Assumptions 1 and 2, the simple wrong model is attentionally stable: the decision maker does not give it up, however much data they accumulate.*

The intuition for this prediction is that a holder of the simple wrong model notices only baseline sales of 100 and discount-day sales of 202, and the model explains both values no matter how many times the pricing schedule repeats. Noticing different sales on two regular-price days would refute the model in favor of the correct model. However, because the simple wrong model treats all regular-price days as the same, the decision maker has no reason to attend to more than one.

Prediction 2 (Counterfactual sales under permanent prices)**.** *Asked to predict daily sales if Retailer A always charged 5.2, holders of the simple wrong model answer 202 and holders of the correct model answer 165.*

The difference at 5.2 follows from how the two models interpret the discount-day spike. Under a permanent price of 5.2, there is no intertemporal substitution and daily sales equal demand $D(5.2)$. A holder of the simple wrong model takes discount-day sales of 202 to be $D(5.2)$ and predicts 202, whereas a holder of the correct model subtracts the 37 purchases shifted from the three dip days and predicts 165.

Prediction 2 provides the basis for our main measure of a decision maker's mental model. We ask respondents in both the manager survey and the online experiments what daily sales

[6]Using the correct model as the formal benchmark is conservative for persistence: in Appendix B.1, any model that survives comparison with the correct model on the noticed data also survives comparison with any other alternative. Even with the correct model as the benchmark, as long as the incumbent model directs attention and explains what is noticed, the decision maker sees no discrepancy that would lead them to look for another model. Finally, when noticed data do refute the incumbent model, comparing directly against the correct model abstracts from how a decision maker searches for or constructs an alternative. In practice, dislodging a wrong model therefore need not move every decision maker to the correct one.

would be if Retailer A always charged 5.2 (Section 3.2). In our main elicitation, respondents choose their prediction from four multiple-choice options: 202, 165, and two answers implied by other wrong models (Appendix B.2). An answer of 192 corresponds to a model in which customers postpone purchases before a discount but do not stock up during the discount for later days. An answer of 100 corresponds to a model in which discounts only shift purchases across days without any demand response to price. We use the respondent's answer as a proxy for their mental model.

Prediction 3 (Biased beliefs about elasticity and optimal price)**.** *Holders of the simple wrong model overestimate demand elasticity in absolute value relative to holders of the correct model.*

This overestimate of demand elasticity comes from mistaking shifted purchases for new demand. In the detergent data, when the price falls from 5.9 to 5.2, a holder of the simple wrong model believes that demand increases from 100 to 202 rather than to 165. Relative to the regular price of 5.9, the implied demand elasticity is -8.60 under the simple wrong model, compared to -5.48 under the correct model.

If fuel consumers substitute intertemporally, the mental model we measure in the detergent task should also predict managers' beliefs about fuel-demand elasticity (Section 3.4). This relationship provides a test of whether the detergent task captures managers' mental models, rather than only their understanding of an abstract setting. In fuel pricing, a manager who holds the simple wrong model likewise overestimates how much lower prices increase demand. At the true profit-maximizing price chosen under the correct model, this manager still perceives a gain from lowering the price. When true demand, costs, and competitors' behavior are held fixed and managers maximize perceived profit under single-peaked profit functions with interior optimal price, this manager charges a lower price than a holder of the correct model (Appendix B). Because this price is suboptimally low under true demand, the manager also earns lower profit.

Prediction 4 (Attention prompts can dislodge the simple wrong model)**.** *An attention prompt that leads the decision maker to notice different sales on two regular-price days dislodges the simple wrong model.*

Because the simple wrong model persists when the decision maker does not attend to the patterns that would refute it, one way to dislodge the model is to direct attention to those patterns. An attention prompt can do so without providing new data or suggesting an alternative model. For example, a prompt that asks the decision maker to compare a baseline day with a dip day can lead them to notice both 100 and 82 at the regular price. Noticing both observations refutes the simple wrong model, because no single regular-price demand value can explain them.

Prediction 5 (Asymmetric effects of encountering a second model)**.** *A decision maker entertains an encountered model with probability $\alpha \in (0,1)$. (i) Entertaining the correct model, or*

another model that directs attention to a dip, dislodges the simple wrong model. (ii) Entertaining the simple wrong model does not dislodge the correct model.

Another way to dislodge the simple wrong model is to encounter a second model, for example when a peer or mentor offers an explanation of the data. A decision maker *entertains* this second model by also attending to the patterns it deems relevant and judging model fit on the combined noticed data. Entertaining the correct model dislodges the simple wrong model, because it leads the decision maker to notice that sales on the dip days differ from baseline sales. By contrast, entertaining the simple wrong model does not dislodge the correct model. A holder of the correct model has already noticed all five sales values, and the simple wrong model directs attention to no new ones. Because the decision maker entertains an encountered model only with probability $\alpha < 1$, encountering the correct model need not dislodge the simple wrong model for every decision maker.

Part (i) does not require the second model to be correct. Any model that directs attention to a dip dislodges the simple wrong model in the same way when entertained. For example, suppose a peer explains the discount-day spike and surrounding dips solely by customers shifting purchases across days, without mentioning new demand, as in the *IS-only model* (Appendix B.2). Entertaining this incomplete model leads a holder of the simple wrong model to notice the dips alongside baseline sales of 100 and discount-day sales of 202. Together, these observations refute both the simple wrong model and the IS-only model, whereas the correct model explains all of them (Appendix B.3).

3 Evidence from the field

We first test our framework among gas station managers. Section 3.1 introduces the retail fuel market, our four datasets, managers' pricing discretion and incentives, and transaction-level evidence that fuel customers substitute intertemporally. In Section 3.2, we explain how our survey measures managers' mental models through the detergent task (Prediction 2). Section 3.3 documents that neglect of intertemporal substitution is widespread and persistent (Prediction 1), and uses managers' open-ended explanations of the data to test model-directed attention (Assumption 1). Finally, Section 3.4 examines how managers' mental models relate to their beliefs about fuel demand, the fuel prices they set, and station profits (Prediction 3).

3.1 Institutional context, data, and intertemporal substitution in fuel markets

Our partner company operates more than 20,000 gas stations nationwide. Stations typically sell both gasoline and diesel and nearly always have a convenience store. Most profit comes

from fuel: on average in our data, fuel accounts for 71 percent of total station profits. The stations are mostly company-owned rather than franchised, and each has a manager with substantial influence over station operations, including pricing. More senior managers also set policies for the stations, and in each of the company's roughly 350 districts, a district manager decides what type and degree of discretion to give station managers.

The market is relatively concentrated, with brand differentiation. Our partner company is one of two major brands in retail gasoline, each with stations across the country and about one third of the market. The rest of the market goes to hundreds of smaller, more local chains, which we call independent competitors. Unlike the independent chains, which must buy fuel on the market, the major brands produce their own fuel. The major brands also position themselves as offering a premium product, and for this reason they typically charge more than independent chains for the same grade of gasoline or diesel.

A government-imposed price ceiling for fuel products, indexed to the world price of oil, is an important feature of the pricing environment. The ceiling is a plausible focal point for coordinating prices, and the two major brands have a policy of generally pricing near it. Independent stations also often set prices that track movements in the ceiling, at a substantial discount. We measure a station's pricing policy by its price ratio, which our partner company computes as the sales-weighted average of each fuel product's price relative to its price ceiling. A price ratio of 1 means that the station prices at the ceiling for all products. The ratio includes all discounts, such as posted-price cuts, coupons and promotions.

3.1.1 Datasets

We obtained the data through a collaboration with the partner company's research arm.[7] The data include performance data, such as profits and prices, and survey data collected through the company's internal survey infrastructure. Because participation in internal surveys is expected and enforced, response rates are high, around 65 percent. The surveys are run by the research arm, a unit independent of senior management, which makes the promise of confidentiality credible.

Our main analysis uses four datasets. The first is a survey of the district managers, 353 of whom responded, close to a 100 percent response rate. It collected information about how much discretion station managers have over fuel pricing and the incentive structure of them, and it elicited the district managers' views on potential pricing mistakes by station managers.

The second dataset is a survey of station managers. The survey was sent to all 20,000 station managers, and the response rate of about 65 percent gives 13,407 responses. The survey measures managers' neglect of intertemporal substitution through the detergent task. To test whether their mental model measured through the detergent task applies to prolonged

[7]IRB approval for this study was granted by Renmin University of China.

discount in the gas station context, we also designed a gas station task where managers need to predict the counterfactual daily sales of a station if they implement a lasting discount. The survey also elicits managers' perceived elasticity of fuel demand by asking them to predict sales change at a typical station and at their own station upon a hypothetical price discount. It also collected information about manager characteristics and work history.

The third dataset is a monthly panel of performance data for the company's gas stations from 2019 to 2024, with fuel and nonfuel profits, sales volume and the average monthly prices of gasoline and diesel. We have the monthly performance data for 26 of the 31 regions in which the company operates.[8] These data cover about 16,000 gas stations. The research arm matched the performance data to survey responses using an internal company identifier, which gives linked data for about 10,000 stations, in line with the 65 percent response rate.

The linked survey and administrative data let us relate managers' mental models of customer behavior, as measured in the survey, to the prices they set and the performance of their stations.

A fourth dataset contains all transactions by loyalty card holders at our partner's stations in one region, recorded to the minute, from 2013 to 2015. These years do not overlap with the manager surveys. The data nevertheless allow a direct test of whether consumers in this market engage in intertemporal substitution.

3.1.2 Manager characteristics, discretion, incentives

Table 2 shows descriptive statistics for station managers and their stations. The median manager is 39 years old, and about 34 percent of managers are female. The modal level of education is a junior-college degree. Managers stay in their jobs for a long time, with a median experience as a station manager of 7 years, but they switch stations periodically, and the median tenure at the current station is about 2 years. With a median of 5 employees, managers have some people management duties, but not for large groups of workers. The median station has 2 competitors in its local market. The company defines the local market by starting from a radius of 2.5 km around the station and then adjusting the number and type of competitors for commuting routes and other factors. In the local market, the median station of our partner company has about 30 percent of fuel sales.

According to the district managers, station managers have substantial influence over fuel prices, although their autonomy varies across districts. In 48 percent of the districts, the district manager reports that station managers can change listed fuel prices directly, within a pre-specified range or without restriction, without submitting a proposal. In the remaining districts, station managers must propose changes to listed prices, and district managers report approving about 34 percent of proposals on average. Even where proposals are required, station managers can influence prices. Their mental models can therefore

[8] Of the remaining five regions, some only record data on a quarterly basis, and others were not made available due to idiosyncratic bureaucratic factors.

Table 2: Descriptive Statistics: Managers and Stations

Manager descriptives		Station descriptives	
Median age	39	Median number of employees	5
Female	34%	Median number of competitors	2
Education level:		Median market share	30%
High school	26%		
Junior college	45%		
College or above	28%		
Median experience (years)	7		
Median tenure at current station (years)	2		

Notes: This table reports descriptive statistics based on the 2024 survey wave. "Experience (years)" represents the total number of years as a station manager, while "tenure at current station (years)" represents the number of years as the station manager at their current assignment. "Competitors" refers to stations from other companies within the local market as defined by the company. "Market share" is a station's share of total fuel sales in its local market as reported by the station manager.

matter for pricing and performance. District managers grant this autonomy despite concerns about station managers' mistakes, because they believe that station managers have local knowledge relevant to pricing.

Station managers receive a base salary and substantial performance pay, which makes up roughly 50 percent of total compensation. Bonuses depend on two key performance indicators, fuel profits and non-fuel (convenience store) performance, each assessed relative to a target and weighted in the bonus. These bonuses give managers an incentive to set fuel prices that maximize station profits.

3.1.3 The relevance of intertemporal substitution in fuel markets

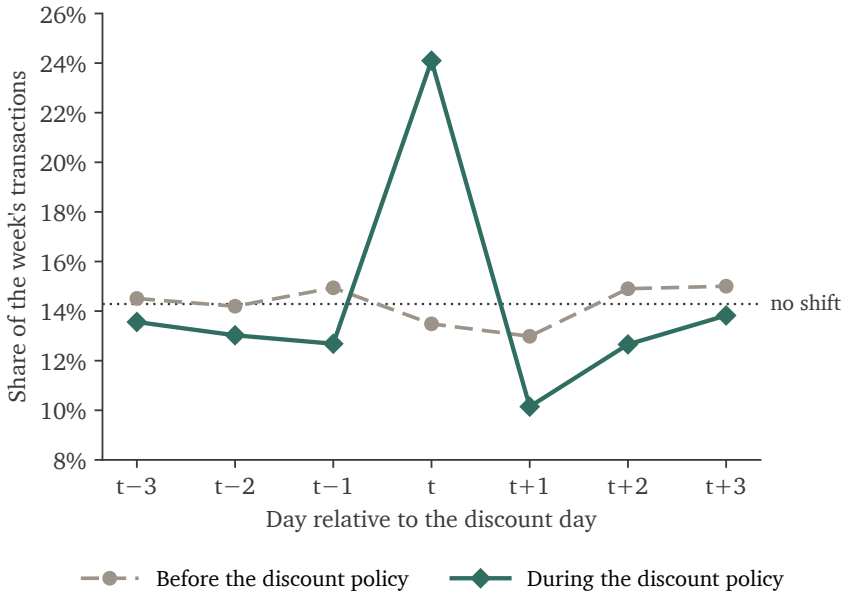
Intertemporal substitution is plausible in fuel markets. Each visit to a station has a fixed cost, which gives consumers an incentive to buy fuel when the tank is nearly empty and then to fill it. Within limits, they can shift the timing of a visit to take advantage of a discount.

To test for intertemporal substitution in this market, we use the transaction data. We look at stations that introduce a regular weekly discount on listed fuel prices and ask how the promotion changes the timing of purchases within the week. Roughly 24 percent of stations try such a promotion at some point, usually with the discount on Saturday or Friday, and some stations discount on both days. We define a station's regular customers as loyalty card holders who bought fuel there at least twice a month in the months before the promotion, and we compare the timing of their purchases before the promotion was introduced and during the promotion.

Figure 1 shows the week around the discount day at stations with one discount day per week. Before the promotion, the day of the week that later becomes the discount day accounts for 13.5 percent of regular customers' weekly transactions, close to the 14.3 percent, one day in seven, that purchases spread evenly across the week would give. During the promotion, its share rises to 24.1 percent. Relative to before the promotion, the shares of the

day before the discount day and of the two days after it fall by 15, 22 and 15 percent, and the shares of the other three days by 7 to 8 percent (Appendix Figure C.1). The week during the promotion thus has the shape of the detergent data, with a spike on the discount day and dips on the day before and the two days after. Fuel volume shows the same pattern. Weekly discounts probably capture only part of the intertemporal substitution in this market, which can also occur around longer or one-time price cuts.

Figure 1: Regular customers' purchases in the week around a weekly discount day



Notes: This figure shows the share of regular customers' weekly transactions that falls on each day of the week around the discount day, before and during a station's weekly discount policy. Day t is the discount day; before the policy, it is the day of the week that later becomes the discount day. The sample includes stations with one discount day per week. Regular customers are loyalty card holders with at least two transactions in the month before a station's weekly discount policy starts and at least four transactions in the two months before. The dotted line marks one day in seven (14.3%), the share of every day if purchases are spread evenly across the week. Appendix Figure C.1 shows fuel volume and the change in each day's share.

3.2 Measuring managers' mental model of consumer response to price discounts

To measure how gas station managers' mental models account for the demand response to price and intertemporal substitution, we presented the detergent data in Table 1. Managers were told that Retailer A sets a regular price of 5.9 per kg in local currency and a discount price of 5.2 on one randomly selected day each week (Days 5 and 11 in the data). The instructions state that Competitor B does not respond, as reflected in its constant price in the table, and that marginal cost is 4.3 per kg. Sales fall from a baseline of 100 to 90 before each discount, rise to 202 on the discount day, and fall to 82 and 91 afterwards before returning to 100. All sales values are in units of 10,000 kg. Managers were then asked:

"If retailer A always charges 5.2 per kg, what will the daily sales volume be?"

They were given four response options (in 10,000 kg): 192, 165, 100, 202.

Each option is the prediction of one of the four models in Section 2. Because price alone determines sales under the simple wrong model, managers with this model take discount-day sales of 202 as demand at a permanent price of 5.2. The pre-discount-only model allows customers to postpone purchases until the discount but omits purchases brought forward from later days, and managers with this model subtract the pre-discount shortfall of $100 - 90 = 10$ from 202 to predict 192, without accounting for the post-discount sales differences.

The correct model accounts for both the demand response to price and all purchases shifted toward the discount. Subtracting the three shortfalls, $(100 - 90) + (100 - 82) + (100 - 91) = 37$, from discount-day sales gives demand of $202 - 37 = 165$ at a permanent price of 5.2. Under the IS-only model, demand is independent of price, and the response to a temporary discount comes entirely from changes in the timing of purchases. Managers with this model use baseline sales of 100 to predict sales at either permanent price. Under the noticing strategy in the framework, they notice the dip values but only that discount-day sales exceed baseline sales, and they therefore do not compare the observed discount-day sales of 202 with those the model implies.

We use the four answers as proxies for managers' mental models, because guessing or another model can produce the same number. Managers may also infer an explanation from the response menu. A manager uncertain about 202 could choose 192, for example, without noticing the pre-discount dip. Each option has an interpretation under the framework, but none uniquely identifies the manager's model. To check whether the main findings depend on the response menu, we also examine managers' explanations of the data, and in the online experiments we run a version in which subjects type their prediction into an empty box.

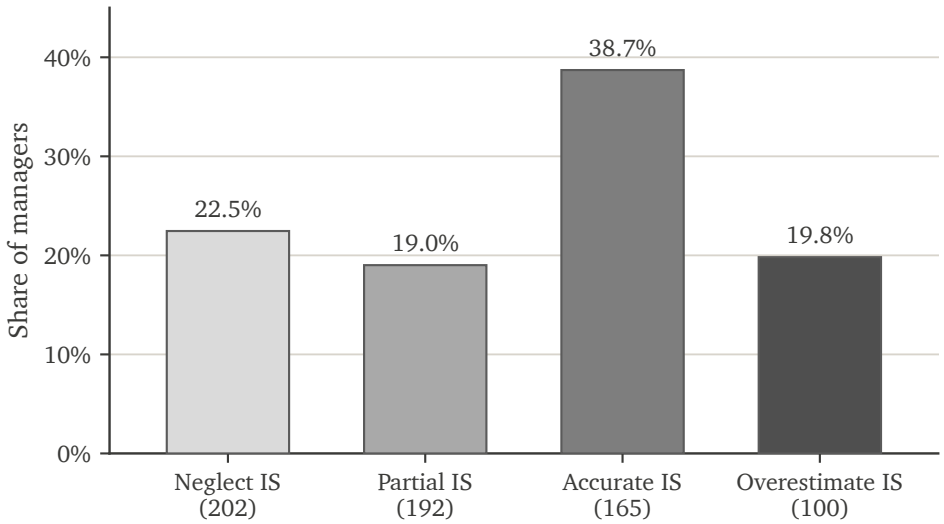
The detergent task is a controlled scenario with a known data-generating process. Managers are told that the data are representative, cover all possible situations and show all the variables that may affect profit. They all face the same information, without the unobserved local conditions or competitor responses that might justify different predictions at their own stations. The benchmark of 165 follows from conservation of purchases under the structure of demand and intertemporal substitution in Section 2, although the instructions do not uniquely identify that structure, and an answer does not establish which model produced it.

3.3 Results from the field: Persistence of wrong mental models, and mechanisms

3.3.1 Prevalence of neglect of intertemporal substitution among station managers

Figure 2 shows the distribution of managers' predictions of daily sales if the retailer permanently charged the discounted price in the detergent task.

Figure 2: Distribution of responses to the detergent task



Notes: This figure shows the distribution of managers' answers to the detergent task, which asks managers to predict daily sales if the retailer permanently charges the discounted price. An answer of 202 neglects intertemporal substitution (IS), 192 underestimates intertemporal substitution, 165 is the correct answer, and 100 attributes the entire sales response to intertemporal substitution. The sample includes $N = 13{,}407$ managers.

Managers' predictions differ widely. Of all managers, 22.5 percent predict that sales would stay at 202, the level on discount days, which is the prediction of the simple wrong model. Another 19.0 percent predict 192, consistent with underestimating intertemporal substitution. The correct answer, 165, is given by 38.7 percent, and 19.8 percent predict 100, the prediction of the IS-only model.

Managers who neglect intertemporal substitution should also view price cuts as effective at attracting new customers and hence at raising profits. We therefore check whether the prediction in the detergent task is related to thinking that the detergent retailer could raise its total profits by charging the low price of 5.2 more often. Managers were asked: "What pricing policy do you think would lead to the highest total profits over the 14 day periods for Retailer A?" The options were "The retailer's current pricing policy," "A policy of always charging 5.9," "A policy of always charging 5.2" and "Charging the price of 5.2 more frequently than the current pricing policy, but not always 5.2." We define choosing the second option as favoring a higher price, and choosing the third or fourth option as favoring a lower price.

The answers are consistent with this link. Appendix Figure D.2 shows that favoring a lower price for the detergent seller becomes less common across the predictions 202, 192, 165 and 100. Managers who predict 165 or 100 also favor a higher price more often than managers who predict 202 or 192 (Appendix D).

Together, 41.5 percent of managers neglect or underestimate intertemporal substitution, predicting 202 or 192. The mistake persists in two senses. Managers make it with the detergent data in front of them, although the data show the pattern of intertemporal substitution twice and with no noise. Station managers also see high-frequency sales data at work, many of them for years, and yet experience helps little. The correlation between neglecting intertemporal substitution and years of experience as a station manager is statistically significant but small. Appendix Figure D.3 shows that the share of managers who predict 202 is 24 percent among managers with up to two years of experience and 21 percent among managers with 16 or more years.

3.3.2 Evidence on model-directed attention as a mechanism for manager neglect of intertemporal substitution

Which mental models do managers who neglect IS hold, and why do they neglect IS despite the data and their experience? To answer these questions, we asked managers to explain the detergent data in an open-ended question, directly after they saw the table:

> "Looking at days 4-7 and days 10-13, why do you think sales show this pattern before and after the days when price is 5.2? Please answer in complete sentences."

An explanation reveals which patterns in the data a manager noticed, even when it does not describe them. The measure can understate noticing, since a manager may notice a pattern and still give an explanation that neither states nor implies it.

To classify responses, we first read a random sample of 1,000 responses and developed a rubric based on the different categories of explanations found in the sample (Appendix E). The rubric includes a category for explanations based on intertemporal substitution, which attribute the dips to customers changing the timing of their purchases to take advantage of the discount. Other categories include explanations that simply state that sales increase when prices are low, and more complicated wrong explanations, such as saying that consumer demand is cyclical and the retailer uses discounts at the troughs of the cycle to stimulate demand. The rubric also classifies, for each explanation, whether it reveals that the manager noticed the pre-discount dip, the discount-day spike and the post-discount dip. A pattern counts as noticed when the explanation, taken as a whole, could not plausibly have been written by a manager who had not noticed it. Only 7.2 percent of managers describe a dip, but 33.2 percent give an explanation that reveals one. Most do so through a cause that only makes sense if the dips are there: customers waiting for the discount imply lower

sales before it, and customers having stocked up imply lower sales after it. A large language model with extended reasoning applied the rubric to each of the 13,407 responses separately, writing a justification for each pattern before coding it. The language model saw only the explanation, not the manager's prediction. It classified each response twice in separate requests, and the two classifications agree on whether the explanation reveals a dip in 96.9 percent of cases.

Answers differ in which patterns in the data they mention, as the following two answers show (our translations). A manager who predicted 202 wrote:

> "The price reduction stimulated customers' purchasing desire, leading to increased sales."

A manager who predicted 165 wrote:

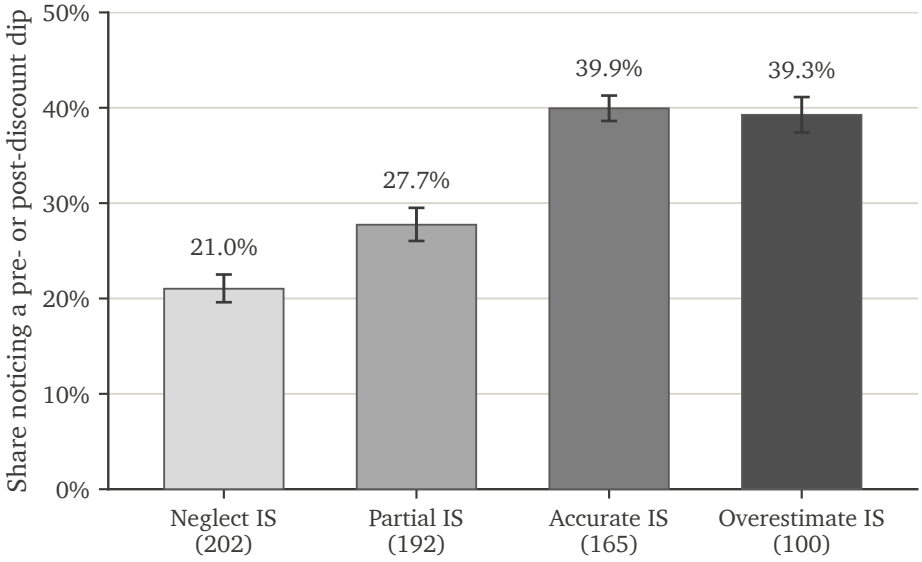
> "Customer traffic decreased the day before and two days after the promotion, as purchases were concentrated on the discount day."

The first answer mentions only the high sales at the low price. The second mentions the pre-discount dip and the post-discount dip and explains them by customers concentrating their purchases on the discount day.

As shown in Appendix Figure F.4, the most frequent type of explanation is one that reveals only the discount-day spike, given by 45.1 percent of managers. Of these, 59.7 percent say nothing more than a version of low price leads to high sales. Overall, 17.8 percent of managers explain the detergent data by intertemporal substitution. This share is consistent with the low rates at which managers notice the dips.

Our conceptual framework links holding the wrong model to not noticing the dips. Figure 3 shows, for each prediction in the detergent task, the share of managers whose explanation reveals that they noticed a dip: the pre-discount dip, the post-discount dip or both. Managers who predict 202, our main indicator of the wrong model, are the least likely to notice a dip: 21.0 percent of them do, compared to 39.9 percent of managers who predict 165 and 36.8 percent of managers who make any other prediction (both $p < 0.01$). The difference is specific to the dips. Managers who predict 202 notice the discount-day spike as often as managers who predict 165 (78.9 compared to 79.7 percent, a difference that is not significant). This pattern is consistent with model-directed attention: managers whose prediction reveals the wrong model are less likely to notice the patterns the wrong model deems irrelevant, but not the pattern it explains. This could help explain why managers have not learned about intertemporal substitution from seeing data. Managers who predict 192 notice a dip at the rate of 27.7 percent, significantly higher than managers who predict 202 ($p < 0.01$). Managers who predict 100, our indicator of the IS-only model, notice the dips as often as managers who predict 165 (39.3 compared to 39.9 percent, a difference that is not significant).

Figure 3: Noticing of the dips in manager explanations, by prediction



Notes: This figure shows the share of managers whose explanation of the detergent data reveals that they noticed the pre-discount dip or the post-discount dip, by prediction in the detergent task. A pattern counts as noticed when the explanation could not plausibly have been written by a manager who had not noticed it, and Appendix E gives the rubric. The sample includes $N = 13{,}407$ managers. Error bars show 95% confidence intervals. The p-values in the text are from two-sample tests of the difference in shares.

3.4 Consequences of neglecting intertemporal substitution among station managers

The detergent task concerns one-day discounts for laundry detergent. If the prediction in this task reflects how managers think about consumer response to price cuts, it should be related to how they think about a lasting fuel price cut. If it also reflects the model they use to set prices, Prediction 3 implies that managers who predict 202 perceive demand as more elastic and, under otherwise optimal pricing, set lower prices and obtain lower profits.

In the lasting price cut task, managers see two weeks of diesel prices and sales at one gas station. The station cuts its price on March 5 and keeps it low, and sales jump on that day and then fall below their earlier level. Managers predict daily sales for March 2 to 6 had the cut started on March 2. Under intertemporal substitution, the sales spike would move to March 2, with lower sales afterwards. We group the predictions by k-means clustering and compare the share of managers in the profile with this shape. Managers who underestimate intertemporal substitution (predict 202 or 192) are in this profile less often than managers who do not (predict 165 or 100), 11.2 versus 16.2 percent ($p < 0.01$), and the difference holds with controls for manager characteristics (including cognitive skills) and region fixed effects (Appendix G.1).

Because the lasting price cut task follows the detergent task, these differences could also come from the detergent data managers have just seen. Before the detergent task, and

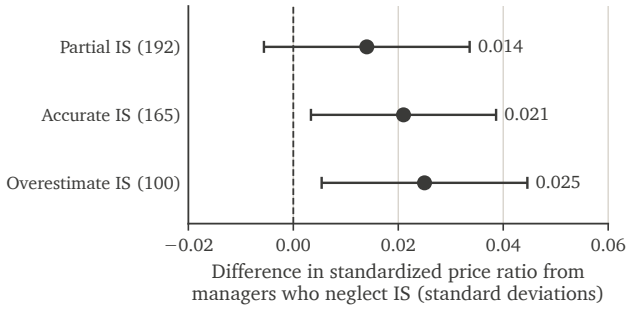
without seeing any data, managers consider a typical station that announces a gasoline price cut a week ahead. The station keeps the cut for four weeks. Under intertemporal substitution, purchases delayed until the cut fall in its first two weeks, and sales in weeks 3 and 4 are lower. Managers who predict 165 or 100 expect lower sales in weeks 3 and 4 more often than managers who predict 202, with the same controls ($p < 0.01$, Appendix G.2).

Managers also answer three questions that measure perceived elasticity, all asked before the detergent task. In the first, the price ceiling falls by 0.3 per liter, a change that affects all stations, and managers predict sales at a typical station of our partner company. In the second, a typical station alone cuts its price by 0.3 per liter, which can also attract customers from rival stations. The third asks about the same cut at the manager's own station, the scenario closest to the prices they set. Perceived elasticity is the percentage change in predicted sales divided by the percentage change in price, in absolute value. Managers who predict 202 have the highest mean perceived elasticity in each scenario (Appendix Figure G.10). Relative to all other managers, their perceived elasticity is 8.8 percent higher for their own station, 8.0 percent higher for the market price and 6.2 percent higher for a typical station alone (all $p < 0.01$). The differences remain significantly different from 0 at the 5 percent level controlling for manager characteristics and region fixed effects (Appendix G.3).

Station managers have substantial influence over fuel prices, and their bonuses are tied partly to fuel profits (Section 3.1). To test whether managers who predict 202 set lower prices, we analyze the standardized monthly price ratio (Section 3.1) for the months between January 2019 and October 2024 at approximately 10,000 stations matched to our survey responses, keeping only the months in which the station was run by the manager who answered the survey. We estimate three specifications that add controls in steps: (1) panel-month-by-district fixed effects only, which compare stations in the same district and month, where the same district-level manager sets the discretion given to station managers, (2) adding market conditions (competitor counts by type, location, 24-hour operation, rental status), and (3) further adding manager characteristics (cognitive skills, gender, age, experience, strategic sophistication). Figure 4 shows specification (3). Relative to managers who predict 202, the standardized monthly price ratio is 0.021 standard deviations higher for managers who predict 165 and 0.025 higher for managers who predict 100 (both $p < 0.05$). The estimates for managers who predict 165 and 100 remain statistically significant in specifications (1) and (2). The price difference between managers who predict 165 and 202 is comparable to the price difference associated with facing one fewer competitor in the local market.

What these lower prices imply for profit cannot be read off the correlation between price and profit, because managers may cut prices when profits are low. We therefore evaluate the price gaps with estimated demand. Han et al. (2025a) use the same administrative data to estimate a constant-elasticity residual demand curve for our partner company's stations that includes both consumers buying less fuel and consumers switching to competing sta-

Figure 4: Mental models of intertemporal substitution and actual prices



Notes: This figure reports coefficient estimates from regressing the standardized monthly price ratio on indicators for the prediction in the detergent task, with managers who neglect intertemporal substitution (predicting 202) as the omitted category. The specification includes panel-month-by-district fixed effects, market conditions (competitors, location type, 24-hour operation, rental status), and manager characteristics (cognitive skills, gender, age, experience, strategic sophistication). The sample is station-months under the manager who answered the survey, January 2019 to October 2024. Standard errors are clustered at the station level. Error bars show 95% confidence intervals.

tions. They instrument price with the government-imposed price ceiling, which is indexed to world oil prices and moves the transfer price that determines each station's marginal cost, and obtain an elasticity of 1.65 (Appendix H). Marginal cost is observed directly in the internal transfer price rather than inferred from prices under the assumption that firms price optimally, an assumption that is problematic when managers hold wrong models of demand. At the average station, we convert the coefficients in Figure 4 into price gaps per liter and compare daily fuel profit at the average price and at the average price minus the gap (Appendix H.2). The price gaps are associations across managers conditional on the controls, and the calculation is static: it compares managers who hold different models rather than changing the model of any one manager. Relative to managers who predict 165, the correct answer, the lower prices of managers who predict 202 imply 3.3 percent lower daily fuel profit at the average station, and relative to managers who predict 100, 3.9 percent lower. Linear demand, with an elasticity of 1.63, gives the same 3.3 and 3.9 percent (Appendix H.3). We do not report a profit gap for managers who predict 192, whose price gap is not statistically significant.

4 Evidence from online experiments

The previous section shows that managers who predict 202 are less likely than other managers to notice the dips. To test whether their ignorance of the dips around the discount days is caused by their mental models or is merely correlated with them, we turn to online experiments, in which we randomly assign different models to subjects. In the experiments, subjects on Prolific see the detergent data shown to the managers (Table 1) after reading a "model", a short description of the relationship between sales and prices written by a subject

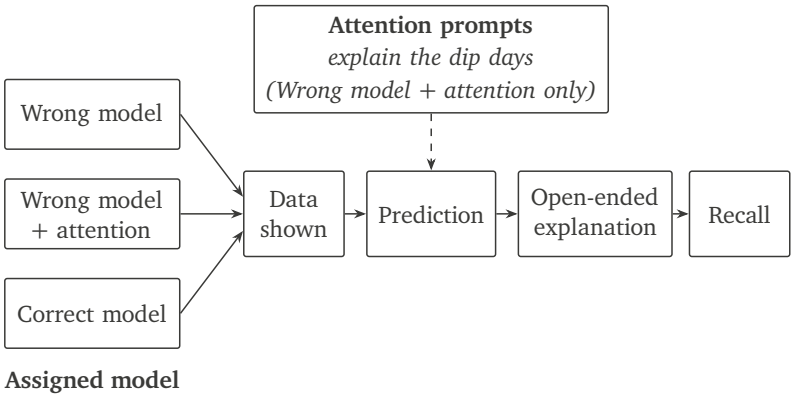
in our initial pilot. Study 1 randomizes which model they read, a description of the wrong model or a description of the correct model. It tests whether the wrong model directs attention away from the dips (Assumption 1), whether subjects who read it predict 202 with the data in view (Prediction 1), and whether directing attention to the dips dislodges it (Prediction 4). Study 2 gives every subject both descriptions, one after the other, and tests whether the effect of the second description depends on which description came first (Prediction 5). Both studies were pre-registered on AsPredicted (#300186 and #300209).

4.1 Study 1

4.1.1 Design

Study 1 has three treatments: *Wrong model*, *Correct model* and *Wrong model + attention*. The treatments differ in the assigned model and, in *Wrong model + attention*, in two attention prompts that subjects answer before the prediction. Otherwise all subjects see the same data and answer the same questions in the same order. Figure 5 shows the timeline.

Figure 5: Timeline of Study 1



Notes: This figure shows the timeline of Study 1. Subjects are randomized into three treatments: *Wrong model*, *Wrong model + attention* and *Correct model*. The treatments differ in the randomly assigned model and in whether attention prompts are presented; the dashed arrow marks the step that only *Wrong model + attention* receives. All three treatments see the same data and answer the same questions in the same order.

The instructions describe the data as the manager survey does. Subjects learn that the data come from a market for laundry detergent, that the study designers created the data from a set of rules that mimic the forces in the real industry, that the data are representative and include every variable that affects Retailer A's sales, and that Retailer A announces its pricing schedule to customers at the beginning of each week. They are asked to infer the rules from the data.

The treatment is the assigned model. Before the data appear, subjects learn that others in an earlier part of the study each saw data from many more weeks and wrote down a description of the rules relating Retailer A's price to its sales. These descriptions are kept in

a bag, and each subject draws one without looking. The page adds that having seen more data gives the writer an advantage in understanding patterns, but does not guarantee that the description is correct. Subjects in *Wrong model* and *Wrong model + attention* draw a description of the wrong model:

> "When the price is low, sales are greater than when the price is high. This has to do with the rule of supply and demand. When the prices are low demand increases."

Subjects in *Correct model* draw a description of the correct model:

> "When retailer A offers a low price on just one day of the week, sales surge on that discount day but drop on the surrounding days, while the other days stay at their usual level. This happens because customers postpone purchases to wait for the discount and then have already stocked up afterward, briefly reducing demand."

Both descriptions are explanations written by subjects in our initial pilot, who saw a series of weekly datasets with different pricing schedules and described the rule relating price to sales. The description of the wrong model was written by a pilot subject who predicted 202 and the description of the correct model by one who predicted 165, and each is typical of the explanations of pilot subjects with that prediction.

In *Wrong model + attention*, two attention prompts appear below the data table before the prediction:

> "What do you think explains the pattern in Retailer A's sales from Day 3 to Day 4, and from Day 9 to Day 10?"
>
> "What do you think explains the pattern in Retailer A's sales from Day 6 to Day 8, and from Day 12 to Day 14?"

The first prompt points at the pre-discount dip and the second at the post-discount dip, and subjects are asked to answer each prompt in complete sentences. The prompts tell subjects which days to explain, but they do not say what happens on those days or suggest a reason. They only serve to direct subjects' attention to the dip days.

With both the assigned model and the data in view, subjects answer the prediction question from the manager survey: "Suppose Retailer A set its price to 5.2 every single day (a permanent discount), with Retailer B's price unchanged. What would Retailer A's daily sales be?" They choose among the four answers of the manager survey and earn $1.00 if they choose the correct answer, 165. Subjects then distribute 100 points across the four answers according to how likely they think each one is. This allocation is not incentivized, and we use the points placed on 202 as a measure of confidence in the wrong model. On the next page, without the data, subjects are asked to describe, in complete sentences, the

relationship between Retailer A's price and its sales and the mechanism that produces it. One of the two answers is drawn and earns $1.00 if it is graded as correct or mostly correct.

We measure attention by recall.[9] Right after the written explanation, and still without the data, subjects fill in Retailer A's sales for Days 1 to 7, or tick "I don't recall" for any day. They start with $0.35, gain $0.10 for each correct value and lose $0.05 for each incorrect value, while "I don't recall" costs nothing. We code whether a subject recalls a pattern rather than an exact number, because model-directed attention concerns whether a day is noticed as different from a baseline day, and this coding is pre-registered. A subject recalls *the discount-day spike* if the value entered for Day 5 is above 100, *the pre-discount dip* if the value entered for Day 4 is below 100, and *the post-discount dip* if the value entered for Day 6 or Day 7 is below 100. As a placebo, we code whether a subject enters a value below 100 for any of Days 1 to 3, when sales were 100.

The study ends with nine Raven matrices as a measure of cognitive skills, followed by a demographic survey that asks for gender, age and education, among other characteristics. An abbreviated nine-item version of Raven's progressive matrices is a reliable proxy for the full 60-item test (Bilker et al., 2012). Subjects earn $0.05 for each correct answer.

We recruited US adults on Prolific in July 2026. Of the 600 subjects who completed the study, we exclude those who ticked "I don't recall" for all seven days in the recall task, which we see as a sign of low effort, and those whose answers were flagged as generated by an AI tool. Both exclusions are pre-registered. They leave 549 subjects: 177 in *Wrong model*, 180 in *Wrong model + attention* and 192 in *Correct model*. The median subject took 15 minutes.

4.1.2 The assigned model and attention to the data

We first test whether the assigned model causally directs subjects' attention to different data patterns as predicted by the conceptual framework. In particular, we examine whether the wrong model directs attention away from the pre-discount dip and the post-discount dip compared to the correct model, while not lowering attention to the discount-day spike.

Table 3 shows how the typical recalled week of sales differs between the two assigned models. In particular, it reports, for each of the first seven days, the median value recalled by subjects, excluding entries of "I don't recall" and those below 20 (the actual minimum sales is 82) and above 400 (the actual maximum is 202). Subjects in *Correct model* recall

[9]We assume that the assigned model affects what subjects encode, and not how they store or retrieve what they have encoded. The assumption is consistent with Gagnon-Bartsch, Rabin and Schwartzstein (2026), where the model directs attention at the moment the data are seen. Recall is elicited after the data are removed, so decay pushes recall of every pattern down and works against finding a difference across treatments. The treatment effect on recall is thus a potentially conservative measure of the treatment effect on attention. Recall is also measured with error, and we complement it with two other measures. In an earlier pilot we also measured attention through the "top-down" approach, by hiding each sales value behind a button that revealed it for one second when clicked, so that clicks on a day count as attention to that day. Appendix I.1 reports the results: subjects assigned the wrong model click the dip days less than subjects assigned the correct model. Further, the causal effect of attention is measured by the attention prompts, which change attention without relying on either measure of it.

a week close to the data, with medians of 92, 90 and 93.5 on Days 4, 6 and 7. Thus, a typical subject in the *Correct model* treatment recall both pre-discount and post-discount dips. In *Wrong model*, the median is 202 on Day 5 and 100 on every other day. A typical subject in this treatment thus instead only recalls one sales value at each price, consistent with model-directed attention.

Table 3: The typical recalled week in Study 1

	Data shown		Median recalled sales	
Day	Price	Sales	*Wrong model*	*Correct model*
1	5.9	100	100	100
2	5.9	100	100	100
3	5.9	100	100	100
4	5.9	**90**	100	92
5	5.2	202	202	202
6	5.9	**82**	100	90
7	5.9	**91**	100	93.5

Notes: This table reports, for each of Days 1 to 7, the price and sales shown to subjects and the median value entered in the recall task by subjects who did not tick "I don't recall" for that day. Entries below 20 or above 400, mostly prices typed into the sales boxes, are excluded. Shaded cells mark the three dip days, Days 4, 6 and 7. The sample includes $N = 177$ subjects in *Wrong model* and $N = 192$ in *Correct model*.

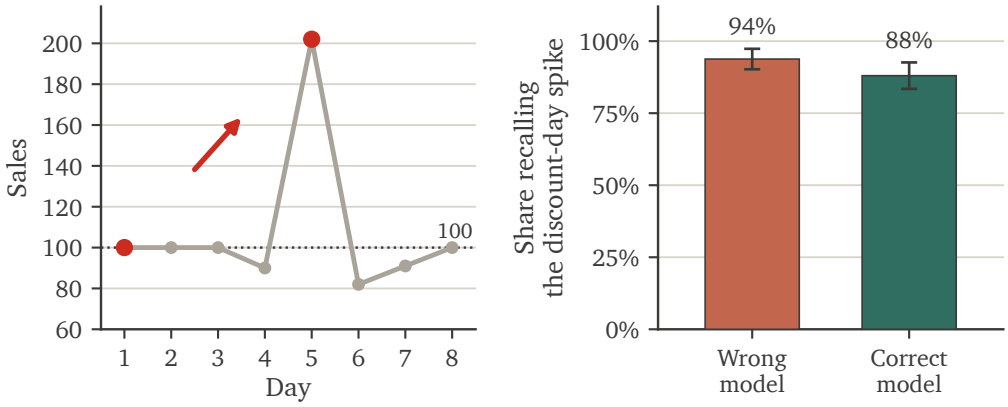
The wrong model lowers recall only of the patterns it deems irrelevant. Figure 6 shows the share of subjects who recall each pattern. Recall of the pre-discount dip is 27 percent in *Wrong model*, compared to 41 percent in *Correct model*, and recall of the post-discount dip is 27 compared to 55 percent (both $p < 0.01$). Almost everyone recalls the discount-day spike, 94 percent in *Wrong model* compared to 88 percent in *Correct model* ($p < 0.10$). Also, we see no treatment differences in recalling the placebo days, Day 1 to Day 3, which has sales of exactly 100. Most subjects in both treatments recall the sales to be 100, and 18 percent in *Wrong model* and 20 percent in *Correct model* enter a value below 100 for one of the first three days.

One concern with the high recall of the dips in *Correct model* is that the description of the correct model itself mentions the dips, and subjects may recall the description rather than the data. A more detailed look at the exact recalled values suggests otherwise. Subjects in *Correct model* enter the exact value of at least one dip day more often than subjects in *Wrong model*, 29 compared to 19 percent ($p < 0.05$). Also, this account cannot explain the higher share of subjects who recall 202 exactly on the discount day but no value below 100 for all regular days in *Wrong model* than in *Correct model* ($51\% vs. 20\%, p < 0.01$), which can be explained by model-directed attention.

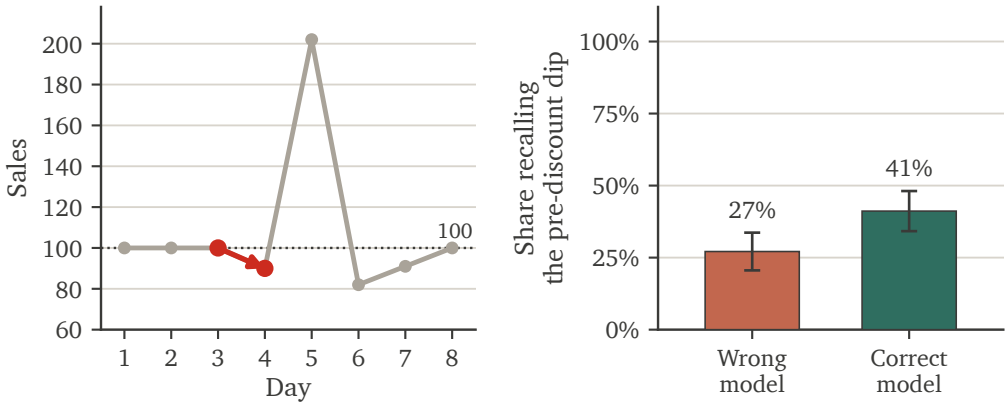
Result 1. *The wrong model lowers recall of the pre-discount and the post-discount dips, which it deems irrelevant, but does not lower recall of the discount-day spike, which it explains.*

Figure 6: Recall of the three patterns in Study 1: *Wrong model* and *Correct model*

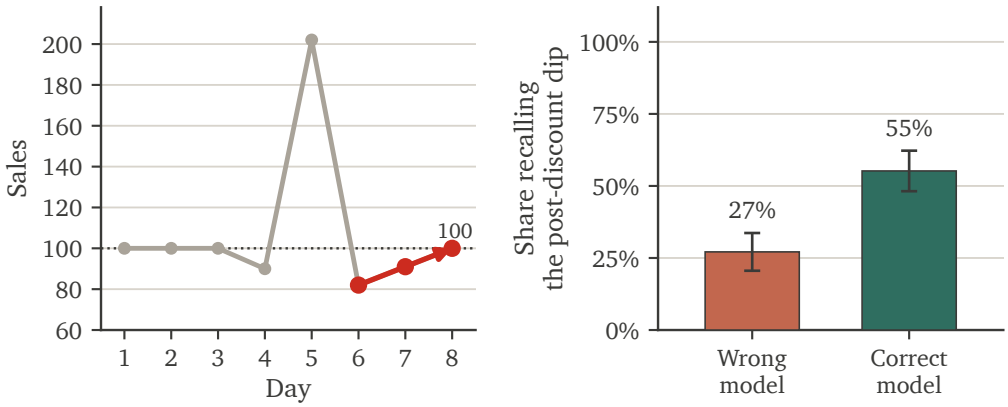
(a) The discount-day spike



(b) The pre-discount dip



(c) The post-discount dip

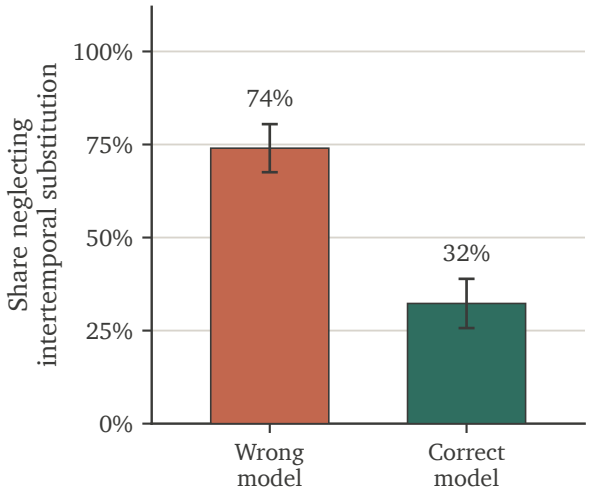


Notes: This figure shows how often subjects recall each of the three patterns. Panel (a) shows the discount-day spike, Panel (b) the pre-discount dip and Panel (c) the post-discount dip. In each panel, the left plot shows sales on Days 1 to 8 and marks one pattern with an arrow, and the right plot shows the share of subjects who recall that pattern. A subject recalls the discount-day spike if the value entered for Day 5 is above 100, the pre-discount dip if the value entered for Day 4 is below 100, and the post-discount dip if the value entered for Day 6 or Day 7 is below 100. Ticking "I don't recall" counts as not recalling the pattern. The sample includes $N = 177$ subjects in *Wrong model* and $N = 192$ in *Correct model*. Error bars show 95% confidence intervals.

4.1.3 The assigned model and predictions

The wrong model also changes what subjects predict. Figure 7 shows that 74 percent of subjects in *Wrong model* predict 202, compared to 32 percent in *Correct model*, a difference of 42 percentage points ($p < 0.01$). Both treatments see a table in which sales fall below 100 around each discount. Subjects in *Wrong model* place 47 of their 100 points on 202 on average, compared to 23 in *Correct model* ($p < 0.01$). In *Correct model*, 41 percent choose 165 and 20 percent choose 100; in *Wrong model*, 15 percent choose 165.

Figure 7: Share neglecting intertemporal substitution in Study 1: *Wrong model* and *Correct model*



Notes: This figure shows the share of subjects whose best guess of daily sales under a permanent price of 5.2 is 202, the answer that neglects intertemporal substitution. The correct answer is 165. The sample includes $N = 177$ subjects in *Wrong model* and $N = 192$ in *Correct model*. Error bars show 95% confidence intervals.

A menu of four answers could suggest a model to subjects who would not otherwise have one. In a follow-up pilot of Study 1 with 229 subjects, subjects type their prediction into an empty box (Appendix I). The wrong model again raises the share typing exactly 202, from 23 to 48 percent, and the share typing 200 or more, from 33 to 67 percent (both $p < 0.01$). It again lowers recall of the two dips, but not recall of the discount-day spike.

Result 2. *The wrong model raises the share of subjects who neglect intertemporal substitution.*

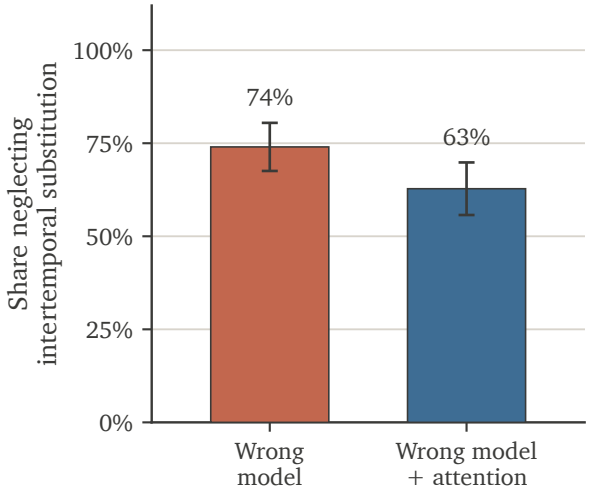
4.1.4 Directing attention to the dips

The attention prompts in *Wrong model + attention* restore recall of the dips. Appendix Figure I.12 compares this treatment to *Wrong model*. Recall of the pre-discount dip rises from 27 to 49 percent and recall of the post-discount dip from 27 to 59 percent (both $p < 0.01$). Both shares are close to those in *Correct model*, 41 and 55 percent, and neither difference from *Correct model* is significant. The prompts do not raise the placebo, 23 compared to 18 percent. Recall of the discount-day spike moves in the opposite direction to recall of the dips:

it is 94 percent in *Wrong model*, 88 percent in *Correct model* and 84 percent in *Wrong model + attention*. The assigned model and the attention prompts change which days subjects recall.

This treatment also addresses the possibility that subjects fill in the week from their model at the time of recall. Both treatments read the wrong description, and among subjects who predict 202, 67 percent in *Wrong model + attention* recall at least one dip, compared to 37 percent in *Wrong model* ($p < 0.01$). Subjects who give the wrong model's answer recall the dips when the prompts directed their attention to those days.

Figure 8: Share neglecting intertemporal substitution in Study 1: *Wrong model* and *Wrong model + attention*



Notes: This figure shows the share of subjects whose best guess of daily sales under a permanent price of 5.2 is 202, the answer that neglects intertemporal substitution. The correct answer is 165. The sample includes $N = 177$ subjects in *Wrong model* and $N = 180$ in *Wrong model + attention*. Error bars show 95% confidence intervals.

The attention prompts also lower the share of subjects who predict 202. Figure 8 shows that the share falls from 74 percent in *Wrong model* to 63 percent in *Wrong model + attention*, a reduction of 11 percentage points ($p < 0.05$). Subjects in *Wrong model + attention* also place fewer points on 202, 39 compared to 47 ($p < 0.05$), and with controls for age, gender and Raven score the reduction in the share predicting 202 is 13 percentage points ($p < 0.01$). Asking about these days tells subjects that they are worth explaining, but the prompts state no pattern and no explanation. Because the assigned model is the same in both treatments, we interpret the reduction as the effect of noticing the dips.

The prompts bring recall of the dips to the level of *Correct model*, but not the prediction. In *Wrong model + attention*, 63 percent of subjects predict 202, compared to 32 percent in *Correct model* ($p < 0.01$). The subjects who move off the wrong model do not move to 165: the share choosing 165 is 20 percent, compared to 15 percent in *Wrong model*, a difference that is not significant, while the share choosing 100 rises from 2 to 7 percent ($p < 0.05$). Appendix B.4 offers a potential explanation for why noticing the dips need not change the answer. A subject who attributes the dips to the calendar, with some days simply selling less, holds the predictor-neglect model, which contains no intertemporal substitution and can

still give 202 as the answer.

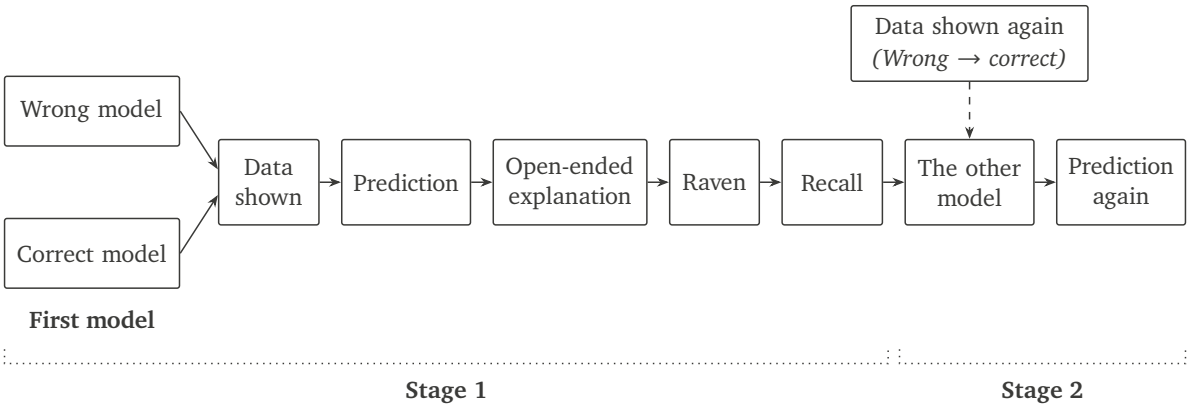
Result 3. *Directing attention to the dips lowers the share of subjects who neglect intertemporal substitution.*

4.2 Study 2

4.2.1 Design

Study 2 asks what happens when subjects who have already formed a view of the data encounter the other model. The study has two stages, and Figure 9 shows the timeline. From the first page, subjects know that they will draw one description from the bag now and a second description from the same bag later. The second description therefore cannot be mistaken for a response to their own answer.

Figure 9: Timeline of Study 2



Notes: This figure shows the timeline of Study 2. Subjects are randomized into three treatments: Wrong model first, Wrong model first with data reshown, and Correct model first, in the ratio 2:2:1.

Stage 1 repeats Study 1 without the attention prompts. Subjects draw a first description, see the data, predict sales under a permanent price of 5.2, distribute 100 points across the four answers and write the explanation. They then answer the nine Raven matrices and, after that, recall the first week of sales, with the same incentives as in Study 1. The Raven block places time and an unrelated task between the data and the recall. In stage 2, subjects draw a second description, which is always the other one: subjects assigned the wrong model first now read the correct description, and subjects assigned the correct model first now read the wrong description. Only the second description is shown. Subjects then choose among the same four answers again and distribute 100 points again, without writing a second explanation. They are told that one of the two predictions is drawn for payment.

Subjects are randomized into three treatments in the ratio 2:2:1. In two treatments, subjects are assigned the wrong model first, and in one of these the data table is shown again next to the correct description in stage 2. In the third treatment, subjects are assigned

the correct model first and see no data in stage 2. Showing the data again tests whether subjects judge the correct model against their memory of the data. For a subject who recalls the discount-day spike but not the dips, the correct model fits the recalled week no better than the wrong model. If subjects judge the correct model against their memory, showing the data again should lower the share predicting 202, with a larger effect among subjects who recall the discount-day spike but not the dips.

We again recruited US adults on Prolific. We apply the same two pre-registered exclusions as in Study 1 to the 1,003 subjects who completed the study. This leaves 896 subjects: 720 assigned the wrong model first (367 without and 353 with the data shown again) and 176 assigned the correct model first. The median subject took 19 minutes.

4.2.2 Encountering the other model

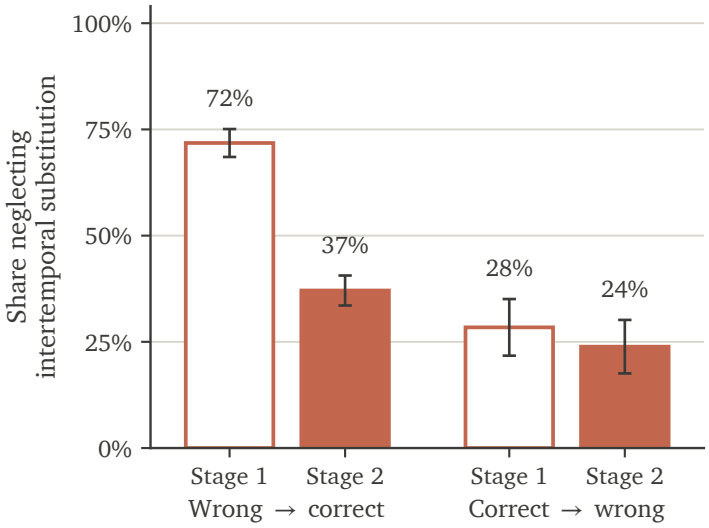
Stage 1 replicates Study 1. Among subjects assigned the wrong model first, 72 percent predict 202, compared to 28 percent among those assigned the correct model first ($p < 0.01$). Even though recall now comes after the Raven block, subjects assigned the wrong model first recall the discount-day spike far more often than the dips: 86 percent recall the discount-day spike, compared to 28 percent for the pre-discount dip and 30 percent for the post-discount dip ($p < 0.01$). Compared to subjects assigned the correct model first, they recall the post-discount dip less often, 30 compared to 52 percent ($p < 0.01$), and the pre-discount dip less often, 28 compared to 34 percent ($p < 0.10$), while recall of the discount-day spike is 86 percent in both groups.

Showing the data again next to the correct description does not change the share predicting 202. Without the data, the share predicting 202 falls by 35 percentage points; with the data, it falls by 34 percentage points, and it is 37 percent at stage 2 in both treatments. The 95 percent confidence interval of the difference between the two changes runs from -6 to 9 percentage points and excludes effects of 10 percentage points in either direction. Showing the data again also has no larger effect among subjects who recall the discount-day spike but not the dips. Because the correct description states in words that sales drop on the dip days, a subject assigned the wrong model first can correct the prediction without returning to the data or to the memory of it. Study 2 therefore does not show that the correction runs through the dips that subjects noticed in stage 1. We pool the two treatments below.

Predictions change in one order only, consistent with Prediction 5. Figure 10 shows the share predicting 202 in each stage. Among subjects assigned the wrong model first, the share falls from 72 to 37 percent after they read the correct description. Among subjects assigned the correct model first, it falls from 28 to 24 percent after they read the wrong description. The difference between the two changes is 30 percentage points ($p < 0.01$). Individual changes in prediction show the same asymmetry. Of the 517 subjects who predict 202 after reading the wrong description, 51 percent move off the wrong model after

encountering the correct model. Of the 126 subjects who do not predict 202 after reading the correct description, 4 move to the wrong model after encountering the wrong model. In the framework, a holder of the simple wrong model gives it up when the correct model is entertained and directs attention to data that refute it.

Figure 10: Share neglecting intertemporal substitution in each stage of Study 2



Notes: This figure shows the share of subjects who predict 202, the answer that neglects intertemporal substitution, in stage 1 (hollow bars) and in stage 2 (solid bars). Wrong → correct: subjects assigned the wrong model first and the correct model second, pooling those with and without the data shown again. Correct → wrong: subjects assigned the correct model first and the wrong model second. Error bars show 95% confidence intervals.

Result 4. *Encountering the correct model after being assigned the wrong model first significantly moves subjects off the wrong model; however, encountering the wrong model after being assigned the correct model first has no effect on moving subjects towards the wrong model.*

The description of the correct model that subjects read in stage 2 is incomplete. It explains the discount-day spike and the dips by customers postponing purchases and stocking up, and it does not mention that the low price also brings new demand. Taken literally, the only mechanism it offers is intertemporal substitution, as in the IS-only model of Section 2, which predicts 100. Among subjects assigned the wrong model first, reading it nevertheless raises the share choosing 165 from 14.4 to 35.4 percent, an increase of 21 percentage points, while the share choosing 100 rises by 6 percentage points, from 7.5 to 13.2 percent. The difference between the two increases is 15 percentage points ($p < 0.01$). In the framework, the IS-only model dislodges the simple wrong model in the same way as the correct model, because it directs attention to the dips (Section 2). A subject who held the wrong model has already noticed discount-day sales of 202, and noticing the dips alongside them refutes the IS-only model as well: the 37 purchases missing from the three surrounding days raise discount-day sales only to 137. Of the four models, only the correct model fits all five

sales values (Appendix B.3). The framework does not specify which model a decision maker adopts after giving one up, and Study 2 does not show that subjects take this route, since showing the data again changes nothing. It does show that an incomplete description of the correct model, and in that sense a wrong model, dislodges the wrong model and moves more subjects to 165 than to any other answer.

Subjects assigned the wrong model first move off it after encountering the correct model, but the correction is not full. At stage 2, both groups have read both descriptions and differ only in the order in which they read them. Appendix Figure I.13 shows the distribution of predictions in each stage. After both descriptions, 37 percent of subjects assigned the wrong model first predict 202, compared to 24 percent of those assigned the correct model first ($p < 0.01$), and 35 percent choose 165, compared to 49 percent ($p < 0.01$). The subjects who move off the wrong model do not all move to 165: of the 265 who move off it, 51 percent choose 165, 29 percent choose 192 and 20 percent choose 100. If subjects simply adopted the more detailed description, the correct description would move more subjects than the wrong one, but the two groups would end in the same place once both had read both descriptions. The persistence of the first model indicates that being assigned the wrong model first has a lasting effect even after encountering the correct model.

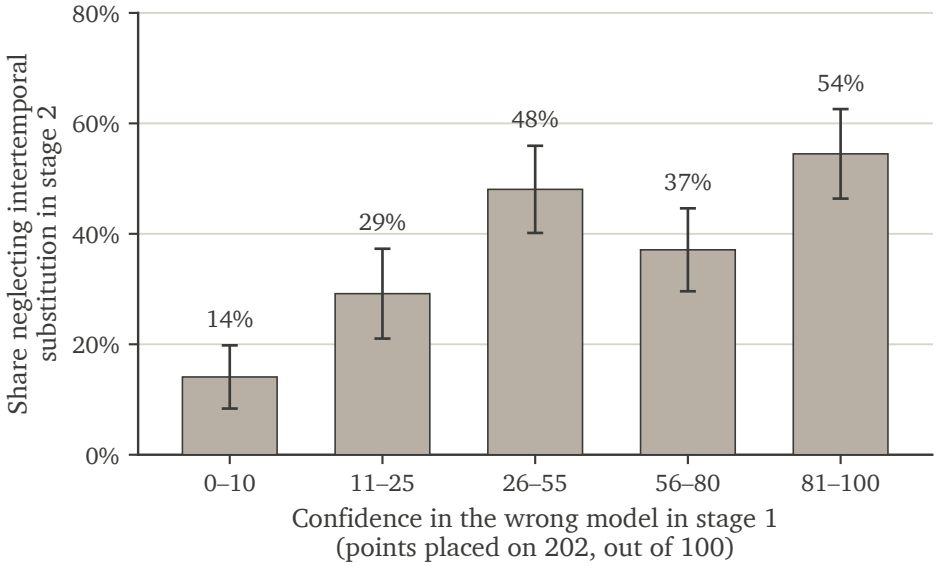
Result 5. *There is an order effect when both models are assigned: being assigned the wrong model first leads to a higher level of neglect of intertemporal substitution than being assigned the correct model first.*

In the framework, a decision maker entertains an encountered model only with some probability (Prediction 5). A subject who is certain of the wrong model may see nothing to learn from the correct one, and confidence in the first model is therefore one proxy for not entertaining the second. Figure 11 plots, for the 720 subjects assigned the wrong model first, the share who predict 202 in stage 2 by the points they allocated to 202 in stage 1. The share is 14 percent among subjects who placed at most 10 points on 202 and 54 percent among those who placed more than 80 points on it. Each additional 10 points on 202 is associated with a share that is 3.6 percentage points higher ($p < 0.01$). Subjects who were more confident in the wrong model are more likely to neglect intertemporal substitution after encountering the correct model.

5 Conclusion

Traditional economic theory assumes that a decision maker who sees enough data will end up with correct beliefs. This paper studies a way in which wrong beliefs can persist even when the data are abundant and arrive exogenously. We show that this can occur when the decision maker holds a wrong mental model from a broad class of models that are silent about what they cannot explain. The mechanism is model-directed attention: because the

Figure 11: Confidence in the wrong model and the prediction in stage 2 of Study 2



Notes: This figure shows the share of subjects who neglect intertemporal substitution (predict 202) in stage 2, by the points they placed on 202 in stage 1, for subjects assigned the wrong model first ($N = 720$). Subjects are divided into five groups of roughly equal size by the points they placed on 202: 0 to 10 points ($N = 142$), 11 to 25 ($N = 120$), 26 to 55 ($N = 154$), 56 to 80 ($N = 159$) and 81 to 100 ($N = 145$). Error bars show 95% confidence intervals.

mental model directs attention towards the patterns it explains, the decision maker neglects precisely the data features that would refute the model. What the decision maker notices does not contradict the model.

We document this in controlled settings in which we know the true data-generating process and the evidence that refutes a simple wrong model is in plain view. Our working example is a mental model of a fundamental economic relationship, how consumers respond to prices. The simple wrong model we study neglects that the response to a price cut can include regular customers shifting when they buy, in addition to new demand. The sign of this intertemporal substitution in data on prices and sales is a dip in sales right before and after a temporary discount. Among roughly 13,000 gas station managers, more than 20 percent neglect intertemporal substitution entirely when presented with a noise-free dataset that shows these dips twice, and the neglect is only weakly related to experience. Those who neglect it are less likely to notice the dips when explaining the data, perceive fuel demand as more elastic and set lower prices, which imply lower profits. Online experiments with Prolific subjects provide causal evidence. Holding the data fixed, being assigned the wrong model lowers recall of the dips and increases neglect of intertemporal substitution, while prompts that direct attention to those days restore recall and reduce neglect. Encountering a description of the correct model later dislodges the wrong model, though not completely, while encountering the wrong model after the correct one has no effect. The description has this effect although it is incomplete: one wrong model, in the sense of an incomplete one, dislodges another.

The mechanism is not specific to pricing or to intertemporal substitution. What makes a wrong model persist is structural rather than topical: the model fits well the patterns it explains and is silent about the rest. This property can identify persistent wrong models ex ante. Neglect of intertemporal substitution is itself an instance with reach beyond retail pricing. Anticipated policy changes generate the same signature. Purchases are pulled forward ahead of an announced tariff, a subsidy expiration or a tax change, which produces a surge and then a trough that do not reflect the underlying demand response. A model in which quantities respond only to prices attributes the surge to strong demand and the trough to weak demand, and commentary built on that model misjudges the policy in a predictable direction. The same silence can appear wherever timing is a margin of adjustment and the mental model neglects the timing dimension.

Two implications follow for how such wrong models might be corrected. First, providing more data is unlikely to help, since the constraint lies in what the model gives a reason to examine rather than in what is available. Second, our two interventions do different work. Directing attention dislodges the wrong model without installing the right one, while a description of the correct model does more. In our framework, a second model dislodges the wrong one by directing attention to the patterns the wrong model cannot explain, and it can do so without being complete. Sharing a model can help even when the model shared is incomplete. This may be one reason why organizations encourage their members to share ideas that are not yet fully formed.

Several questions remain open. In our online experiments we assign mental models through the descriptions of earlier subjects, to mimic social learning. In our field setting, managers report learning from mentors and peers early in their careers, which suggests that wrong models may be transmitted and therefore correlated within organizations rather than idiosyncratic across them. If so, shared reporting conventions and common training may sustain a shared silence that competition does not discipline. How wrong models spread, and which features of an environment make a wrong model persist, seem to us the natural next questions.

References

Ambuehl, Sandro and Heidi C. Thysen, "Choice with Competing Models: An Experimental Study," ECON Working Paper 458, Department of Economics, University of Zurich 2025. Revised November 2025.

_ , **Rahul Bhui, and Heidi C. Thysen**, "Mental Models of Causal Structure in Economics and Psychology," 2026. arXiv:2603.29070, April 2026.

Anderson, Richard C. and James W. Pichert, "Recall of Previously Unrecallable Information Following a Shift in Perspective," *Journal of Verbal Learning and Verbal Behavior*, 1978, *17* (1), 1–12.

Andre, Peter, Carlo Pizzinelli, Christopher Roth, and Johannes Wohlfart, "Subjective Models of the Macroeconomy: Evidence From Experts and Representative Samples," *The Review of Economic Studies*, 2022, *89* (6), 2958–2991.

_ , Ingar Haaland, Christopher Roth, Mirko Wiederholt, and Johannes Wohlfart, "Narratives about the Macroeconomy," *The Review of Economic Studies*, 2026. Advance access, published online 18 February 2026.

_ , Philipp Schirmer, and Johannes Wohlfart, "Mental Models of the Stock Market," *The Quarterly Journal of Economics*, 2026. Advance access, published online 25 July 2026.

Aragones, Enriqueta, Itzhak Gilboa, Andrew Postlewaite, and David Schmeidler, "Fact-Free Learning," *American Economic Review*, 2005, *95* (5), 1355–1368.

Ba, Cuimin, "Robust Misspecified Models," *American Economic Review*, 2026, *116* (4), 1340–1379.

Barron, Kai and Tilman Fries, "Narrative Persuasion," Discussion Paper SP II 2023–301r, WZB Berlin Social Science Center 2024. January 2023, revised April 2024.

Bénabou, Roland and Jean Tirole, "Mindful economics: The production, consumption, and value of beliefs," *Journal of Economic Perspectives*, 2016, *30* (3), 141–64.

Berk, Robert H., "Limiting Behavior of Posterior Distributions when the Model is Incorrect," *The Annals of Mathematical Statistics*, 1966, *37* (1), 51–58.

Bilker, Warren B, John A Hansen, Colleen M Brensinger, Jan Richard, Raquel E Gur, and Ruben C Gur, "Development of abbreviated nine-item forms of the Raven's standard progressive matrices test," *Assessment*, 2012, *19* (3), 354–369.

Bohren, J. Aislinn and Daniel N. Hauser, "Misspecified Models in Learning and Games," *Annual Review of Economics*, 2025, *17*, 427–451.

Boizot, Christine, Jean-Marc Robin, and Michael Visser, "The demand for food products: an analysis of interpurchase times and purchased quantities," *The Economic Journal*, 2001, *111* (470), 391–419.

Brewer, William F. and James C. Treyens, "Role of Schemata in Memory for Places," *Cognitive Psychology*, 1981, *13* (2), 207–230.

Cai, Jing and Adam Szeidl, "Interfirm Relationships and Business Performance," *Quarterly Journal of Economics*, 2018, *133* (3), 1229–1282.

Charles, Constantin and Chad Kendall, "Causal Narratives," 2025. Working paper, November 2025. Earlier version circulated as NBER Working Paper 30346 (2022).

Cho, In-Koo and Kenneth Kasa, "Learning and Model Validation," *The Review of Economic Studies*, 2015, *82* (1), 45–82.

Coglianese, John, Lucas W Davis, Lutz Kilian, and James H Stock, "Anticipation, tax avoidance, and the price elasticity of gasoline demand," *Journal of Applied Econometrics*, 2017, *32* (1), 1–15.

DellaVigna, Stefano and Matthew Gentzkow, "Uniform pricing in US retail chains," *The Quarterly Journal of Economics*, 2019, *134* (4), 2011–2084.

Doherty, Michael E., Clifford R. Mynatt, Ryan D. Tweney, and Michael D. Schiavo, "Pseudodiagnosticity," *Acta Psychologica*, 1979, *43* (2), 111–121.

Drew, Trafton, Melissa L.-H. Võ, and Jeremy M. Wolfe, "The Invisible Gorilla Strikes Again: Sustained Inattentional Blindness in Expert Observers," *Psychological Science*, 2013, *24* (9), 1848–1853.

Ellison, Glenn, "Bounded rationality in industrial organization," *Econometric Society Monographs*, 2006, *42*, 142.

Enke, Benjamin, "What You See Is All There Is," *The Quarterly Journal of Economics*, 2020, *135* (3), 1363–1398.

_ **and Florian Zimmermann**, "Correlation Neglect in Belief Formation," *The Review of Economic Studies*, 2019, *86* (1), 313–332.

_ **and Thomas Graeber**, "Cognitive uncertainty," *The Quarterly Journal of Economics*, 2023, *138* (4), 2021–2067.

Esponda, Ignacio and Demian Pouzo, "Berk–Nash Equilibrium: A Framework for Modeling Agents with Misspecified Models," *Econometrica*, 2016, *84* (3), 1093–1130.

_ **and Emanuel Vespa**, "Endogenous Sample Selection: A Laboratory Study," *Quantitative Economics*, 2018, *9* (1), 183–216.

_ **,** _ **, and Sevgi Yuksel**, "Mental Models and Transfer Learning," *AEA Papers and Proceedings*, 2023, *113*, 659–664.

_ **,** _ **, and** _ , "Mental Models and Learning: The Case of Base-Rate Neglect," *American Economic Review*, 2024, *114* (3), 752–782.

Fréchette, Guillaume R., Emanuel Vespa, and Sevgi Yuksel, "Extracting Models From Data Sets: An Experiment," 2025. Working paper, version of October 28, 2025, SSRN 5026751.

Frick, Mira, Ryota Iijima, and Yuhta Ishii, "Welfare Comparisons for Biased Learning," *American Economic Review*, 2024, *114* (6), 1612–1649.

Fudenberg, Drew and David K. Levine, "Self-Confirming Equilibrium," *Econometrica*, 1993, *61* (3), 523–545.

_ **and Giacomo Lanzani**, "Which Misspecifications Persist?," *Theoretical Economics*, 2023, *18* (3), 1271–1315.

_ , _ , **and Philipp Strack**, "Limit Points of Endogenous Misspecified Learning," *Econometrica*, 2021, *89* (3), 1065–1098.

Gabaix, Xavier, "A sparsity-based model of bounded rationality," *The Quarterly Journal of Economics*, 2014, *129* (4), 1661–1710.

_ , "Behavioral inattention," in "Handbook of behavioral economics: Applications and foundations 1," Vol. 2, Elsevier, 2019, pp. 261–343.

Gagnon-Bartsch, Tristan, Matthew Rabin, and Joshua Schwartzstein, "Channeled Attention and Stable Errors," August 2026. Working paper.

Galperti, Simone, "Persuasion: The Art of Changing Worldviews," *American Economic Review*, 2019, *109* (3), 996–1031.

Goldfarb, Avi and Mo Xiao, "Who thinks about the competition? Managerial ability and strategic entry in US local telephone markets," *American Economic Review*, 2011, *101* (7), 3130–3161.

_ **and** _ , "Transitory Shocks, Limited Attention, and a Firm's Decision to Exit," *Quantitative Marketing and Economics*, 2024, *22* (3), 223–255.

Golman, Russell, David Hagmann, and George Loewenstein, "Information avoidance," *Journal of economic literature*, 2017, *55* (1), 96–135.

Graeber, Thomas, "Inattentive Inference," *Journal of the European Economic Association*, 2023, *21* (2), 560–592.

Grass, Paul, Philipp Schirmer, and Malin Siemers, "Sticky Models," ECONtribute Discussion Paper 355, University of Bonn and University of Cologne 2026. Updated version dated June 19, 2026; also CRC TR 224 Discussion Paper No. 763; SSRN 5384226.

Han, Yi, David Huffman, and Yiming Liu, "Minds, models and markets: How managerial cognition affects pricing strategies," Technical Report, Working Paper 2025a.

Hanna, Rema, Sendhil Mullainathan, and Joshua Schwartzstein, "Learning Through Noticing: Theory and Evidence from a Field Experiment," *The Quarterly Journal of Economics*, 2014, *129* (3), 1311–1353.

He, Kevin and Jonathan Libgober, "Misspecified Learning and Evolutionary Stability," *Journal of Economic Theory*, 2025, *230*, 106082.

Heidhues, Paul and Botond Kőszegi, "Behavioral industrial organization," *Handbook of Behavioral Economics: Applications and Foundations 1*, 2018, *1*, 517–612.

_ , Botond Kőszegi, and Philipp Strack, "Unrealistic Expectations and Misguided Learning," *Econometrica*, 2018, *86* (4), 1159–1214.

_ , _ , and _ , "Convergence in Models of Misspecified Learning," *Theoretical Economics*, 2021, *16* (1), 73–99.

Hendel, Igal and Aviv Nevo, "Sales and consumer inventory," *The RAND Journal of Economics*, 2006a, *37* (3), 543–561.

_ and _ , "Measuring the implications of sales and consumer inventory behavior," *Econometrica*, 2006b, *74* (6), 1637–1673.

Hong, Harrison, Jeremy C. Stein, and Jialin Yu, "Simple Forecasts and Paradigm Shifts," *The Journal of Finance*, 2007, *62* (3), 1207–1242.

Hong, Lu and Scott E. Page, "Groups of Diverse Problem Solvers Can Outperform Groups of High-Ability Problem Solvers," *Proceedings of the National Academy of Sciences*, 2004, *101* (46), 16385–16389.

Hortaçsu, Ali and Steven L Puller, "Understanding strategic bidding in multi-unit auctions: a case study of the Texas electricity spot market," *The RAND Journal of Economics*, 2008, *39* (1), 86–114.

_ , Fernando Luco, Steven L Puller, and Dongni Zhu, "Does strategic ability affect efficiency? Evidence from electricity markets," *American Economic Review*, 2019, *109* (12), 4302–4342.

Jenkins, Herbert M. and William C. Ward, "Judgment of Contingency Between Responses and Outcomes," *Psychological Monographs: General and Applied*, 1965, *79* (1), 1–17. Whole No. 594.

Karni, Edi and Marie-Louise Vierø, ""Reverse Bayesianism": A Choice-Based Theory of Growing Awareness," *American Economic Review*, 2013, *103* (7), 2790–2810.

Kendall, Chad and Ryan Oprea, "On the Complexity of Forming Mental Models," *Quantitative Economics*, 2024, *15* (1), 175–211.

Klayman, Joshua and Young-Won Ha, "Confirmation, Disconfirmation, and Information in Hypothesis Testing," *Psychological Review*, 1987, *94* (2), 211–228.

Lanzani, Giacomo, "Dynamic Concern for Misspecification," *Econometrica*, 2025, *93* (4), 1333–1370.

Levy, Gilat, Ronny Razin, and Alwyn Young, "Misspecified Politics and the Recurrence of Populism," *American Economic Review*, 2022, *112* (3), 928–962.

Li, Shanjun, Joshua Linn, and Erich Muehlegger, "Gasoline Taxes and Consumer Behavior," *American Economic Journal: Economic Policy*, 2014, *6* (4), 302–342.

List, John A and Charles F Mason, "Are CEOs expected utility maximizers?," *Journal of Econometrics*, 2011, *162* (1), 114–123.

_ , Ian Muir, Devin Pope, and Gregory Sun, "Left-digit bias at lyft," *Review of Economic Studies*, 2023, *90* (6), 3186–3237.

Maćkowiak, Bartosz, Filip Matějka, and Mirko Wiederholt, "Rational Inattention: A Review," *Journal of Economic Literature*, 2023, *61* (1), 226–273.

Montiel Olea, José Luis, Pietro Ortoleva, Mallesh M. Pai, and Andrea Prat, "Competing Models," *Quarterly Journal of Economics*, 2022, *137* (4), 2419–2457.

Ortoleva, Pietro, "Modeling the Change of Paradigm: Non-Bayesian Reactions to Unexpected News," *American Economic Review*, 2012, *102* (6), 2410–2436.

Pesendorfer, Martin, "Retail sales: A study of pricing behavior in supermarkets," *The Journal of Business*, 2002, *75* (1), 33–66.

Schwartzstein, Joshua and Adi Sunderam, "Using Models to Persuade," *American Economic Review*, 2021, *111* (1), 276–323.

Simons, Daniel J. and Christopher F. Chabris, "Gorillas in Our Midst: Sustained Inattentional Blindness for Dynamic Events," *Perception*, 1999, *28* (9), 1059–1074.

Sims, Christopher A., "Implications of Rational Inattention," *Journal of Monetary Economics*, 2003, *50* (3), 665–690.

Slade, Margaret E, "Conjectures, firm characteristics, and market structure," *International Journal of Industrial Organization*, 1986, *4* (4), 347–369.

Smedslund, Jan, "The Concept of Correlation in Adults," *Scandinavian Journal of Psychology*, 1963, *4* (1), 165–173.

Spiegler, Ran, "Bayesian Networks and Boundedly Rational Expectations," *The Quarterly Journal of Economics*, 2016, *131* (3), 1243–1290.

Strulov-Shlain, Avner, "More than a penny's worth: Left-digit bias and firm pricing," *Review of Economic Studies*, 2023, *90* (5), 2612–2645.

Tadelis, Steven, Christopher Hooton, Utsav Manjeer, Daniel Deisenroth, Nils Wernerfelt, Nick Dadson, and Lindsay Greenbaum, "Learning, Sophistication, and the Returns to Advertising: Implications for Differences in Firm Performance," Technical Report, National Bureau of Economic Research 2023.

Wason, Peter C., "On the Failure to Eliminate Hypotheses in a Conceptual Task," *Quarterly Journal of Experimental Psychology*, 1960, *12* (3), 129–140.

Zimmermann, Florian, "The dynamics of motivated beliefs," *American Economic Review*, 2020, *110* (2), 337–61.

Online Appendix

A Related literature from psychology

Our paper complements a body of research in psychology showing in various ways that what a person is doing determines what the person sees and remembers (Anderson and Pichert, 1978; Brewer and Treyens, 1981; Simons and Chabris, 1999; Drew et al., 2013). In the canonical demonstration of inattentional blindness, viewers who have been instructed to count the passes made by one of two intermingled basketball teams, and who must track and filter a moving display to do so, fail to report a person in a gorilla suit walking slowly through the middle of it (Simons and Chabris, 1999). In the corresponding demonstration for memory, readers of a story about two boys spending the day at one boy's house recall different details of the same text depending on whether they were asked to read it from the point of view of a burglar or of a prospective homebuyer, and they produce details they had omitted the first time when the perspective is switched and they are asked to recall it again (Anderson and Pichert, 1978).

It also complements research demonstrating systematic biases in how humans verify the support for causal relationships using data (Smedslund, 1963; Jenkins and Ward, 1965; Doherty et al., 1979; Klayman and Ha, 1987). In demonstrations of covariation neglect, subjects who are handed complete data on cases (for each patient, whether a symptom was present and whether a diagnosis was present) judge whether the two are related largely on the strength of the cases in which both appear, giving little weight to the cases that bear equally on the question, so that their judgments end up nearly uncorrelated with the true contingency (Smedslund, 1963). In one canonical demonstration of the positive test strategy, by contrast, subjects must discover a rule the experimenter has in mind, and the only evidence available is evidence they generate themselves: they submit triples of numbers of their own choosing and learn of each only whether it satisfies the rule. They almost always submit triples that their current hypothesis says should qualify, rather than ones that would expose it as wrong: having inferred "ascending by twos" from the starting example 2, 4, 6, they try 8, 10, 12 instead of 3, 9, 11. Because the rule is merely "ascending," the data they assemble consist entirely of confirmations, and they announce the wrong rule with confidence (Wason, 1960; Klayman and Ha, 1987).

What this literature has not established is that a belief about how the world works, assigned from outside, determines which parts of complete and freely inspectable statistical data get noticed, and that the resulting error persists, to some extent resists correction by encountering the correct model, and has consequences beyond judgment for actual economic decisions.

B Additional results on the conceptual framework

B.1 Noticing and the first three predictions

Throughout we condition on the announced price path, with $H = D(5.9)$ and $L = D(5.2)$. A model specifies a common structure for a set of candidate demand processes and a prior over their unknown parameters. Write π_S for the simple wrong model, π_C for the correct model, π_P for the pre-discount-only model and π_T for the IS-only model. Parameters take integer values consistent with nonnegative sales, and each model's prior assigns positive mass to every admissible parameter vector. In particular, the true process has positive prior mass μ^* under π_C. Because the parameter spaces are countable, likelihoods are probability masses. On permanent-price paths, we set $d_t = \varnothing$ and $\delta(\varnothing) = 0$.

For one cycle in the table, denote baseline sales by b and sales at distances $-1, 0, 1, 2$ by x_{-1}, x_0, x_1, x_2. The displayed values are

$$(b, x_{-1}, x_0, x_1, x_2) = (100, 90, 202, 82, 91).$$

A noticed sales value includes its price and, when the model distinguishes it, its distance to the discount day. Baseline sales provide the regular-price reference. We describe the distinctions noticed in a completed table without assuming an order in which its rows are read. We also distinguish a value implied by the model from an observation separately noticed in the table.

Model-directed attention without a reading order. Write R_M for the representation of the table under model π_M. The noticing described in the main text is represented by

$$\begin{aligned} R_S &= (b, x_0), \qquad R_P = (b, x_{-1}, x_0), \\ R_C &= (b, x_{-1}, x_0, x_1, x_2), \\ R_T &= (b, x_{-1}, x_1, x_2, c_b(x_0)), \end{aligned}$$

where the IS-only model censors exact discount-day sales above baseline:

$$c_b(x) = \begin{cases} x, & x \le b, \\ \mathsf{high}, & x > b. \end{cases}$$

Here high records only that sales exceed baseline, so histories with different discount-day sales can have the same representation. Histories sharing a representation form one noticing cell (Gagnon-Bartsch, Rabin and Schwartzstein, 2026). These maps specify the noticing strategies we study under Assumption 1; model-directed attention permits other partitions as well.

Claim 1 (Identification from noticed data)**.** *For each of the four models, its representation identifies all of its demand and substitution parameters under that model. The displayed data therefore contain sufficient information for parameter learning, without an additional assumption that learning succeeds.*

Proof of Claim 1. Whenever the noticed observations are consistent with the model, they determine a unique parameter vector. The two noticed sales values give $(H, L) = (b, x_0)$ under π_S. Under π_P, the pre-discount shortfall identifies substitution, giving $(H, L, a) = (b, x_0 + x_{-1} - b, b - x_{-1})$. Under π_C, the shortfall on each surrounding day identifies $a_j = b - x_j$, $j \in \{-1, 1, 2\}$, and conservation then gives $H = b$ and $L = x_0 + x_{-1} + x_1 + x_2 - 3b$. For π_T, $H = b$ and the same three differences identify its substitution amounts, which imply $x_0 = 4b - x_{-1} - x_1 - x_2 = 137 > b$. The implied discount-day sales agree with the category high noticed for the actual 202, so identification does not require noticing exact discount-day sales. Each wrong model therefore fits its representation of the displayed data even though it fails to fit the full table. □

The identified parameters determine the permanent-price prediction. The representations are therefore sufficient in the sense of Gagnon-Bartsch, Rabin and Schwartzstein (2026, Definition 1); no separate behavioral condition is needed. Here sufficiency applies at the end of inspection: predictions are elicited after the table is available, there are no payoff-relevant decisions during inspection, and repeated decision dates index completed tables or cycles. Sufficiency for forecasts at intermediate dates would require a separate argument. The representations need not be minimal under their Definition 2, since b alone determines the IS-only model's permanent-price prediction, and $x_0 + x_{-1} + x_1 + x_2 - 3b$ determines the correct model's prediction without distinguishing every component.

Let n^t be the observations noticed by t and $P(n^t \mid \pi)$ their probability under model π, integrating over its prior. Because the true deterministic process assigns these observations probability one and has positive prior mass under the correct model, $\mu^* \leq P(n^t \mid \pi_C) \leq 1$. A likelihood ratio against π_C therefore reaches zero, or has limit inferior zero, exactly when the ratio against the truth does, which establishes the equivalence to the comparison in Gagnon-Bartsch, Rabin and Schwartzstein (2026, Definition 3). Under Assumption 2, the correct model is the known alternative for comparison.

Proof of Prediction 1. A candidate process in π_S, specified by (H, L), assigns sales H to every regular-price day and L to every discount day. Under the specified noticing strategy, the decision maker notices only b and x_0, the values on Days 1 and 5, which fix $(H, L) = (100, 202)$. This parameter vector fits every observation the strategy notices, including under repetitions of the schedule. Its positive prior mass $\beta_S > 0$ therefore bounds the probability of every finite noticed history under π_S from below by β_S. Since the probability under π_C is

at most one,

$$\frac{P(n^t \mid \pi_S)}{P(n^t \mid \pi_C)} \geq \beta_S > 0$$

at every t. Under the specified strategy, the model is therefore never rejected, either at a finite date or through a likelihood ratio tending to zero. Stability of π_S depends on the strategy. Another sufficient strategy, which separately notices a baseline and a dip, refutes π_S. □

The bound also holds against any alternative model over these discrete outcomes, whose likelihood is at most one. If that likelihood is zero, the positive numerator gives a ratio of $+\infty$. The main text's use of censoring follows the logic of Gagnon-Bartsch, Rabin and Schwartzstein (2026, Appendix B.2). Our deterministic, price-conditioned process does not satisfy the conditions of their Proposition B.2, so stability here follows from the positive prior mass argument above, without directly applying their stationary Definition B.1.

Proof of Prediction 3. The detergent example below gives the elasticity comparison. The same reasoning applies to fuel demand if managers use the same models in both settings and mistake purchases shifted toward fuel discounts for new demand. Applying the comparison across settings requires this condition in addition to the information in the table.

For the pricing comparison, consider a common family of feasible pricing policies indexed by x in an interval, where a higher x gives a lower average posted price. The index allows for deeper price cuts, more frequent discounts, or other policies that lower the average posted price. Hold true demand, costs and competitors' behavior fixed, and evaluate profits over the same horizon, including intertemporal substitution. Let $V(x)$ be actual profit, correctly perceived under the correct model, and $\widehat{V}_S(x)$ be profit perceived under the simple wrong model. Both functions are differentiable and strictly single-peaked, with unique interior maximizers x_C and x_S, and each manager maximizes profit under their own model. Suppose neglect of IS overstates the marginal gain from lowering prices at the correct-model optimum:

$$\widehat{V}_S'(x_C) > V'(x_C) = 0.$$

At the correct-model optimum, a manager with the simple wrong model still perceives a gain from lowering prices. Single-peakedness therefore implies $x_S > x_C$, giving a lower average posted price and, because x_C uniquely maximizes actual profit, $V(x_S) < V(x_C)$. With market conditions held fixed, the pricing choice accounts for the profit loss. The strict comparison requires the assumed marginal distortion for the policy managers choose, which the two-price elasticity comparison alone does not establish for every policy. It also uses interiority: discrete choices or binding constraints can leave chosen policies unchanged.

For a price-level choice, the condition follows from overstating demand elasticity at the correct price. Let $D^*(p)$ denote true demand and $\widehat{D}_S(p)$ demand perceived under the simple wrong model. Both are positive, continuously differentiable and downward-sloping on

a common feasible price interval $I \subset (c, \infty)$, where $c \geq 0$ is marginal cost, and for this comparison we assume that the holder of the correct model knows true permanent-price demand throughout I. Define the absolute price elasticities

$$\varepsilon^*(p) = -\frac{pD^{*\prime}(p)}{D^*(p)}, \qquad \widehat{\varepsilon}_S(p) = -\frac{p\widehat{D}_S'(p)}{\widehat{D}_S(p)}.$$

Suppose $\widehat{\varepsilon}_S(p) > \varepsilon^*(p)$ on I. True and perceived profits are $\Pi^*(p) = (p-c)D^*(p)$ and $\widehat{\Pi}_S(p) = (p-c)\widehat{D}_S(p)$. Each is strictly increasing up to a unique interior maximizer and strictly decreasing thereafter. Denote these maximizers by p_C and p_S, respectively. Both decision makers maximize profits under their own model, face the same feasible prices, and know the same marginal cost.

At the correct-model optimum, the first-order condition gives

$$\frac{p_C - c}{p_C}\varepsilon^*(p_C) = 1.$$

The derivative of perceived profit at that same price is therefore

$$\widehat{\Pi}_S'(p_C) = \widehat{D}_S(p_C)\left[1 - \frac{p_C - c}{p_C}\widehat{\varepsilon}_S(p_C)\right] < 0.$$

Perceived profit is decreasing at the correct-model optimum, so single-peakedness implies $p_S < p_C$. Both decision makers face the same actual profit function, uniquely maximized at p_C, and hence $\Pi^*(p_S) < \Pi^*(p_C)$. The difference in the model used to choose the price thus lowers actual profit. The strict elasticity inequality at p_C suffices for the argument, with the inequality throughout I providing a sufficient condition. This extension specifies perceived demand between the two observed prices, where the table alone does not identify it.

For discount frequency, fix a regular price p_h and a discount price p_l, with $p_h > p_l > c$. Suppose promotions do not overlap and each shifts $A > 0$ units to the discount day from regular-price days. True demand is D_h at the regular price and D_l at the discount price. The true profit gain from one additional discount day, including the associated dips, is

$$B = (p_l - c)D_l - (p_h - c)D_h - (p_h - p_l)A.$$

The simple wrong model counts $D_l + A$ as demand and ignores the dips, giving a perceived gain

$$\widehat{B} = (p_l - c)(D_l + A) - (p_h - c)D_h, \qquad \widehat{B} - B = (p_h - c)A > 0.$$

Holding these quantities and any other correctly perceived policy costs fixed, the simple wrong model overvalues each additional discount by $(p_h - c)A$. This is the positive marginal distortion required for the policy comparison. Because more frequent discounts lower the average posted price, that comparison applies even when the policy change is not a permanent

price cut.

In the numerical example, under π_S, $\widehat{D}(5.9) = 100$ and $\widehat{D}(5.2) = 202$. Under π_C, all five values are noticed, and conservation gives

$$\widehat{D}(5.2) = x_0 + x_{-1} + x_1 + x_2 - 3b = 165, \qquad \widehat{D}(5.9) = b = 100.$$

The finite-change elasticities, relative to the regular price, are $-(102/100)(5.9/0.7) \approx -8.60$ and $-(65/100)(5.9/0.7) \approx -5.48$, respectively. These are comparisons at the two displayed prices, without extrapolation to a continuous demand curve.

For pricing, let $c < 5.2$ be marginal cost. The true profit change from a permanent cut is

$$(5.2 - c)165 - (5.9 - c)100 = 268 - 65c,$$

whereas a holder of π_S expects $(5.2-c)202-(5.9-c)100 = 460.4-102c$. The latter exceeds the former by $(5.2 - c)37 > 0$. Whenever

$$268/65 < c < 460.4/102,$$

the holder of π_S cuts the price and the holder of π_C does not. The stated cost $c = 4.3$ lies in this interval. True daily profits are 160 at the regular price and 148.5 at the permanent low price, whereas π_S predicts 181.8 at the low price. The strict price and profit ordering therefore also holds in this two-price example. □

Proof of Prediction 4. Under every candidate process in π_S, all regular-price sales equal H. Two separately noticed regular-price values that differ therefore have probability zero under π_S. Days 3 to 4 and 9 to 10 contain 100 and 90, and Days 6 to 8 and 12 to 14 contain 82, 91 and 100. Separately noticing different sales values within any one of these windows therefore refutes π_S, while the correct model admits all these observations. □

The simple wrong model fails to conceptualize the event that two regular-price days have different sales (Gagnon-Bartsch, Rabin and Schwartzstein, 2026, Definition 5). A prompt that exposes this event refines the original noticing partition at the prompt date. Setting the refined strategy equal to the original at all other dates gives the temporary refinement in their Definition 11, and the direct proof above establishes rejection at the prompt date.

B.2 Two other wrong models

The pre-discount-only model. Candidate processes in π_P have parameters (H, L, a) with $a \geq 0$, and

$$q_t = D(p_t) + \delta_P(d_t), \qquad \delta_P(d) = \begin{cases} -a, & d = -1, \\ a, & d = 0, \\ 0, & \text{otherwise.} \end{cases}$$

This model preserves conservation but attributes all substitution to purchases postponed from the pre-discount day. It therefore distinguishes that day from other regular-price days, which it treats alike. Under the specified strategy, the decision maker notices (b, x_{-1}, x_0), identifying

$$H = b = 100, \qquad a = b - x_{-1} = 10, \qquad L = x_0 - a = 192.$$

Discount-day sales are needed to identify L under π_P, since b and x_{-1} determine only H and a. Once the price is permanent, the substitution term is zero and the model predicts sales of 192 at 5.2.

The process with these parameters assigns sales of 100 to both post-discount days. Because the model places them in the same category as baseline days, the decision maker does not notice their actual sales of 82 and 91. The parameter vector $(100, 192, 10)$ thus fits every history noticed under this strategy, and its positive prior mass β_P bounds the probability of each history from below by β_P. As for π_S, this likelihood bound proves stability under repetitions of the schedule. Separately noticing either post-discount sales value alongside baseline sales would instead refute π_P.

The IS-only model. Candidate processes in π_T have parameters (H, a_{-1}, a_1, a_2) with non-negative substitution amounts, and

$$q_t = H + \delta_T(d_t), \qquad \delta_T(d) = \begin{cases} -a_{-1}, & d = -1, \\ a_{-1} + a_1 + a_2, & d = 0, \\ -a_1, & d = 1, \\ -a_2, & d = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Here H is demand at either permanent price and a_j is the quantity shifted to the discount day from the day at distance j. Demand is independent of price, and conservation requires that all additional discount-day sales come from purchases shifted from surrounding days.

The exact values in its representation are (b, x_{-1}, x_1, x_2), which give

$$H = 100, \qquad a_{-1} = 10, \qquad a_1 = 18, \qquad a_2 = 9.$$

Because these observations determine demand and all substitution amounts under π_T, they also determine discount-day sales:

$$\widehat{x}_0 = H + a_{-1} + a_1 + a_2 = 4b - x_{-1} - x_1 - x_2 = 137. \tag{2}$$

The decision maker notices discount-day sales only through $c_b(x_0) = \mathsf{high}$, which is the same for the observed $x_0 = 202$ and the implied 137. These different sales values therefore generate the same representation under π_T, without an arithmetic error in using the noticed observations. Identification uses the baseline and dips jointly, in any reading order. At a permanent price, purchases are no longer shifted toward temporary discounts, and predicted sales equal $H = 100$. Since the fitted parameter vector agrees with every noticed observation, its positive prior mass β_T bounds the likelihood of every noticed history away from zero. The specified strategy is therefore stable under repetitions of the schedule.

Under π_T, discount-day sales are a distinct category even though the baseline and dips determine their exact value. Censoring a model-implied value can thus sustain the model without requiring that value to equal another noticed value, as it does under π_S and π_P. Here censoring is the explicit coarsening c_b. Stability follows from the likelihood bound, without applying the stationary censored-model theorem in their Appendix B.2. Separately noticing 202 together with the other four values refutes π_T, since conservation requires

$$x_{-1} + x_0 + x_1 + x_2 = 4b,$$

while the two sides evaluated at the observed values are 465 and 400. Both the pre-discount-only model and the IS-only model can therefore be refuted by separately noticing sales values already in the table.

Beliefs and prices across the four models. Predicted sales at a permanent price of 5.2 are 202 under π_S, 192 under π_P, 165 under π_C, and 100 under π_T, with finite-change elasticities of approximately -8.60, -7.75, -5.48 and zero. At $c = 4.3$, decision makers with π_S and π_P choose a permanent price of 5.2 because their expected profits of 181.8 and 172.8, respectively, exceed the 160 earned at the regular price. Both earn actual profit of 148.5, whereas decision makers with π_C and π_T choose the regular price and earn 160. Although the IS-only model understates demand elasticity in absolute value, it yields the same price choice as the correct model in this two-price example. A strictly higher price or lower profit under the IS-only model would require additional prices and demand assumptions.

We use the numerical answers as proxies for mental models, since guessing or other models can generate the same answer: the predictor-neglect model below, for example, allows a subject to notice the dips and still predict 202. Similarly, selecting 192 does not establish that the pre-discount dip was noticed. Each model also predicts which sales values the decision maker notices, and attention can therefore be examined separately from the numerical

answer.

B.3 Encounters between models

The encounter protocol. The probability of entertaining an encountered alternative is α. If the alternative is entertained and the relevant data are available, the decision maker temporarily notices the alternative's specified observations together with those already noticed under the incumbent, and evaluates both models on these observations. This common refinement of noticing is an additional assumption about encounters, beyond sufficiency for the original counterfactual question.

In Gagnon-Bartsch, Rabin and Schwartzstein (2026, Definition 13), a perturbed model is a positive-weight mixture of an incumbent and an alternative prior, which by itself does not require noticing their combined observations. We specify inspection at the encounter date τ by letting N'_τ be the common refinement of the incumbent's history partition N_τ and the partition recording the alternative's specified observations, with $N'_t = N_t$ for $t \neq \tau$. This is a temporary-noticing refinement under their Definition 11, defined on the common history space including histories to which the incumbent assigns zero probability. Rejection is assessed at τ using both previously noticed and additional observations. At a later encounter, the conclusion is conditional on the observations the decision maker still remembers and uses, since Definition 11 does not specify subsequent memory. The complete comparisons below require inspection of the combined observations before either model is dismissed, including when the alternative conflicts with previously noticed data. Dismissal before further inspection can prevent the additional refutations, but under either protocol a model contradicted by observations already noticed cannot be adopted.

Proof of Prediction 5. (i) Because π_S assigns the same sales value to every regular-price day, a refinement that reveals both baseline sales and a distinct regular-price dip gives it zero likelihood. Each alternative, π_P, π_T or π_C, induces such a comparison under the stated protocol. (ii) A holder of π_C has noticed all five sales values, including baseline sales and a distinct regular-price dip. These values give π_S zero likelihood, and the decision maker cannot adopt it on entertaining it. The correct model fits all five values and survives the encounter. □

Which models survive the combined observations? When the encounter involves π_S and π_P, the decision maker notices (b, x_{-1}, x_0). The different regular-price sales refute π_S, while π_P fits these observations at $(H, L, a) = (100, 192, 10)$ and is the only model in the pair that survives. An encounter involving π_C and any wrong model instead leads to noticing all five sales values. Because each wrong model contradicts at least one of these observations, only π_C survives that comparison and predicts 165.

An encounter involving π_S and π_T also leads to noticing all five sales values, but here both models are refuted: the different regular-price sales contradict π_S, and the observed discount-day sales of 202 contradict Equation 2 under π_T. Both models are also refuted in an encounter involving π_P and π_T: the post-discount sales contradict π_P, and all five observations together contradict π_T. These comparisons hold in either encounter order because the stated refinement requires inspecting the combined observations before dismissing either model.

An additional wrong model can thus expose a mistake by directing attention to a previously ignored distinction, as in the discussion of fresh eyes in Gagnon-Bartsch, Rabin and Schwartzstein (2026, Sections 6–7).

Profits and the manager's own data. Lower profits play no direct role in the rejection rule. Bohren and Hauser (2025) distinguish criteria based on model fit or perceived predictive accuracy (Cho and Kasa, 2015; Montiel Olea et al., 2022; Lanzani, 2025; Ba, 2026) from criteria based on payoff performance or welfare (Levy et al., 2022; Fudenberg and Lanzani, 2023; Frick et al., 2024; He and Libgober, 2025). Assumption 2 uses the former, applied to noticed data. Gagnon-Bartsch, Rabin and Schwartzstein (2026, Proposition 6) show that wrong models can admit stable attentional strategies with arbitrarily large payoff costs in some choice environments.

The framework also takes the price path to be exogenous, which the table satisfies, whereas a station manager's own pricing decisions affect the data generated. In such settings a wrong model may persist because the observations generated by its induced actions fit it, as in self-confirming equilibrium (Cho and Kasa, 2015). This mechanism is distinct from overlooking refuting observations in a fixed table.

B.4 The predictor-neglect model

Section 4.1.4 notes that a subject can notice the dips and still predict 202 if they attribute the dips to the calendar, with some days simply selling less. Some managers give a related explanation of the detergent data, that consumer demand is cyclical (Section 3.3.2). The predictor-neglect model formalizes this explanation. It conditions on the day but not on a day's position relative to the discount. Its candidates are

$$q_t \;=\; D(p_t) + b_t,$$

where $\{b_t\}$ is a deterministic profile of day-specific demand that is unrestricted across days and is a property of the calendar rather than of the price path. Nothing is left to noise, and nothing about the profile is tied to where the discounts fall. For any chosen values $H = D(5.9)$ and $L = D(5.2)$, the effects $b_t = q_t - D(p_t)$ fit the table, which therefore does not identify either demand value or their difference. Consider a decision maker who takes

the baseline-day value, 100, as $D(5.9)$, and 202 as $D(5.2)$. The calendar effect is then zero on baseline days and on the two dates on which a discount happened to fall. The fitted effects are then -10 on Days 4 and 10, -18 on Days 6 and 12, and -9 on Days 7 and 13.

Claim 2 (Non-refutation on the displayed table)**.** *Fix the fourteen dates in the table and the two demand values. Under a prior that assigns positive mass to every vector of fourteen integer day effects, noticed data have positive probability under the model, however finely attention to this table is directed. Under the reading above its answer to the counterfactual is 202 on any date with no calendar effect, which is the answer that neglects intertemporal substitution, and* $202 + \bar{b}$ *averaged over a week, where* $\bar{b} = -37/7$*, or about 196.7.*

Proof of Claim 2. For every displayed date, set $b_t = q_t - D(p_t)$. This vector assigns the entire table probability one. Under a full-support prior over the fourteen integer day effects the noticed data therefore have positive probability whatever distinctions attention is directed to. The counterfactual statement follows from reading the discount-day value as $D(5.2) = 202$ and the day effects as invariant to the price path: at a permanent 5.2, sales on date t are $202 + b_t$; averaging the fitted effects -10, -18 and -9 over the seven days of a week gives $202 - 37/7 \approx 196.7$. □

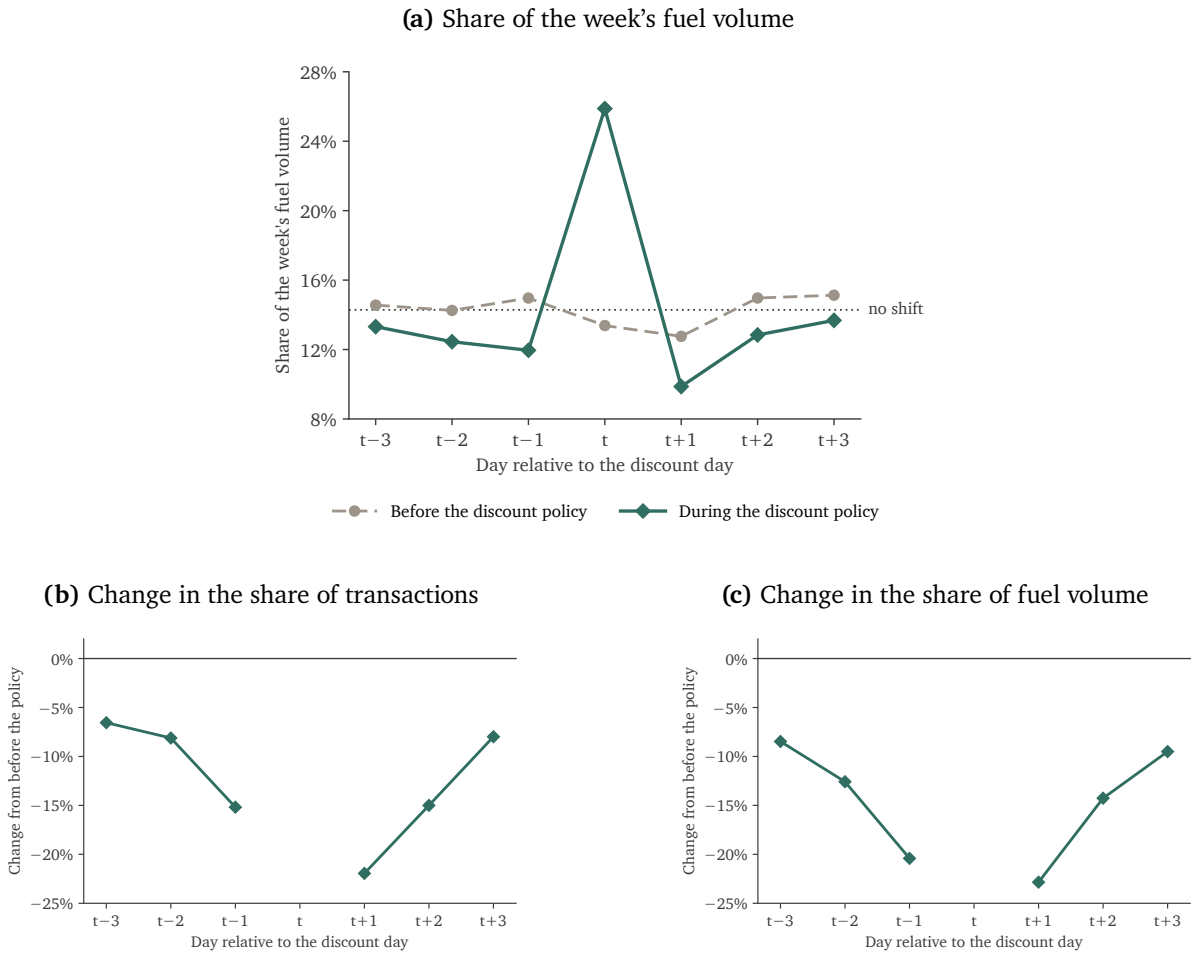
The counterfactual is conditional on the decision maker's reading. For any fitted values H and L, sales at a permanent price of 5.2 on date t would be $q_t + (L - H)\mathbf{1}\{p_t = 5.9\}$. The table does not identify $L - H$. For example, $H = 100$ and $L = 165$, with $b_t = 37$ on discount dates and the observed effects on the other dates, also fit the data and give an average of 165 at a permanent price of 5.2. Predictor neglect alone therefore implies neither an answer of 202 nor an upward-biased average.

Three remarks bound the claim. First, disciplined special cases of the model are refutable. A pattern that repeats by day of the week cannot fit Table 1: the two discounts fall on different days of their respective weeks, the fifth and the fourth, and the dips follow the discount: the third day of the week is a baseline day in one week and low in the other, the sixth is low in both weeks but at different values, and the seventh is low in one and a baseline day in the other. A subject reaching for weekly seasonality is contradicted once they notice the values. It is the unrestricted day-specific version that fits everything. It supplies no intertemporal substitution, and gives the wrong counterfactual under the stated reading. Second, moving the discount changes the dips under the correct model, while calendar effects stay fixed. Using this difference to distinguish the models requires observations under different price paths for the same dates, or restrictions connecting calendar effects across dates. The fourteen dates, each observed once, cannot refute unrestricted day effects. This establishes non-refutation on the displayed table; it does not establish stability under the asymptotic likelihood-ratio criterion as new dates accumulate. Supplying the correct model offers an alternative explanation, but does not by itself refute the predictor-neglect model. Third, idiosyncratic noise would accommodate the values equally well, but the instructions

foreclose it: managers are told that the table shows all the variables that may affect profit, subjects that it shows all the variables that can affect sales, and both that the data are representative of all possible situations. Deterministic day effects are the accommodation that the instructions leave open. They, and not noise, are the relevant caveat to Prediction 4.

C Evidence on the existence of intertemporal substitution in fuel markets

Figure C.1: Regular customers' fuel volume, and the change in each day's share, in the week around a weekly discount day



Notes: This figure shows regular customers' fuel volume and the change in each day's share of the week around the discount day, at stations with one discount day per week. Panel (a) shows the share of the week's fuel volume that falls on each day, before and during the policy. Panels (b) and (c) show the percentage change in each day's share of the week's transactions and fuel volume, during the policy relative to before it, and omit the discount day. Day t is the discount day; before the policy, it is the day of the week that later becomes the discount day. Regular customers and the sample are as in Figure 1. Shares during the policy start from its second week. The dotted line in Panel (a) marks one day in seven (14.3%).

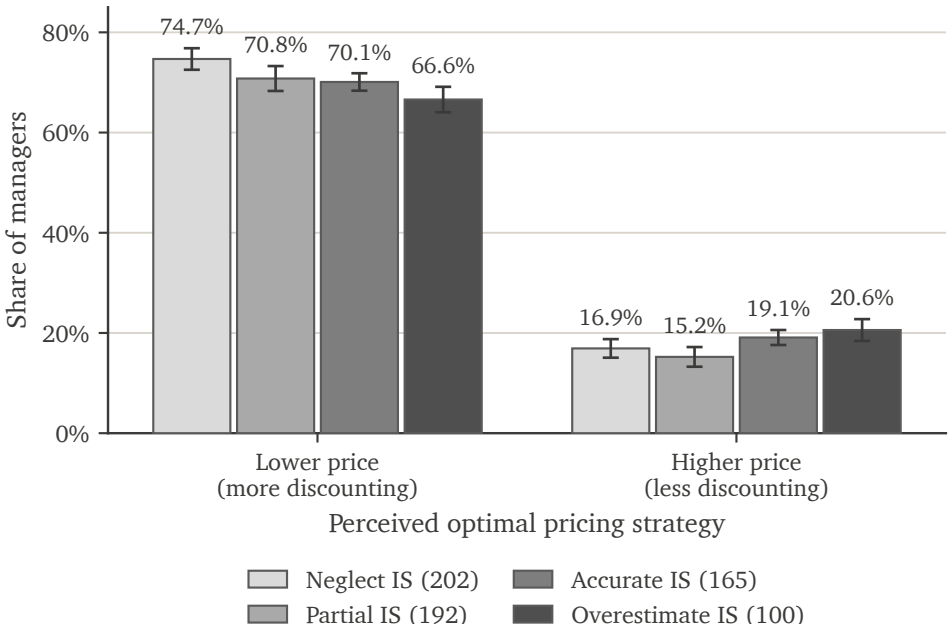
Fuel volume shows the same pattern as transactions (Figure C.1). During the policy, the discount day's share of regular customers' weekly fuel volume rises from 13.4 to 25.9 percent, and the shares of the day before and the day after it fall from 15.0 to 12.0 percent and from 12.8 to 9.9 percent. Relative to before the policy, the share of transactions falls by 15 percent on the day before the discount day and by 22 percent on the day after it, and the share of fuel volume falls by 20 and 23 percent.

D Additional results on predictions in the detergent task

Figure D.2 shows views on optimal pricing by prediction in the detergent task. The share of managers favoring a lower price, with more discounting, declines across the four predictions: 74.7 percent among managers who predict 202, 70.8 percent among managers who predict 192, 70.1 percent among managers who predict 165 and 66.6 percent among managers who predict 100. The share favoring a higher price, with less discounting, is 16.9 and 15.2 percent among managers who predict 202 and 192, and 19.1 and 20.6 percent among managers who predict 165 and 100.

This pattern is consistent with Prediction 3, in which neglecting intertemporal substitution leads managers to overestimate demand elasticity. A manager who sees much of the sales increase from a temporary discount as purchases shifted in time, rather than as new demand, has reason to be more cautious about discounting.

Figure D.2: Mental models of intertemporal substitution and perceived optimal price: four categories

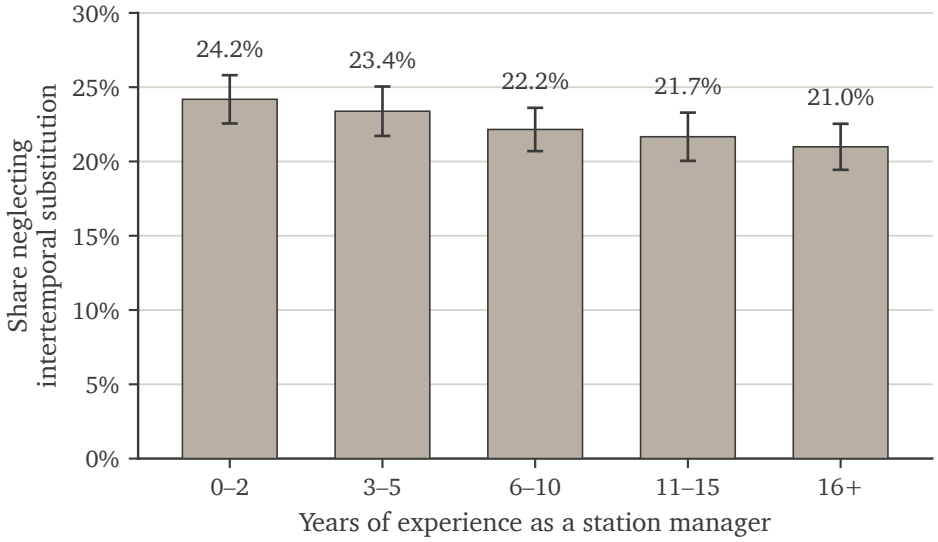


Notes: This figure shows the share of managers choosing each perceived optimal pricing strategy, by their prediction in the detergent task. The lower price strategy involves more discounting and the higher price strategy less discounting. The sample includes the $N = 6{,}840$ managers who answered the pricing question. Error bars show 95% confidence intervals.

E Coding the managers' explanations

We classify each manager's explanation of the detergent data on two dimensions: which of the three patterns in the data the explanation reveals that the manager noticed, and the cause the manager gives. This appendix sets out the rubric. A large language model with extended reasoning applied it to each of the 13,407 explanations in a separate request. Each

Figure D.3: Neglect of intertemporal substitution and experience as a station manager



Notes: This figure shows the share of managers who predict 202 in the detergent task, the answer that neglects intertemporal substitution, by years of experience as a station manager. Error bars show 95% confidence intervals.

request contained the full rubric, one manager's answer in the original language together with a machine translation, and an instruction to write a justification for each pattern before coding it. The rubric describes the research question and tells the coder that we expect many managers not to notice the dips, but no request contained the manager's prediction. To measure how stable the codes are, the model classified each explanation twice, in separate requests that did not contain the other classification. The two classifications agree on whether an explanation reveals a dip in 96.9 percent of cases. All results in the paper use the first classification. Before the final classification, two different language models coded a pilot sample of 120 explanations, and we resolved each of the 25 disagreements between them ourselves, also without seeing the managers' predictions. The rubric below reflects those decisions. Two earlier classifications by other language models, made under earlier versions of the rubric, agree with the final one on whether an explanation reveals a dip. The first covers all 13,407 explanations and agrees in 86 percent of cases. The second covers 5,188 of them and agrees in 90 percent. In both, managers who predict 202 are the least likely to notice a dip and managers who predict 192 the next least likely.

What the coder is told about the data. The rubric first describes the data in Table 1. Baseline sales are 100 at the regular price of 5.9. Sales are 90 on the day before the one-day discount, 202 on the discount day at the price of 5.2, and 82 and 91 on the two days after, and the pattern repeats in the second week while the competitor's price is constant. The rubric then gives the question, which asks the manager to look at Days 4 to 7 and 10 to 13 and explain why sales show this pattern before and after the days when the price is 5.2. Since the question already points at those days and calls what happens on them a pattern,

the rubric reads each answer as an attempt to explain that pattern unless it plainly is not one. The rubric also explains what intertemporal substitution is and what each of the four predictions means as a mental model, because the coder needs that background to tell a manager who noticed the dips from one who did not.

The three patterns and the test. These are lower sales on the day before the discount (90 compared to 100), high sales on the discount day (202) and lower sales on the two days after (82 and 91 compared to 100). A pattern counts as noticed when the explanation, taken as a whole, could not plausibly have been written by a manager who had not noticed it. The manager need not describe it. The coder proceeds in order. It first works out what the manager meant, reading the original answer and treating the machine translation as a hint, since answers are often terse and contain typing errors. It then states in one sentence what the manager believes was happening on the days around the discount, including what customers were doing. Only then does it ask, for each pattern, whether that belief requires the day to differ from an ordinary day, and code the pattern as noticed if it does. For every noticed pattern the coder records whether the explanation states it or only implies it, so that a stricter measure can be computed. Where two readings are equally plausible, the rubric credits the pattern and the coder marks its confidence as low. We chose this direction because it works against our claim that many managers do not notice the dips. Treating every dip in a code marked as low confidence as not noticed lowers the share of managers who notice a dip to 14.9 percent among those who predict 202 and 30.8 percent among those who predict 165 ($p < 0.01$), compared to 21.0 and 39.9 percent under the rubric. The rules that follow are presented to the coder as worked examples of this reasoning rather than as a list to match, with the instruction that its reading of the actual sentence prevails where a rule and the reading conflict.

Causes that imply a pattern. Of the explanations that reveal a dip, 78.5 percent do not describe one; they give a cause that only makes sense if the dips are there. Customers waiting for the discount, or holding off because they know it is coming, imply the pre-discount dip. Customers having stocked up, having bought enough, or needing time before buying again imply the post-discount dip, and usually the discount-day spike as well, because demand can only be used up if much of it was spent on the discount day. An explanation that attributes the pattern to unchanged underlying demand or an unchanged customer base, or that says the days around the discount add up to an ordinary period, describes the discount as rearranging the same demand. This is the IS-only model of Section 2, and holding it requires having noticed the discount-day spike and the dips that offset it. The rubric therefore codes all three patterns. A cycle, habit, saturation or psychology story offered as the reason for the pattern counts for the days it locates. A story that locates no day and denies no difference, such as regularities in consumption, is treated as an account of the whole shape and codes

all three patterns. By contrast, a cycle that names the discount day as the busy one and calls the other days ordinary codes only the discount-day spike.

Concentration and the direction of movement in time. An explanation that customers concentrate their purchases on the discount day explains why that day is high. On its own it does not claim that the surrounding days are below an ordinary day, and it codes only the discount-day spike. What adds a dip is a direction of movement in time. Waiting moves purchases out of the days before and codes the pre-discount dip. Buying earlier than one otherwise would, or demand from the following days gathered onto the discount day, leaves the days after short and codes the post-discount dip. An explanation with both directions codes both dips. This rule withholds only a dip the explanation does not describe: if the same explanation also marks a fall after the discount in one of the ways set out in the next paragraph, the post-discount dip is coded.

Recovery, and the price mechanism in reverse. The days after the discount sell 82 and 91, and the following day is back at 100. The post-discount dip is coded when the explanation places those days below an ordinary day, not merely below the discount day. An explanation in which sales come back, recover, return to normal or take a few days to regain their level does this, since there is nothing to return from unless the days after were below normal. The same holds for a fall after the discount that the explanation marks as unexpected, as worse than before, as temporary, as needing time or as a sharp drop on the next day. It also holds for an episode split into a rush and a later stage in which buyers wait to buy again, and for customers whose capacity or need is used up once the price is back. The price mechanism in reverse, sales falling because the price went back up, places the days after below the discount day but not below an ordinary day, and does not code the post-discount dip. Neither does an explanation that denies any later difference.

The discount-day spike, and what codes no pattern. Any explanation treating the discount day as selling unusually well codes the discount-day spike, including the bare statement that a low price raises sales. Any cause offered for that day's sales, such as word of mouth, spreading awareness, publicity, the depth of the discount or more vehicles coming in, also codes the discount-day spike, as does an account of how customers read the price. An explanation that only names or labels the day without giving a reason codes no pattern, as does an answer with no substantive content. A complaint that promotion was inadequate codes no pattern unless the explanation offers it as the reason some days were low, in which case it locates that dip.

Causes and intertemporal substitution. The coder also assigns one of the causes in Table E.1. An explanation is based on intertemporal substitution when it attributes the dips

to customers changing the timing of their purchases. This classification follows the explanation's account of the dips rather than the cause assigned to it, so an explanation assigned another cause, most often saturation, can be based on intertemporal substitution, and a timing explanation that only places purchases on the discount day is not. Table E.2 gives examples of explanations and their codes, with our translations.

Table E.1: Causes in the rubric

Cause	Description
Timing	Customers change when they buy: they wait for the discount, buy ahead, stock up or concentrate purchases on the discount day, or demand is used up afterwards
New customers	The discount brings extra customers, including customers of the competitor
Price and volume	A low price sells more, with no claim about the surrounding days
Cycle	A consumption, refuelling or purchase cycle, or a calendar effect
Saturation	The market or the customer base is limited, or demand is met or exhausted
Fatigue	The novelty or the effect of the promotion fades
Psychology	Expectations, price perception, or how customers feel about the price
Awareness	Customers learn of the promotion, through publicity or word of mouth
Competitor	The competitor's price or behavior
Random timing	The discount day is random, unannounced or badly timed
Other	A substantive cause that fits none of the above
None	No cause given

Notes: This table reports the causes the coder assigns to each manager's explanation of the detergent data, one per explanation.

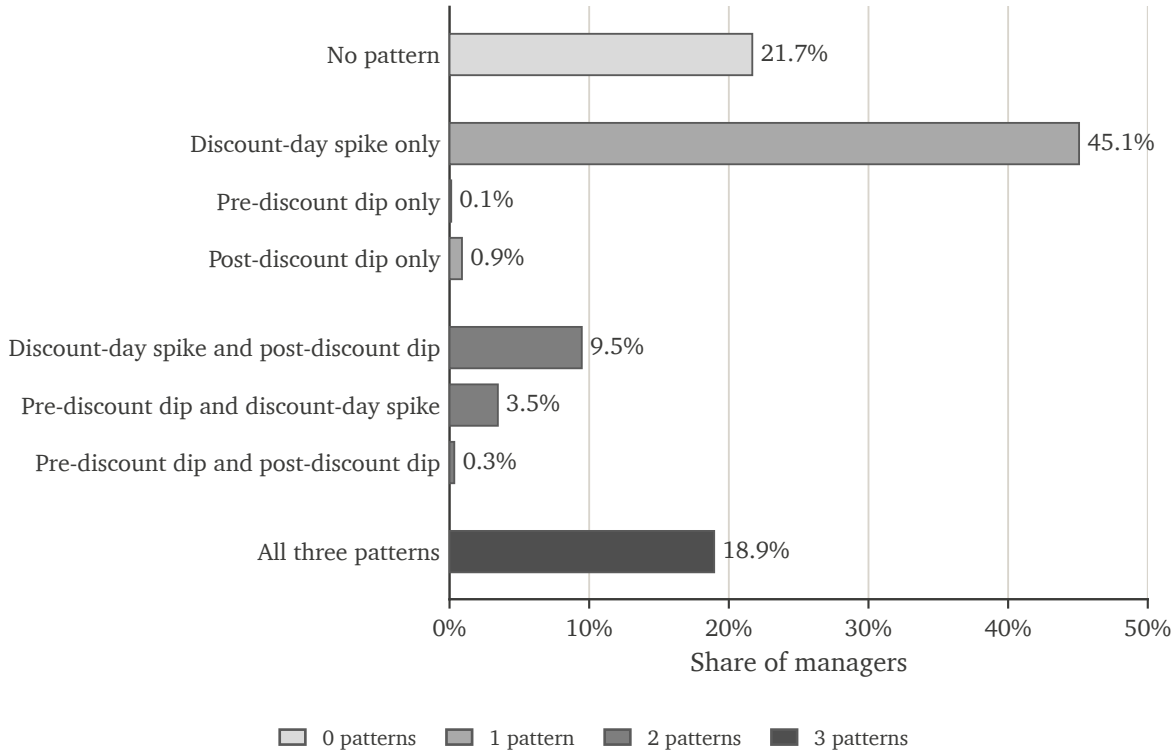
Table E.2: Examples of explanations and their codes

Explanation	Pre-discount dip	Discount-day spike	Post-discount dip	Cause	Intertemporal substitution
"Sales rose after the price cut, and profit rose with them."	0	1	0	Price and volume	No
"Once customers know about the promotion, they choose to shop on the promotion day."	1	1	1	Timing	Yes
"Some customers buy on the promotion day."	0	1	0	Timing	No
"Retailer A's customers mostly wait for the promotion to buy."	1	1	0	Timing	Yes
"The price cut makes customers buy earlier."	0	1	1	Timing	Yes
"Demand is stable, and so is the customer base."	1	1	1	Saturation	Yes
"Added up over the four days around the discount, sales and profit are no different from usual."	1	1	1	Timing	Yes
"Customers have no demand left."	0	1	1	Saturation	Yes
"Regularities in consumption."	1	1	1	Cycle	No
"The first refuelling cycle has passed."	0	0	1	Cycle	No
"Days 5 and 11 are probably weekends, people concentrate their refuelling on those days, and the other days are about level."	0	1	0	Cycle	No
"After the promotion, sales settle back to normal."	0	1	1	Price and volume	No
"Once the price returns to normal, sales come down."	0	1	0	Price and volume	No
"The discount has a clear effect on the day, but once the price is restored, sales actually fall."	0	1	1	Price and volume	No
"After the promotion, sales are the same as usual."	0	1	0	Price and volume	No
"The first stage is a promotional rush, and the second is a stage of buying back."	0	1	1	Timing	Yes
"Customer referrals."	0	1	0	Awareness	No
"Members' day."	0	0	0	None	No
"The promotion was not publicized well."	0	0	0	Awareness	No
"Weak publicity early on lowered gross profit, and later sales rose and profit recovered."	1	1	1	Awareness	No
"Market factors."	0	0	0	None	No
"Sales were stable at first but later declined somewhat."	0	0	0	Other	No

Notes: This table reports examples of managers' explanations of the detergent data, in our translations, with the codes the rubric assigns. A 1 means that the explanation reveals that the manager noticed the pattern. The last column reports whether the explanation is based on intertemporal substitution, that is, whether it attributes the dips to customers changing the timing of their purchases.

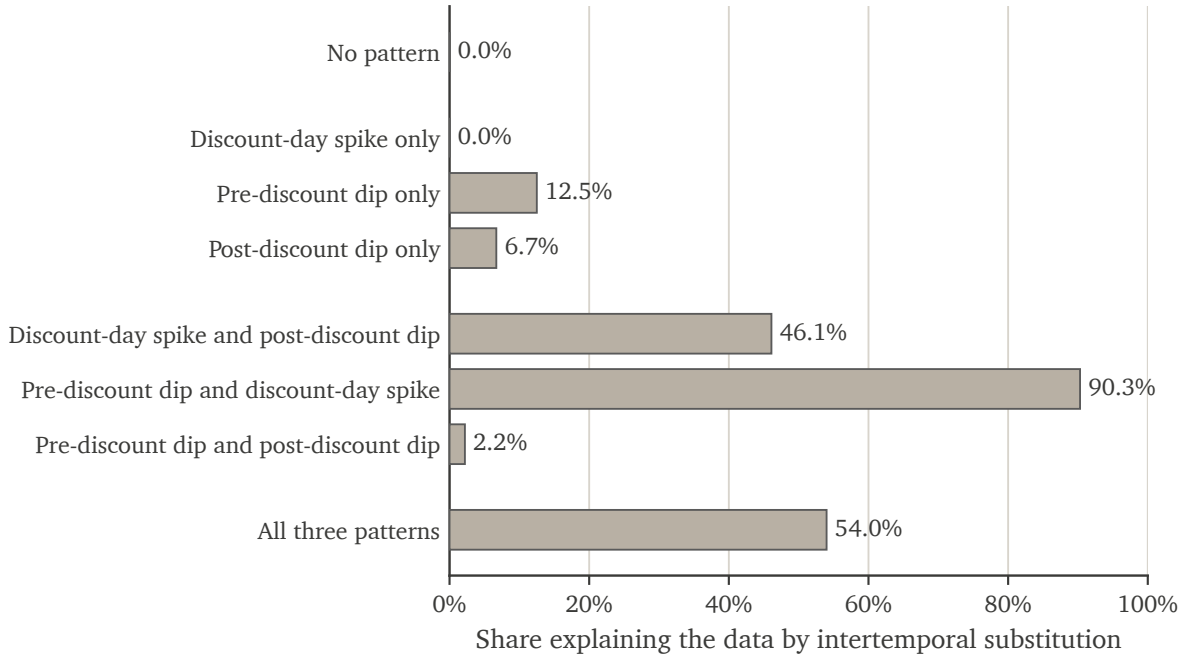
F Additional results on mechanisms underlying manager neglect of intertemporal substitution

Figure F.4: Type and number of data patterns noticed in manager explanations



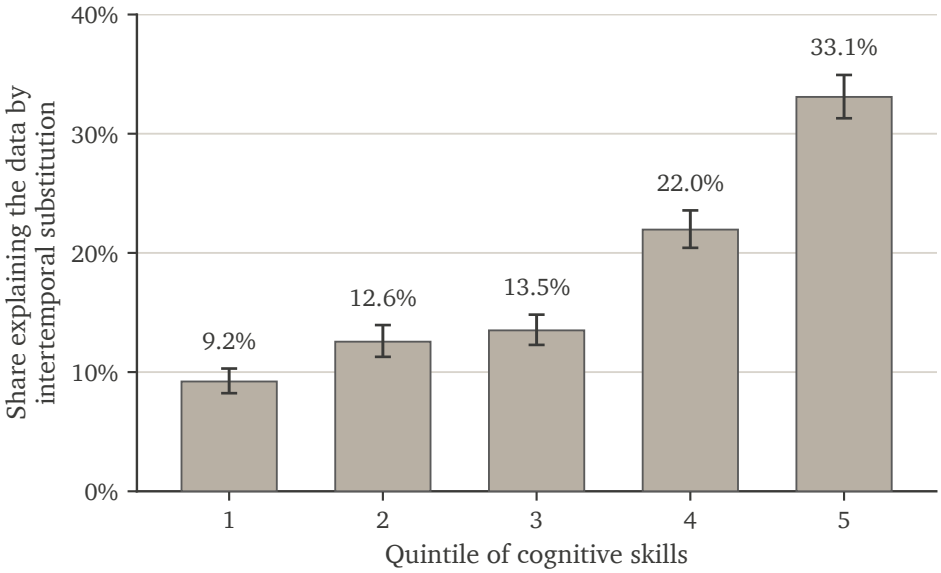
Notes: This figure shows the share of managers whose explanation of the detergent data reveals that they noticed each combination of the three data patterns: the pre-discount dip, the discount-day spike and the post-discount dip. Shading gives the number of patterns noticed, and the bars are grouped by that number. Explanations were classified by a large language model with the rubric in Appendix E. The sample includes $N = 13{,}407$ managers, and the categories partition the sample.

Figure F.5: Frequencies of explanation based on intertemporal substitution, conditional on number and type of patterns noticed



Notes: This figure shows the share of managers whose explanation of the detergent data is based on intertemporal substitution, by which of the three data patterns the explanation reveals that they noticed. An explanation is based on intertemporal substitution when it attributes the dips to customers changing the timing of their purchases. The categories are those of Figure F.4. The shares for no pattern and for the discount-day spike alone are zero by construction, since an explanation based on intertemporal substitution accounts for the dips. The shares for the pre-discount dip alone and for the two dips without the discount-day spike rest on 16 and 46 managers. The sample includes $N = 13{,}407$ managers.

Figure F.6: Frequency of explaining the data by intertemporal substitution, by cognitive skills



Notes: This figure shows the share of managers whose explanation of the detergent data is based on intertemporal substitution, by quintile of cognitive skills. An explanation is based on intertemporal substitution when it attributes the dips to customers changing the timing of their purchases. Cognitive skills are measured by the Raven score, and quintiles are of unequal size because the score is heavily tied. The sample includes $N = 13{,}407$ managers. Error bars show 95% confidence intervals.

G Additional results on the consequences of neglecting intertemporal substitution

This appendix reports the analyses behind Section 3.4 and managers' beliefs about the profitability of price cuts for fuel (Appendix G.4). Throughout, managers are grouped by their prediction in the detergent task: 202 neglects intertemporal substitution, 192 underestimates it, 165 is the correct answer and 100 attributes the entire sales response to intertemporal substitution.

G.1 The lasting price cut task

The lasting price cut task follows the detergent task in the survey. About half of the managers, selected at random, answer it (6,840 managers). Managers see two weeks of diesel prices and sales at one gas station, shown in Table G.3. The station cuts its price from 5.83 to 5.53 on March 5, and sales jump to 11,273 liters that day and then fall below their earlier level. After a second cut, to 5.33 on March 14, sales jump to 16,206 liters and fall to 7,572 liters the next day. Managers then read the question

> "If the station had reduced the price to 5.53 per liter starting from March 2 instead of March 5, what do you think the sales volume would have been from March 2 to March 6?"

and predict sales for each of the five days. In the figures and tables below, Day 1 to Day 5 are March 2 to March 6: Day 1 is the new start of the price cut and Day 4 is its original start. The task comes in five randomly assigned versions, which differ in a statement by another station manager shown with the data; one version shows no statement. Intertemporal substitution implies that the sales spike moves to March 2 and that sales are lower on the days after it. The data are real and noisy, and managers may also consider factors such as competitor responses.

We first summarize the five predictions with a principal component analysis of the standardized predictions. Two components have eigenvalues greater than 1, and Table G.4 reports their scoring coefficients. Component 1 has positive weights on Day 2 to Day 5 and almost no weight on Day 1. Component 2 has a large weight on Day 1 and a negative weight on Day 4. Because of these weights, predictions with high sales on Day 1 and lower sales afterwards, the profile intertemporal substitution implies, have a high score on Component 2 and a low score on Component 1.

We then group managers by k-means clustering of the five standardized predictions. The Calinski-Harabasz criterion selects four clusters, and we use the four-cluster solution with the lowest within-cluster sum of squares across many random starts. Figure G.7 plots the

Table G.3: Gas station data in the lasting price cut task

Date	Price	Price ceiling	Fuel sales	Profit
February 28	5.83	5.83	7,800	7,800
March 1	5.83	5.83	4,275	4,275
March 2	5.83	5.83	6,439	6,439
March 3	5.83	5.83	9,776	9,776
March 4	5.83	5.83	9,293	9,293
March 5	5.53	5.83	11,273	7,891
March 6	5.53	5.83	6,431	4,501
March 8	5.53	5.83	4,676	3,273
March 9	5.53	5.83	2,350	1,645
March 10	5.53	5.83	4,020	2,814
March 11	5.53	5.83	5,875	4,112
March 12	5.53	5.83	6,376	4,463
March 14	5.33	5.83	16,206	8,103
March 15	5.33	5.83	7,572	3,786

Notes: This table reports the 14 days of data that managers see in the lasting price cut task: diesel prices, the price ceiling, sales in liters and profit at one gas station. Prices are per liter, and prices and profit are in local currency.

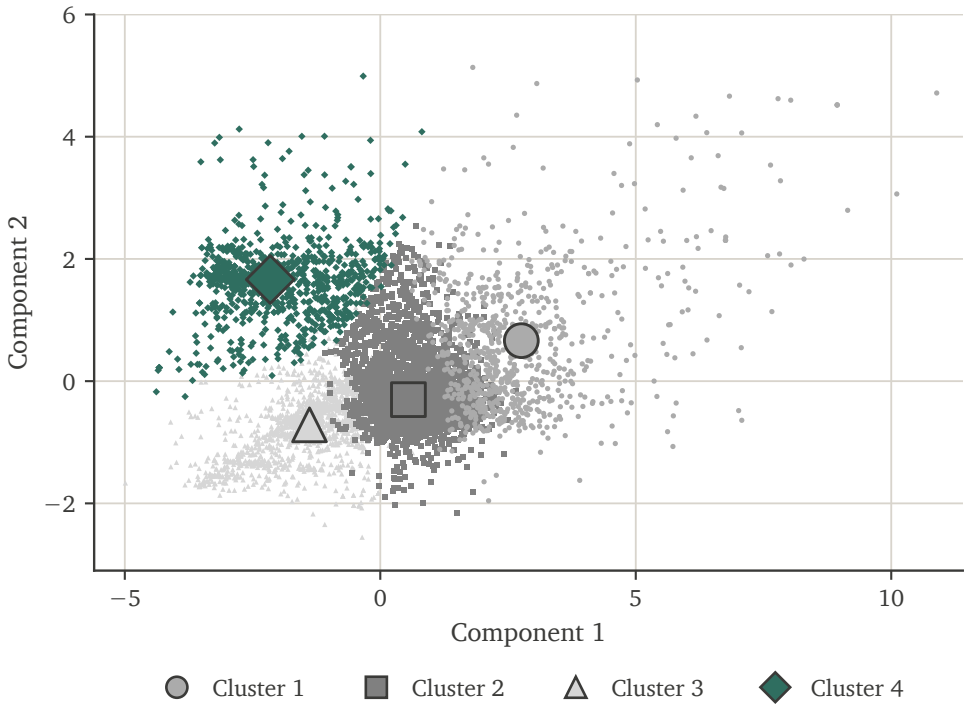
Table G.4: Principal component scoring coefficients for the sales predictions in the lasting price cut task

Variable	Component 1	Component 2
Day 1 sales (z-score)	0.048	0.949
Day 2 sales (z-score)	0.504	0.179
Day 3 sales (z-score)	0.549	-0.021
Day 4 sales (z-score)	0.525	-0.256
Day 5 sales (z-score)	0.408	0.026

Notes: This table reports the scoring coefficients from a principal component analysis of managers' five standardized sales predictions in the lasting price cut task. Day 1 is March 2, the new start of the price cut, and Day 4 is March 5, its original start. The table shows the two components with eigenvalues greater than 1. The sample includes $N = 6{,}840$ managers.

clusters in the space of the two components, and Figure G.8 shows the mean predicted sales of each cluster. We refer to the clusters as sales-prediction profiles and number them Cluster 1 to Cluster 4. Cluster 1 (852 managers) predicts high and mostly rising sales, Cluster 2 (3,453 managers) keeps the original pattern on its dates, and Cluster 3 (1,571 managers) predicts low sales that rise slowly. Cluster 4 (964 managers) moves the sales spike to Day 1 and predicts lower sales afterwards, as intertemporal substitution implies. Its mean predictions are 11,082, 7,335, 5,828, 4,711 and 5,368 liters on Day 1 to Day 5. The mean on Day 1 is close to the 11,273 liters observed on March 5, and 93.3 percent of managers in Cluster 4 put their highest prediction on Day 1.

Figure G.7: Sales-prediction profiles in the space of the first two principal components

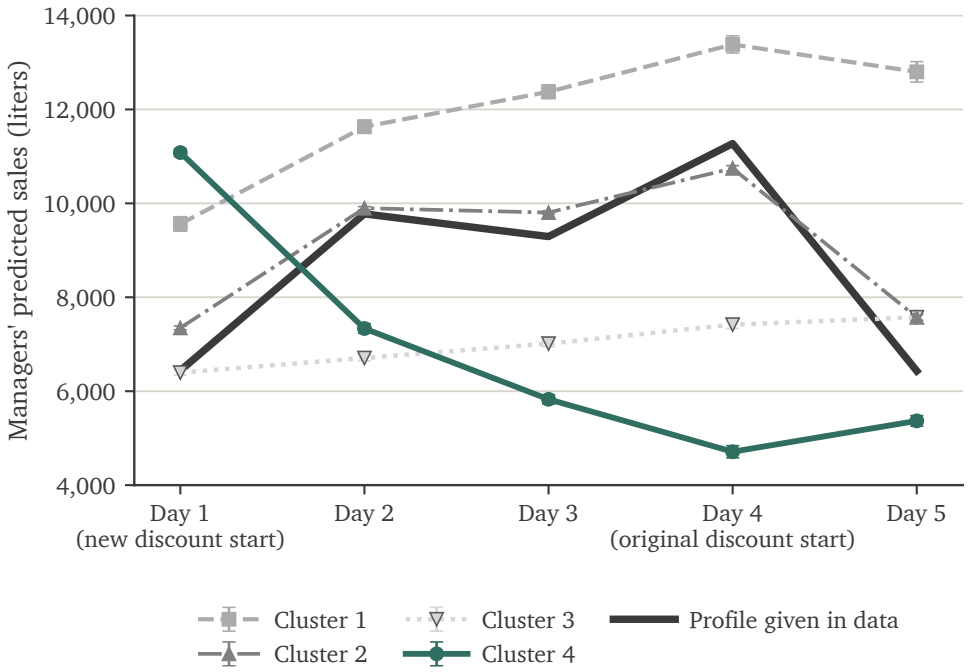


Notes: This figure shows the four sales-prediction profiles of the lasting price cut task, formed by k-means clustering of managers' five standardized daily predictions. Each point is a manager, positioned by the scores on the first two principal components, and the large outlined markers are the cluster centroids. Component 1 has positive weights on Day 2 to Day 5 and almost no weight on Day 1 (March 2), the new start of the price cut. Component 2 has a large weight on Day 1 and a negative weight on Day 4 (March 5), the original start. The sample includes $N = 6{,}840$ managers. Appendix Table G.4 reports the scoring coefficients, and Figure G.8 shows the mean predicted sales of each profile.

Table G.5 reports the share of managers in each profile, by prediction in the detergent task, and Figure G.9 plots the share in Cluster 4. A Pearson chi-squared test rejects that the profile is independent of the prediction ($p < 0.01$). Cluster 1 has 11.3 to 13.0 percent of managers for every prediction.

Table G.6 reports linear probability models of being in Cluster 4. Panel B compares managers who underestimate intertemporal substitution (predict 202 or 192) with managers who do not (predict 165 or 100). Without controls, 11.2 percent of the first group and 16.2

Figure G.8: Mean sales predictions in the lasting price cut task, by sales-prediction profile



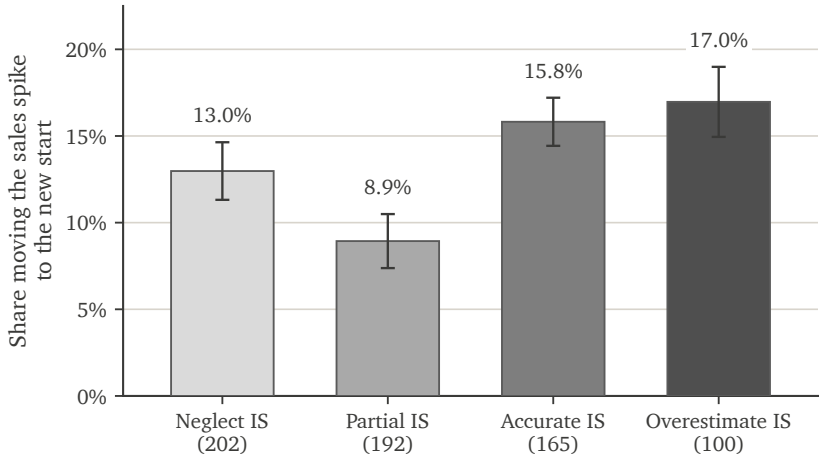
Notes: This figure shows managers' mean predicted sales on each day of the lasting price cut task, by sales-prediction profile. Day 1 is March 2, the new start of the price cut, and Day 4 is March 5, its original start. The profiles are formed by k-means clustering with $k = 4$ on managers' five standardized daily predictions. The heavy line shows the sales observed in the data managers see. The sample includes $N = 6{,}840$ managers. Error bars show 95% confidence intervals. Figure G.7 plots the profiles in the space of the first two principal components.

Table G.5: Sales-prediction profiles in the lasting price cut task, by prediction in the detergent task

	Dependent variable: Share of managers (%)				
	Predicts 202 (1)	Predicts 192 (2)	Predicts 165 (3)	Predicts 100 (4)	All managers (5)
Cluster 1: high, mostly rising sales	12.3	12.6	13.0	11.3	12.5
Cluster 2: original pattern kept	56.5	50.5	47.5	49.4	50.5
Cluster 3: low sales, rising slowly	18.2	28.0	23.7	22.3	23.0
Cluster 4: sales spike moved to the new start	13.0	8.9	15.8	17.0	14.1
Observations	1,572	1,287	2,655	1,326	6,840

Notes: This table reports the percentage of managers in each of the four sales-prediction profiles of the lasting price cut task, by prediction in the detergent task. The profiles are the clusters of a k-means clustering of managers' five standardized daily predictions. Cluster 1 predicts high and mostly rising sales, Cluster 2 keeps the observed pattern with its sales spike on March 5, Cluster 3 predicts low sales that rise slowly, and Cluster 4 moves the sales spike to March 2 and predicts lower sales afterwards. In the detergent task, 202 neglects intertemporal substitution, 192 underestimates it, 165 is the correct answer and 100 attributes the entire sales response to intertemporal substitution. Each column sums to 100 up to rounding. The sample includes $N = 6{,}840$ managers.

Figure G.9: Prediction in the detergent task and moving the sales spike to the new start of a lasting price cut



Notes: This figure shows, by prediction in the detergent task, the share of managers whose predictions in the lasting price cut task fall in Cluster 4, the profile that moves the sales spike to the new start of the price cut and predicts lower sales afterwards. In the detergent task, 202 neglects intertemporal substitution (Neglect IS), 192 underestimates it (Partial IS), 165 is the correct answer (Accurate IS) and 100 attributes the entire sales response to intertemporal substitution (Overestimate IS). The sample includes $N = 6{,}840$ managers. Error bars show 95% confidence intervals. Table G.6 reports the corresponding linear probability models.

percent of the second are in Cluster 4, and column (1) gives a gap of 5.0 percentage points. Column (2) adds version fixed effects, because the statement by another station manager differs across the five versions and may affect predictions, and the gap stays at 5.0 percentage points. Column (3) adds manager characteristics (cognitive skills, age, gender and experience as a station manager), which may relate both to the prediction in the detergent task and to predictions in the lasting price cut task, and region fixed effects. The gap is then 3.3 percentage points. The gap is statistically significant in all three columns ($p < 0.01$).

Panel A reports the four predictions separately, with managers who predict 202 as the omitted category. Of these managers, 13.0 percent are in Cluster 4. Managers who predict 192 are less likely to be in Cluster 4 in every column, by between 2.9 and 4.1 percentage points ($p < 0.05$ in column (3) and $p < 0.01$ in the other columns). Managers who predict 100 are more likely, by 4.0 percentage points without manager characteristics ($p < 0.01$) and 3.4 points with them ($p < 0.01$). The difference for managers who predict 165 is 2.8 percentage points in columns (1) and (2) ($p < 0.05$) and is not statistically significant in column (3).

G.2 Beliefs about an announced price cut

Before the detergent task, and without seeing any data, managers consider a typical station that cuts its listed gasoline price from 6.7 to 6.4, announces the cut one week in advance, and keeps it for two weeks and then for two more weeks. They answer whether weekly

Table G.6: Prediction in the detergent task and moving the sales spike to the new start of a lasting price cut

	Dependent variable: Sales spike moved to the new start		
	(1)	(2)	(3)
Panel A: Four predictions			
Predicts 192	−0.040***	−0.041***	−0.029**
	(0.012)	(0.012)	(0.012)
Predicts 165	0.028**	0.028**	0.014
	(0.011)	(0.011)	(0.011)
Predicts 100	0.040***	0.040***	0.034***
	(0.013)	(0.013)	(0.013)
R^2	0.007	0.007	0.075
Panel B: Predicts 202 or 192			
Predicts 202 or 192	−0.050***	−0.050***	−0.033***
	(0.008)	(0.008)	(0.008)
R^2	0.005	0.005	0.074
Version fixed effects		Yes	Yes
Manager characteristics and region fixed effects			Yes
Mean for managers who predict 202	0.130	0.130	0.130
Observations	6,840	6,840	6,840

Notes: This table reports linear probability models of whether a manager's predictions in the lasting price cut task fall in Cluster 4, the profile that moves the sales spike to the new start of the price cut and predicts lower sales afterwards, as intertemporal substitution implies. In the detergent task, 202 neglects intertemporal substitution, 192 underestimates it, 165 is the correct answer and 100 attributes the entire sales response to intertemporal substitution. In Panel A, the regressors are indicators for predicting 192, 165 and 100, and managers who predict 202 are the omitted category. In Panel B, the regressor is one indicator for predicting 202 or 192, the two predictions that underestimate intertemporal substitution, and managers who predict 165 or 100 are the omitted category. The sample includes $N = 6{,}840$ managers. Version fixed effects are indicators for the five randomly assigned versions of the task, which differ in the statement by another station manager shown with the data (none in one version). Manager characteristics are cognitive skills, age, gender and years of experience as a station manager. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

sales in weeks 3 and 4 will be higher than, the same as, or lower than in weeks 1 and 2. A separate question asks the same about diesel. Under intertemporal substitution, customers delay purchases until the announced cut starts. These delayed purchases fall in weeks 1 and 2, and sales in weeks 3 and 4 are lower.

For gasoline, 9.9 percent of managers who predict 202 expect lower sales in weeks 3 and 4, compared to 10.4 percent of managers who predict 192, 12.7 percent of managers who predict 165 and 12.9 percent of managers who predict 100. Across all managers the share is 11.7 percent, and 63.7 percent expect higher sales. The share expecting higher sales is 67.9 and 67.6 percent among managers who predict 202 and 192, and 62.1 and 58.5 percent among managers who predict 165 and 100. Table G.7 reports linear probability models of expecting lower sales. Relative to managers who predict 202, managers who predict 165 are 2.8 percentage points and managers who predict 100 are 3.0 points more likely to expect lower sales (both $p < 0.01$). With manager characteristics and region fixed effects, the differences are 2.7 and 3.1 points (both $p < 0.01$). The difference for managers who predict 192 is 0.5 points, and 0.7 points with controls, and neither is statistically significant.

For diesel, the shares expecting lower sales are 8.8, 8.0, 10.0 and 9.9 percent among managers who predict 202, 192, 165 and 100. The differences are smaller. Managers who predict 165 are 1.2 percentage points more likely than managers who predict 202 to expect lower sales ($p < 0.10$), and 1.4 points more likely with controls ($p < 0.05$). The differences for managers who predict 100 and 192 are not statistically significant.

G.3 Perceived elasticity

Before the detergent task, managers answer three questions that measure perceived elasticity. In the first, the price ceiling falls by 0.3 per liter in local currency, a change that affects all stations, and managers predict sales at a typical station of our partner company. In the second, a typical station alone cuts its price by 0.3 per liter while the price ceiling is unchanged, and the cut can also attract customers from rival stations. The third asks about the same cut at the manager's own station. In the second and third questions, managers predict weekly sales in the first and the second week of the price cut, and the first two questions give the typical station's weekly sales before the cut. For each question, perceived elasticity is the percentage change in predicted sales divided by the percentage change in price, in absolute value.

Figure G.10 shows mean perceived elasticity by prediction in the detergent task. Managers who predict 202 have the highest mean in each scenario, at 4.77 for the market price, 4.99 for a typical station alone and 8.32 for their own station. In every group, the mean for a typical station alone is higher than the mean for the market price, and the mean for the own station is higher than both.

Relative to all other managers, the perceived elasticity of managers who predict 202 is

Table G.7: Prediction in the detergent task and beliefs about an announced price cut

	Dependent variable: Lower sales in weeks 3 and 4			
	Gasoline		Diesel	
	(1)	(2)	(3)	(4)
Predicts 192	0.005	0.007	−0.008	−0.008
	(0.008)	(0.008)	(0.007)	(0.007)
Predicts 165	0.028***	0.027***	0.012*	0.014**
	(0.007)	(0.007)	(0.007)	(0.007)
Predicts 100	0.030***	0.031***	0.011	0.012
	(0.008)	(0.008)	(0.008)	(0.008)
Manager characteristics and region fixed effects		Yes		Yes
Mean for managers who predict 202	0.099	0.099	0.088	0.088
Observations	13,407	13,407	13,407	13,407
R^2	0.002	0.013	0.001	0.010

Notes: This table reports linear probability models of whether a manager expects weekly sales in weeks 3 and 4 of an announced price cut to be lower than in weeks 1 and 2, the answer implied by intertemporal substitution. Before the detergent task and without seeing any data, managers consider a typical station that cuts its listed gasoline price from 6.7 to 6.4, announces the cut one week in advance and keeps it for two weeks and then for two more weeks, and they answer whether weekly sales in weeks 3 and 4 will be higher than, the same as, or lower than in weeks 1 and 2. Columns (1) and (2) use this question and columns (3) and (4) the corresponding question about diesel. In the detergent task, 202 neglects intertemporal substitution, 192 underestimates it, 165 is the correct answer and 100 attributes the entire sales response to intertemporal substitution. The regressors are indicators for predicting 192, 165 and 100, and managers who predict 202 are the omitted category. The sample includes $N = 13{,}407$ managers, each of whom answered both questions and the detergent task. Manager characteristics are cognitive skills, age, gender and years of experience as a station manager. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

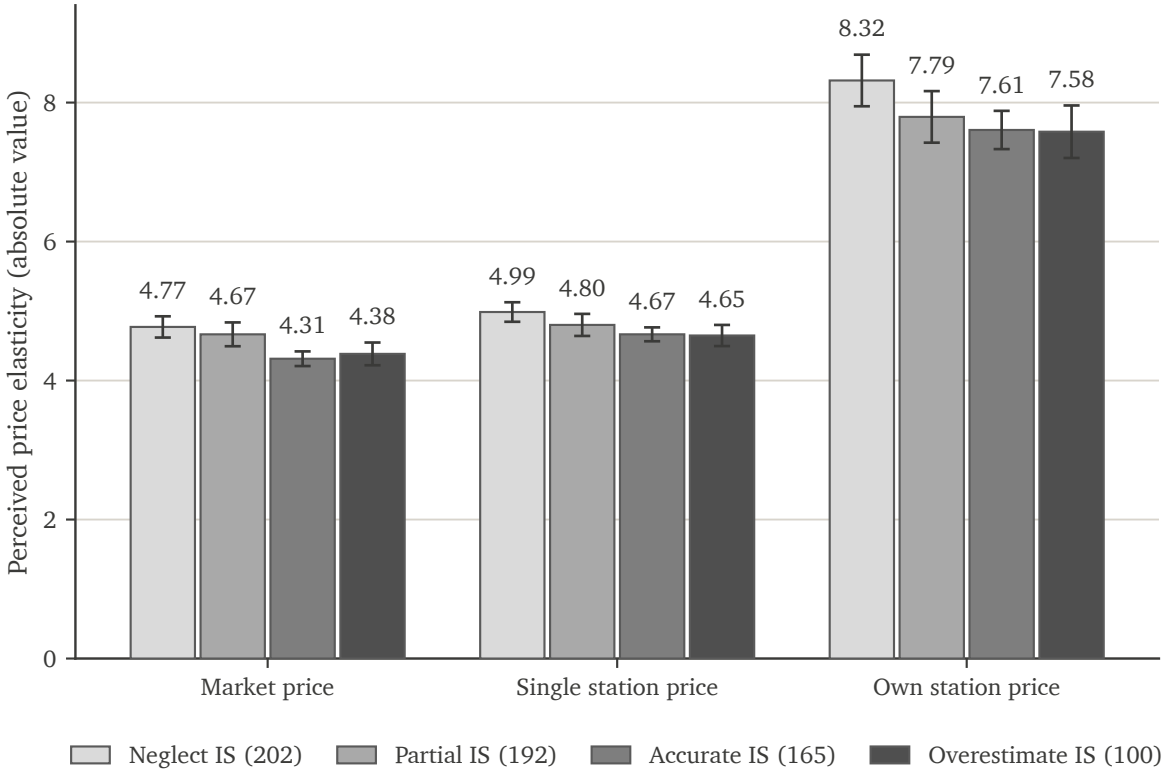
higher by 0.35 for the market price (standard error 0.09, $p < 0.01$), which is 8.0 percent of the other managers' mean. The gap is 0.29 for a typical station alone (standard error 0.08, $p < 0.01$), or 6.2 percent, and 0.67 for their own station (standard error 0.21, $p < 0.01$), or 8.8 percent. With manager characteristics and region fixed effects, the three gaps are 0.28 (standard error 0.09, $p < 0.01$), 0.21 (standard error 0.08, $p < 0.05$) and 0.60 (standard error 0.21, $p < 0.01$). The own-station measure is available for 9,482 managers. It requires the manager's own sales before the cut and is not computed when predicted sales after the cut are below that level.

The two questions about a typical station set perceived elasticity to zero when a manager predicts sales below the level before the price cut, and managers who predict 202 give such answers less often. We therefore also restrict the comparison to managers with a positive perceived elasticity. The gaps for a typical station are then smaller and not statistically significant once manager characteristics and region fixed effects are included, while the gap for the own station remains.

G.4 Beliefs about the profitability of price cuts

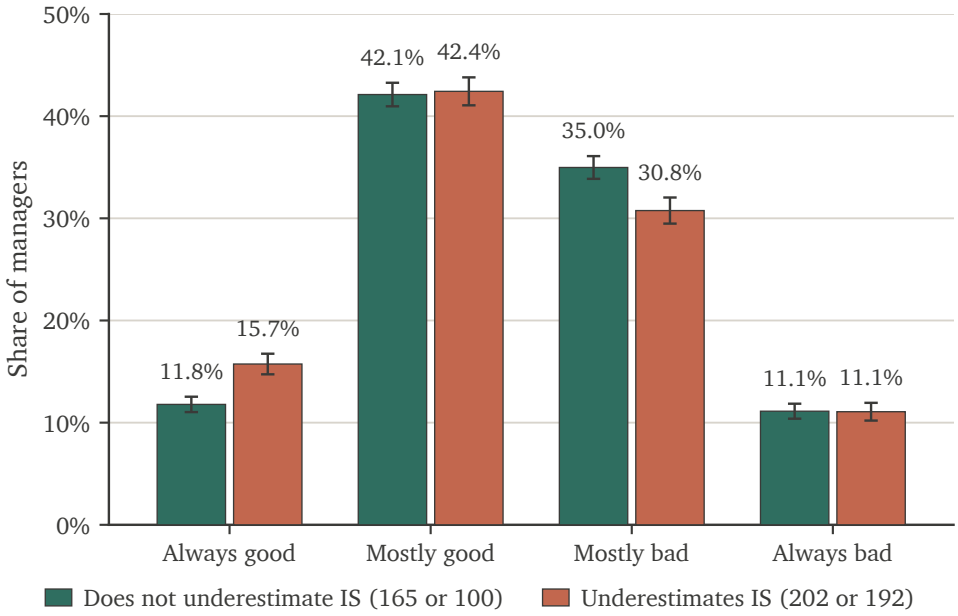
The survey also asks managers whether cuts of listed prices are always good, mostly good, mostly bad or always bad for fuel profits. Figure G.11 shows the distribution of answers for managers who underestimate intertemporal substitution (predict 202 or 192) and for managers who do not (predict 165 or 100). About 16 percent of managers who predict 202 or 192 answer that price cuts are "always good" for fuel profits, compared to 12 percent of managers who predict 165 or 100, whereas about 35 percent of managers who predict 165 or 100 answer "mostly bad", compared to 31 percent of managers who predict 202 or 192. The shares answering "mostly good" and "always bad" are about the same in the two groups.

Figure G.10: Mental models of intertemporal substitution and perceived price elasticities



Notes: This figure shows managers' mean perceived price elasticities of demand, in absolute value, by prediction in the detergent task, for three scenarios on a common axis. *Market price* is the predicted demand response at a typical station when the price ceiling falls by 0.3 per liter in local currency. *Single station price* is the predicted response when a typical station alone cuts its price by 0.3 per liter while the ceiling is unchanged. *Own station price* is the same cut at the manager's own station. IS is intertemporal substitution. The sample includes $N = 13{,}407$ managers for the market price and single station price scenarios and $N = 9{,}482$ managers for the own station price scenario. Error bars show 95% confidence intervals.

Figure G.11: Mental models of intertemporal substitution and beliefs about the profitability of price cuts



Notes: This figure shows the distribution of managers' beliefs about whether cuts of listed prices are good or bad for fuel profits, split by whether managers underestimate intertemporal substitution in the detergent task (predict 202 or 192) or not (predict 165 or 100). Error bars show 95% confidence intervals.

H Demand estimation and profit calculations

This appendix reports the demand estimates behind the profit calculations of Section 3.4 and the steps that turn the coefficients of Figure 4 into changes in daily fuel profit at an average station. The demand estimates, and the price scale used in Appendix H.2, are from Han et al. (2025a), which uses the same administrative data from our partner company.

H.1 Estimating the residual demand elasticity

We model each station as facing a downward-sloping residual demand curve that includes both consumers buying less fuel and consumers switching to competing stations. The elasticity thus incorporates competitors' responses in the local market. Following a standard approach in the gasoline demand literature (Li et al. 2014; Coglianese et al. 2017), we assume constant elasticity of demand and specify residual demand for station s as $q_s = A_s \cdot p_s^{-\varepsilon}$, where q_s is quantity sold, p_s is own price, $\varepsilon > 1$ is the price elasticity of demand and A_s is a station-specific demand shifter. Because we do not need to separate the direct response of demand from the cross-price response for this calculation, we adopt a single-instrument strategy that omits competitor price from the estimation and assume instead that the cross-price response is incorporated into ε.

To estimate the elasticity, we take advantage of the fact that our partner company is vertically integrated. Under the company's internal accounting rules, the transfer price that determines each station's marginal cost moves one-for-one with the government-imposed price ceiling, which is indexed to world oil prices and adjusted every ten working days. We therefore use the ceiling as a cost shifter that is exogenous to local demand shocks. Let Ceiling_{jt} denote the price ceiling for fuel product j that is in force during the current pricing cycle t. Taking logs of the demand equation yields the estimating equation

$$\ln q_{sjt} = a_{sj} - \varepsilon \ln p_{sjt} + \delta_{sjm} + \phi_{sjw} + \psi_{sjy} + \eta_{sjt},$$

where q_{sjt} is the quantity sold in period t at station s of product j, p_{sjt} is the corresponding own price, a_{sj} is a station-product fixed effect, δ_{sjm} are month fixed effects, ϕ_{sjw} are day-of-week fixed effects, ψ_{sjy} are year fixed effects and η_{sjt} is an error term. We treat $\ln \text{Ceiling}_{jt}$ as the excluded instrument for the endogenous price regressor $\ln p_{sjt}$.

The price ceiling changes every ten working days, which creates variation at a frequency distinct from our time fixed effects. Day-of-week fixed effects absorb systematic differences across days within a week, month fixed effects capture seasonal patterns and year fixed effects control for annual trends. Because the ten-working-day adjustment cycle spans several days of the week and does not align with calendar month boundaries, we can include all of these fixed effects and still retain variation in the instrument that identifies the demand

parameters.

We estimate an average elasticity of $\hat{\varepsilon} = 1.65$. As a check on the functional form, Appendix H.3 reports an alternative specification with linear demand, which separately identifies own-price and cross-price responses and yields an average elasticity of 1.63 in absolute value.

H.2 Profit calculations

Using the constant elasticity demand function of Appendix H.1, we calculate what the price gaps between groups of managers imply for daily fuel profit at an average station. We back out the demand shifter for an average station from the observed mean price and quantity, $\bar{A} = \bar{q} \cdot \bar{p}^{\hat{\varepsilon}}$, and compute quantities at counterfactual prices as $q = \bar{A} \cdot p^{-\hat{\varepsilon}}$, with $\hat{\varepsilon} = 1.65$.

Producer surplus is revenue minus variable cost,

$$PS = p \cdot q - MC \cdot q = (p - MC) \cdot q,$$

and changes in producer surplus equal changes in profit, because the difference between profit and producer surplus, namely fixed costs, is differenced out. Marginal cost is observed directly in the internal transfer price rather than inferred from prices. Let p_h and p_l denote the higher and the lower price, with q_h and q_l the corresponding quantities. The difference in producer surplus is

$$\Delta PS = PS_h - PS_l = (p_h - MC) \cdot q_h - (p_l - MC) \cdot q_l.$$

We set p_h equal to the average price $\bar{p}$ and p_l equal to p_h minus the estimated price gap, compute the corresponding quantity at each price, and report the percentage change in producer surplus as $\frac{\Delta PS}{PS_h}$.

The coefficients in Figure 4 are in standard deviations of the standardized monthly price ratio, which Section 3.1 defines, while the profit calculation needs a price gap in local currency per liter. In Han et al. (2025a), a coefficient of 0.017 standard deviations of the same price ratio corresponds to a price gap of 0.02 in local currency per liter at an average station, and we apply this scale to our coefficients. Relative to managers who predict 202, managers who predict 165 have a coefficient of 0.021 standard deviations ($p < 0.05$), a price gap of about 0.025 per liter, and managers who predict 100 have a coefficient of 0.025 standard deviations ($p < 0.05$), a price gap of about 0.029 per liter. Because the coefficient for managers who predict 192, 0.014 standard deviations, is not statistically significant, we report no price gap and no profit gap for that group. The price gaps are associations across managers conditional on the controls, not causal effects of holding a model, and the calculation compares managers who hold different models rather than changing the model of any one

manager. With these price gaps, the lower prices of managers who predict 202 imply 3.3% lower daily fuel profit at an average station relative to managers who predict 165, and 3.9% lower daily fuel profit relative to managers who predict 100. The calculation is static.

H.3 Linear demand

The constant elasticity specification of Appendix H.1 does not separate the own-price response from the cross-price response, and linear demand, which separates the two, provides a check on the functional form. We use the conjectural variations framework for differentiated products (Slade 1986), in which residual demand for station i is approximated by

$$q_i = f_i(p_i, p_{-i}, z) = a + b_i p_i + c_i p_{-i} + g(z),$$

where q_i is the quantity of fuel products, gasoline and diesel, sold at station i, p_i is own price, p_{-i} is the average competitor price and $g(z)$ are demand shifters. To calculate how a price change affects producer surplus, we need the response of demand to price, $\frac{dq_i}{dp_i} = b_i + c_i\theta_i$, where θ_i is the competitor price response.

We use two exogenous cost shifters, the government-imposed price ceiling in the *current* pricing cycle, $Ceiling_t$, and the ceiling from the *previous* cycle, $Ceiling_{t-1}$, to estimate the own-price response b_i and the cross-price effect c_i. Two institutional features identify these responses. First, the price ceiling is updated every ten working days following a predetermined formula indexed to world oil prices. Second, our partner company's internal accounting rule mandates that marginal cost changes *one-for-one* with the current ceiling, which also means that the previous ceiling has little influence on the current price of our partner company's stations once the current ceiling is controlled for. Independent competitors buy refined oil from independent refineries. Their marginal costs therefore adjust to the ceiling with a delay and depend on both the current ceiling and the ceiling in the previous cycle.

The estimating equation for station s and fuel product j is

$$q_{sjt} = a_{sj} + b\, p_{sjt} + c\, p_{-sjt} + \delta_{sjm} + \phi_{sjw} + \psi_{sjy} + \eta_{sjt},$$

where q_{sjt} is the quantity sold in period t at station s of product j, p_{sjt} and p_{-sjt} are the corresponding own and competitor prices, a_{sj} is a station-product fixed effect, δ_{sjm} are month fixed effects, ϕ_{sjw} are day-of-week fixed effects, ψ_{sjy} are year fixed effects and η_{sjt} is an error term.

The first-stage regressions show the expected differential impact of the two cost shifters. For our partner company's stations' own prices, the coefficient on $Ceiling_t$ is 0.98 (s.e. 0.001) and the coefficient on $Ceiling_{t-1}$ is 0.01 (0.001), consistent with near one-for-one

pass-through of current costs and a negligible influence of the lagged ceiling. For competitor prices the pattern differs. The coefficient on $Ceiling_t$ is 0.73 (0.001), a pass-through that is large but incomplete, and the coefficient on $Ceiling_{t-1}$ is 0.20 (0.001), which shows the influence of previous-cycle costs.

The second-stage coefficients from the instrumental variables regression yield the own-price and cross-price elasticities evaluated at the mean price and quantity,

$$\bar{b}\,\frac{\bar{p}}{\bar{q}} = -2.07, \qquad \bar{c}\,\frac{\bar{p}}{\bar{q}} = 1.15.$$

An event study of competitor responses to price cuts that are not driven by a change in the price ceiling gives an average competitor price response of $\bar{\theta} = 0.38$. Combining the competitor price response with the demand parameters yields the effective price elasticity when a manager changes price while the price ceiling is fixed,

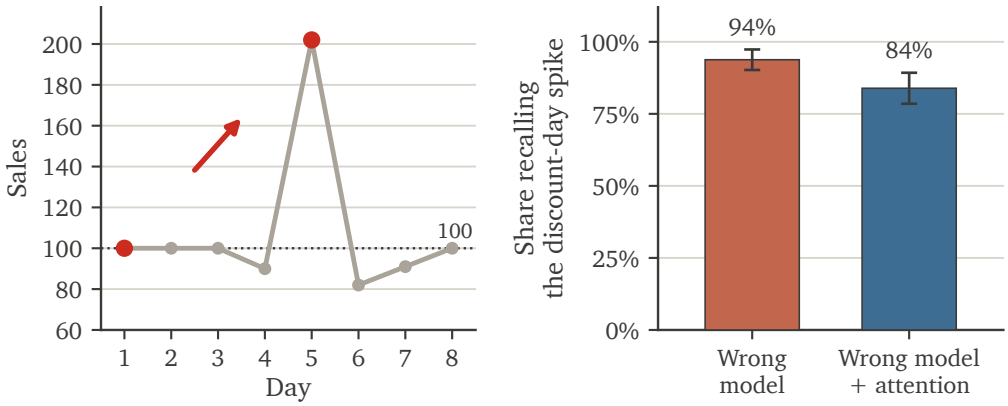
$$\frac{d\bar{q}}{d\bar{p}}\,\frac{\bar{p}}{\bar{q}} = (\bar{b} + \bar{c}\,\bar{\theta})\frac{\bar{p}}{\bar{q}} = -2.07 + 0.38 \times 1.15 = -1.63.$$

With this elasticity, the same price gaps of about 0.025 and about 0.029 in local currency per liter imply losses in daily fuel profit of 3.3% and 3.9%.

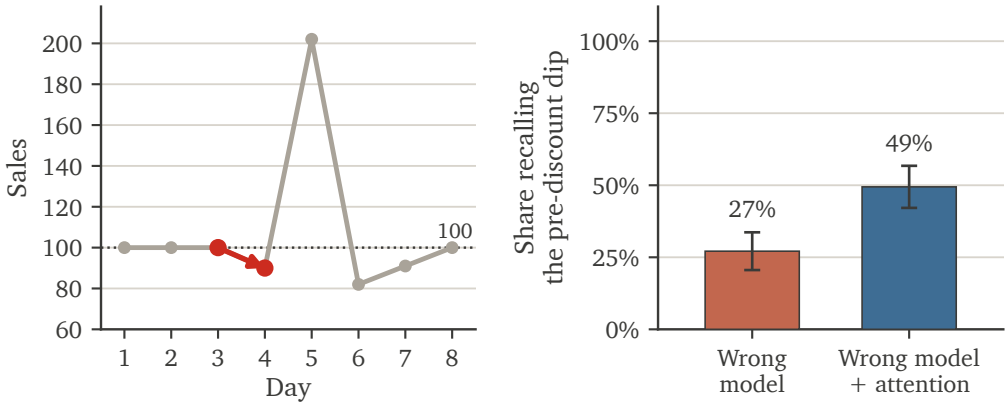
I Additional details and results from online experiments

Figure I.12: Recall of the three patterns in Study 1: *Wrong model* and *Wrong model + attention*

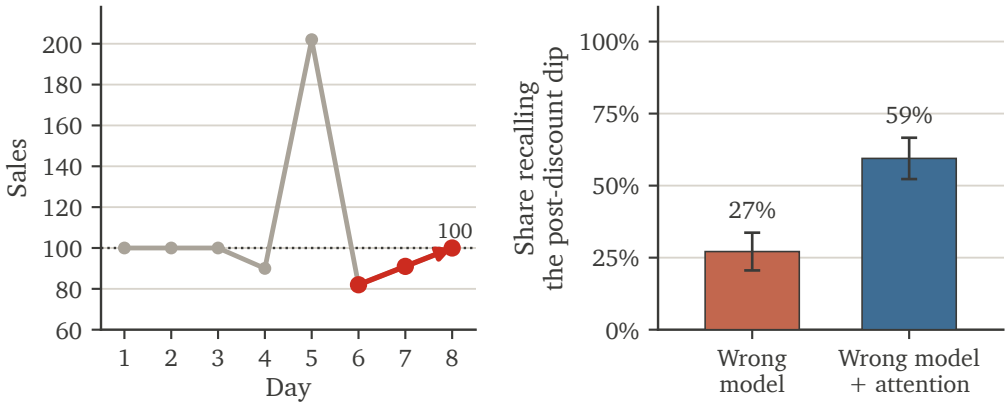
(a) The discount-day spike



(b) The pre-discount dip

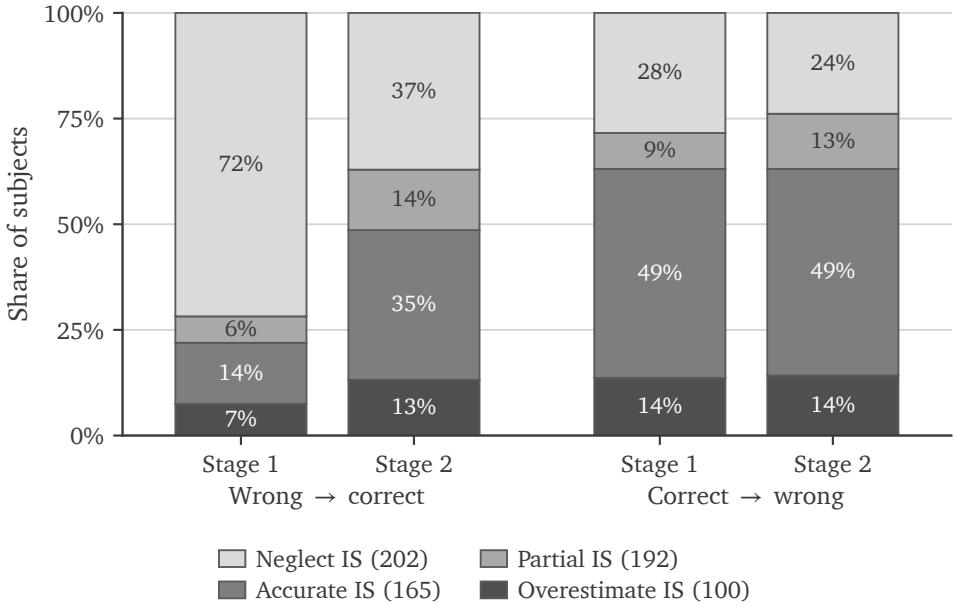


(c) The post-discount dip



Notes: This figure shows how often subjects recall each of the three patterns. Panel (a) shows the discount-day spike, Panel (b) the pre-discount dip and Panel (c) the post-discount dip. In each panel, the left plot shows sales on Days 1 to 8 and marks one pattern with an arrow, and the right plot shows the share of subjects who recall that pattern. The patterns are coded as in Figure 6. In *Wrong model + attention*, subjects answer two attention prompts about Days 3 to 4 and 9 to 10 and about Days 6 to 8 and 12 to 14 before the prediction. The sample includes $N = 177$ subjects in *Wrong model* and $N = 180$ in *Wrong model + attention*. Error bars show 95% confidence intervals.

Figure I.13: Distribution of predictions in each stage of Study 2



Notes: This figure shows the distribution of best guesses of daily sales under a permanent price of 5.2, by group and stage. The correct answer is 165, and 202 neglects intertemporal substitution. Wrong → correct: subjects assigned the wrong model first and the correct model second ($N = 720$). Correct → wrong: subjects assigned the correct model first and the wrong model second ($N = 176$).

Samples and balance. Table I.8 reports the analysis samples and subject characteristics in each treatment. The exclusions are similar across treatments. In Study 1, 17, 11 and 13 subjects in *Wrong model*, *Wrong model + attention* and *Correct model* ticked "I don't recall" for all seven days, and 4, 6 and 0 were flagged as using an AI tool. In Study 2, the numbers are 29, 35 and 21 for ticking "I don't recall" on all days and 11, 9 and 2 for AI use, in the treatment without the data shown again, the treatment with the data shown again and the treatment that assigns the correct model first. Subjects in *Wrong model* of Study 1 are older on average and less likely to be women than subjects in the other two treatments. Controlling for age, gender and Raven score, the effect of the wrong model on the share predicting 202 is 43 percentage points and the effect of the attention prompts is −13 percentage points (both $p < 0.01$), compared to 42 and −11 percentage points without controls. Among all 600 subjects who completed Study 1, the two effects are 38 and −9 percentage points ($p < 0.01$ and $p < 0.05$). Among all 1,003 subjects who completed Study 2, the difference between the changes in the two orders is 31 percentage points ($p < 0.01$), and showing the data again changes the share predicting 202 in stage 2 by less than one percentage point.

Predictions typed into an empty box. In Study 1, subjects choose their prediction from four answers, and the menu could suggest a model to a subject who would not otherwise have one. We therefore ran a follow-up pilot of Study 1 in August 2026 in which subjects type the prediction into an empty box and earn $1.00 if the value is within 10 of 165. The

Table I.8: Sample size and subject characteristics by treatment

	N	Age	Female	Raven score
Study 1				
Wrong model	177	40.8	0.46	4.65
Wrong model + attention	180	37.6	0.60	4.48
Correct model	192	38.6	0.59	4.55
p-value, equal means		0.05	0.02	0.76
Study 2				
Wrong → correct, no data in stage 2	367	39.5	0.57	4.50
Wrong → correct, data shown again	353	39.5	0.58	4.57
Correct → wrong	176	39.1	0.60	4.62
p-value, equal means		0.95	0.89	0.80

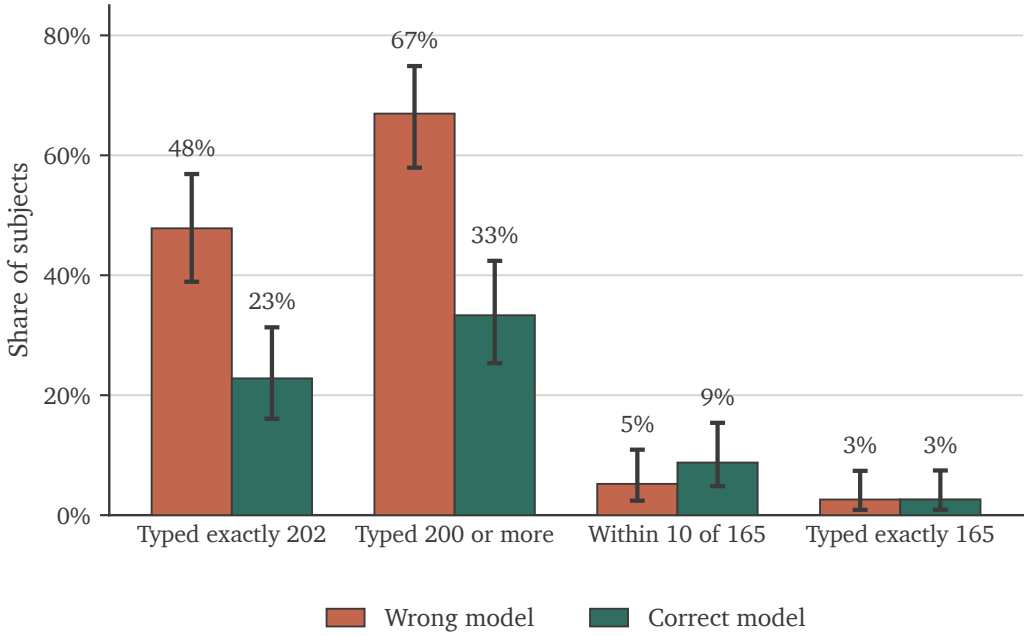
Notes: This table reports the analysis samples after exclusions. Age and gender are taken from subjects' Prolific profiles, and Female is the share reporting female. Raven score is the number of correct answers to nine Raven matrices. The p-values are from F-tests of equal means across the treatments of each study.

pilot has two treatments, *Wrong model* with 115 subjects and *Correct model* with 114. The pages, the descriptions, the recall task and the Raven matrices are those of Study 1, and a question about confidence in the typed prediction, which is not incentivized, replaces the allocation of 100 points. Figure I.14 shows the predictions. The wrong model raises the share of subjects typing exactly 202 from 23 to 48 percent, and the share typing 200 or more from 33 to 67 percent (both $p < 0.01$). About 12 percent of subjects in *Wrong model* type 200 or 203, values the menu does not offer. The wrong model also lowers recall of the pre-discount dip by 13 percentage points and recall of the post-discount dip by 16 percentage points (both $p < 0.05$), while recall of the discount-day spike and the placebo differ between the treatments by less than 4 percentage points. Few subjects in either treatment type a value close to 165: 5 percent in *Wrong model* and 9 percent in *Correct model* are within 10 of it, and the difference is not significant.

I.1 Attention measured by clicking

Recall measures attention after the data are gone. An earlier pilot measured attention while subjects worked with the data. Subjects saw the same table of 14 days, with every sales value hidden behind a button. Clicking a button revealed that day's sales for one second, after which the value was hidden again, and a subject who wanted to compare two days had to click repeatedly. We count the clicks on a day as attention to that day. The dip days are the days of the pre-discount and post-discount dips, Days 4, 6, 7, 10, 12 and 13. The baseline days, on which sales are 100, are Days 1, 2, 3, 8, 9 and 14. Subjects were randomly assigned the wrong model or the correct model, as in Study 1, and the pilot had no attention prompts. We use the treatments in which clicking is free, and leave out a further treatment

Figure I.14: Predictions typed into an empty box



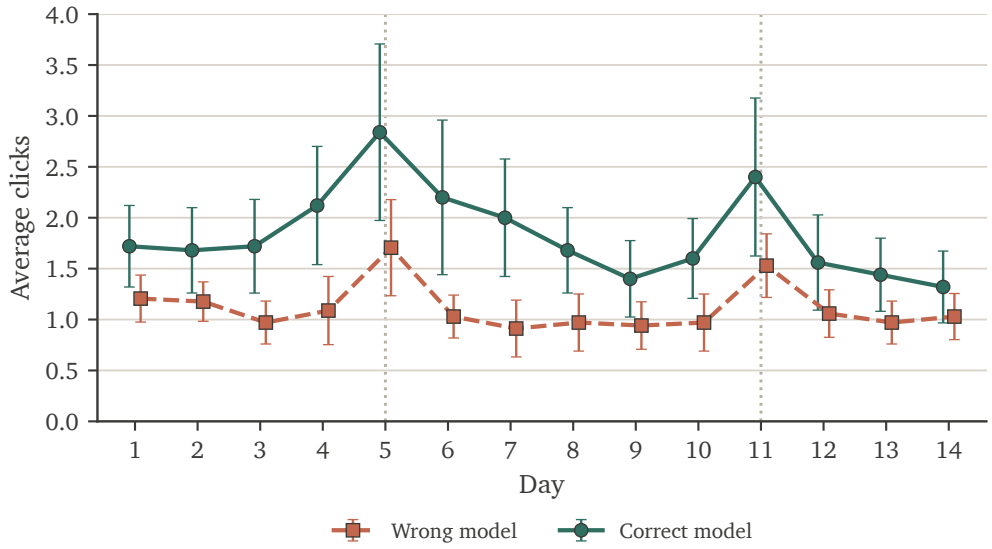
Notes: This figure reports results from a follow-up pilot of Study 1 in which subjects type their prediction of daily sales under a permanent price of 5.2 into an empty box. Each pair of bars gives the share of subjects whose typed prediction meets the condition below it. Subjects earn $1.00 if the typed value is within 10 of 165. Error bars show 95% confidence intervals. *Wrong model* $N = 115$, *Correct model* $N = 114$.

that charged $0.05 per click.

Figure I.15 plots the average number of clicks on each day. Subjects assigned the wrong model click the dip days less than subjects assigned the correct model, 1.01 compared to 1.82 clicks per day ($p < 0.01$). They also click less on every other kind of day, 1.62 compared to 2.62 on the discount days and 1.05 compared to 1.59 on the baseline days (both $p < 0.05$), and they click 15.6 times in total compared to 25.7 ($p < 0.01$). Part of the difference is therefore in how much of the table subjects inspect at all. The contrast between the dip days and the baseline days separates the two. Subjects assigned the correct model click the dip days 0.23 times per day more than the baseline days, while subjects assigned the wrong model click them 0.04 times per day less, and the difference between the two contrasts is 0.28 clicks per day ($p < 0.05$). The share of a subject's clicks that falls on the dip days is 0.36 under the wrong model and 0.39 under the correct model, a difference that is not significant.

The pilot is small, and clicking imposes a cost of attention that the recall measure does not. We report it because it measures attention while the data are on the screen rather than afterwards, and because it points the same way as the recall results in Section 4.1.2.

Figure I.15: Clicks on each day by assigned model



Notes: This figure shows the average number of clicks on each day's sales value, by assigned model, in the pilot treatments in which every value is hidden and a click reveals one value for one second. The dip days are Days 4, 6, 7, 10, 12 and 13, and the dotted lines mark Days 5 and 11, the discount days. Error bars show 95% confidence intervals. The sample includes $N = 34$ subjects assigned the wrong model and $N = 25$ assigned the correct model, and leaves out the treatment that charged for clicks.